

Programme Structure and Syllabi for B. Tech. Civil Engineering (2023 batch onwards)



**Department of Civil Engineering
Punjab Engineering College
(Deemed to be University)
Chandigarh, India**

First Year

Semester-I

Course Code	Course Name	L	T	P	Credits
OR2301	Orientation	-	-	-	1
OR2302	Introduction to Discipline Engineering	1	0	0	1
MA2301	Calculus	3	0	2	4
PY2302	Mechanics and Optics	3	0	2	4
GS2301	Introduction to Environmental Sciences	1	0	0	1
HS2351	Communication Skills	2	0	2	3
ES2301	Introduction to Computer Programming	3	0	2	4
ES2306	Strength of Materials	3	0	2	4
	Total Credits:				22

Semester-II

Course Code	Course Name	L	T	P	Credits
MA2302	Linear Algebra, Differential Equations and Vector Calculus	3	0	2	4
CH2302	Applied Chemistry – II (Civil & Production)	3	0	2	4
ES2302	Engineering Drawings with CAD Software	2	0	4	4
GS2302	Universal Human Values	1	0	0	1
ES2303	Skill Development Workshop	0	0	4	2
ES2307	Introduction to Product Design	0	0	4	2
ES2304	Introduction to Mechatronics	3	0	2	4
	Total Credits:				21

Second Year

Semester-III

Course Code	Course Name	L	T	P	Credits	Course Type
CEN3001	Geo-informatics	3	0	2	4	Core Course
CEN3002	Structural Analysis –I	3	0	2	4	Core Course
CEN3003	Fluid Mechanics	3	0	2	4	Core Course
CEN3004	Building Material and Building Construction	3	0	2	4	Core Course
CEN3005	Transportation Engineering	3	0	2	4	Core Course
Total Credits:					20	

Semester-IV

Course Code	Course Name	L	T	P	Credits	Course Type
	HSM – II	3	0	0	3	
CEN4001	Design of concrete structure-I	3	0	2	4	Core Course
CEN4002	Structural analysis –II	3	0	0	3	Core Course
CEN4003	Soil Mechanics – T&P	3	0	2	4	Core Course
CEN4004	Irrigation Engineering & Open Channel Hydraulics	3	0	0	3	Core Course
CEN4005	Environmental Engineering–I	3	0	0	3	Core Course
CEN4006	Railway, Airport & Tunnel Engineering	3	0	0	3	Core Course
Total Credits:					23	

Third Year

Semester-V

Course Code	Course Name	L	T	P	Credits	Course Type
CEN5001	Design of steel structure	3	0	0	3	Core Course
CEN5002	Design of concrete structures –II	3	0	0	3	Core Course
CEN5003	Environmental Engineering- II	3	0	2	4	Core Course
CEN5004	Hydrology and design of Hydraulic Structures	3	0	0	3	Core Course
CEN5005	Foundation Engineering	3	0	0	3	Core Course
CEP5101	Minor Project	0	0	8	4	Core Course
	Total Credits:				20	

Semester-VI

Course Code	Course Name	L	T	P	Credits	Course Type
CEN6001	Internship Part-I	0	0	12	6	Core Course
CEN6002	Internship Part-II	0	0	4	2	Core Course
CEN6003	Internship Part-III	0	0	8	4	Core Course
	Total Credits:				12	
	OR					
	Optional Course Work					
	Course Name	L	T	P	Credits	Course Type
CEE1012, CEE1016, CEE1019, CEE1021	Deptt. Elective Course-V	3	1	0	4	Elective Course
CEO1001	Disaster Management	3	1	0	4	Open Elective Course
CEP6001	Project Work	0	0	8	4	Core Course
	Total				12	

Fourth Year

Semester-VII

Course Code	Course Name	L	T	P	Credit	Course Type
	HSM – IV	3	0	0	3	
CEE1005-10	Departmental Elective Course – I	3	1	0	4	Elective Course
CEE1011-15	Departmental Elective Course – II	3	1	0	4	Elective Course
CEO1001	Disaster Management	3	1	0	4	Open elective Course
CEO1002	Green Buildings	3	1	0	4	Open Elective Course
CEP7001	Major Project – I	0	0	8	4	Core Course
	Total Credits:				23	

Semester-VIII

Course Code	Course Name	L	T	P	Credit	Course Type
	HSM – III	3	0	0	3	
CEE1001-02, CEE1016-19	Departmental Elective Course – III	3	1	0	4	Elective Course
CEE1003-04, CEE1020-23	Departmental Elective Course – IV	3	1	0	4	Elective Course
CEO1003	Clean Technology	3	1	0	4	Open Elective Course
	Proficiency – I	0	0	2	2	
CEP8001	Major Project – II*	0	0	8	4	Core Course
	Total Credits:				21	

HONOURS DEGREE IN CIVIL ENGINEERING

Course Code	Course Name	L	T	P	Credits	Course Type
CEH1001	Honours Project-I	0	0	6	3	Honours
CEH1002	Honours Project-II*	0	0	8	4	Honours
CEH1003	Honours Project-III**	0	0	10	5	Honours
Total Credits					12	

MINOR SPECIALIZATION IN CIVIL ENGINEERING

Course Code	Course Name	L	T	P	Credits	Course Type
CEM1001	Structure Analysis and Design	3	0	2	4	Minor
CEM1002	Intelligent Transportation Systems	3	1	0	4	Minor
CEM1003	Water and Waste Water Engineering	3	1	0	4	Minor
CEM1004	Minor Specialization Project-I	0	0	6	3	Minor
CEM1005	Minor Specialization Project-II*	0	0	6	3	Minor
Total Credits					18	

MINOR SPECIALIZATION IN SUSTAINABLE INFRASTRUCTURE ENGINEERING

Course Code	Course Name	L	T	P	Credits	Course Type
CEM2001	Green Buildings	3	0	2	4	Minor
CEM2002	Environmental Impact Assessment	3	1	0	4	Minor
CEM2003	Road Safety	3	1	0	4	Minor
CEM2004	Minor Specialization Project-I	0	0	6	3	Minor
CEM2005	Minor Specialization Project-II*	0	0	6	3	Minor
Total Credits					18	

- To get Honours, the student will have to complete additional 12 credits of discipline electives.
- Minor specialization will be given to a student who earns 18 credits from the basket of MSC offered by one department (outside the parent department).

Minimum credits for award of degree = 162

Credits to be earned for degree with Honours = 162 + 12 = 174

Credits to be earned for degree with Minor Specialization = 162 + 18 = 180

- * It is proposed that 'A+' grade should only be given to students who have at least one paper accepted/published in refereed Journal or full length papers published in peer reviewed

conferences organized by IISC/IIT/NIT/IIIT/Premier R&D organizations/ Professional societies or any patent published or first 3 position holders in any reputed national hackathons or project competitions or participation in International hackathons or project competitions.

** It is proposed that 'A+' grade should only be given to students who have at least one paper accepted/published in SCI/SCIE/SSCI/Web of Science/SCOPUS Indexed Journals or any patent published or first 3 position holders in any reputed national hackathons or project competitions or participation in International hackathons or project competitions.

*** The following ESC courses are proposed to be mandatory for all branches: -

- i. Computer Programming (ESC-I)
- ii. Engineering Drawing (ESC-II)
- iii. Workshop (IoT/ECE/SIEMENS etc.) (ESC-III)

HONOURS / MINOR SPECIALIZATION PROGRAMME

Students with good academic performance (having CGPA ≥ 8.5 for Honours and CGPA ≥ 7 for Minor specialization) and desirous of excelling further in academics have the following opportunities:

- a) **Honours:** To get Honours in the parent discipline, a student will have to earn additional 12 credits (over and above 162 credits) in the parent department.
- b) **Minor Specialization:** To get Minor specialization, a student will have to complete 18 credits (over and above 162 credits) by doing courses outside the parent department during the entire duration of the programme in the institute.

Honours/ Minor specialization will be awarded to a student on the recommendation of the DAPC of the parent department. A student may do Honours, Honours with Minor Specialization OR Minor Specialization only.

1. Department Core Courses (DCC)

Course Code	Course Name	L	T	P	Credits	Course Type
CEN3001	Geo-informatics	3	0	2	4	Core Course
CEN3002	Structural analysis –I	3	0	2	4	Core Course
CEN3003	Fluid Mechanics	3	0	2	4	Core Course
CEN3004	Building Material and Building Construction	3	0	2	4	Core Course
CEN3005	Transportation Engineering	3	0	2	4	Core Course
CEN4001	Reinforced Concrete Design -I	3	0	2	4	Core Course
CEN4002	Structural analysis –II	3	0	0	3	Core Course
CEN4003	Soil Mechanics	3	0	2	4	Core Course
CEN4004	Irrigation Engineering and Open Channel Hydraulic	3	0	0	3	Core Course
CEN4005	Environmental Engineering – I	3	0	0	3	Core Course
CEN4006	Railway, Airport & Tunnel Engineering	3	0	0	3	Core Course
CEN5001	Design of steel structure	3	0	0	3	Core Course
CEN5002	Reinforced concrete design–II	3	0	0	3	Core Course
CEN5003	Environmental Engineering- II	3	0	2	4	Core Course
CEN5004	Hydrology and Hydraulic Structures	3	0	0	3	Core Course
CEN5005	Foundation Engineering	3	0	0	3	Core Course

2. Department Elective Courses (DEC)

Department Elective Courses – I

Course Code	Course Name	L	T	P	Credits	Course Type
CEE1005	Concrete Technology	3	1	0	4	Elective Course
CEE1006	Multi Modal Urban Transportation System	3	1	0	4	Elective Course
CEE1007	Construction Project Management and Economics	3	1	0	4	Elective Course
CEE1008	Solid Waste Management	3	1	0	4	Elective Course
CEE1009	Engineering Geology	3	0	2	4	Elective Course
CEE1010	Ground Water Hydraulics	3	1	0	4	Elective Course

Department Elective Courses – II

Course Code	Course Name	L	T	P	Credits	Course Type
CEE1011	Design of Steel Structures - II	3	1	0	4	Elective Course
CEE1012	Road Safety	3	1	0	4	Elective Course
CEE1013	Hydro Power Engineering	3	1	0	4	Elective Course
CEE1014	Unit Processes in water and waste water treatment	3	1	0	4	Elective Course
CEE1015	Disaster Management	3	1	0	4	Elective Course

Department Elective Courses – III

Course Code	Course Name	L	T	P	Credits	Course Type
CEE1001	Seismic design of Structures	3	1	0	4	Elective Course
CEE1016	Traffic Engineering	3	1	0	4	Elective Course
CEE1017	Stochastic Hydrology	3	1	0	4	Elective Course
CEE1018	Environmental Pollution Management	3	1	0	4	Elective Course
CEE1002	Remote Sensing and GIS	3	0	2	4	Elective Course
CEE1019	Green Building Materials and Techniques	3	1	0	4	Elective Course

Department Elective Courses – IV

Course Code	Course Name	L	T	P	Credits	Course Type
CEE1020	Bridge Engineering	3	1	0	4	Elective Course
CEE1003	Transportation Planning & Management	3	1	0	4	Elective Course
CEE1021	Water Resources Planning and Management	3	1	0	4	Elective Course
CEE1004	Environmental Impact Assessment	3	1	0	4	Elective Course
CEE1022	Reinforced Soil Analysis and Design	3	1	0	4	Elective Course
CEE1023	Clean Technology	3	1	0	4	Elective Course

Department Elective Courses – V

Course Code	Course Name	L	T	P	Credits	Course Type
CEE1001	Seismic design of Structures	3	1	0	4	Elective Course
CEE1002	Remote Sensing and GIS	3	0	2	4	Elective Course
CEE1003	Transportation Planning & Management	3	1	0	4	Elective Course
CEE1004	Environmental Impact Assessment	3	1	0	4	Elective Course

3. Open Elective Courses (OEC)

Course Code	Course Name	L	T	P	Credits	Course Type
CEO 1001	Disaster Management	3	1	0	4	Open Elective Course
CEO 1002	Green Buildings	3	1	0	4	Open Elective Course
CEO 1003	Clean Technology	3	1	0	4	Open Elective Course

Open Elective Course (VI Semester)

Course Code	Course Name	L	T	P	Credits	Course Type
CEO 1001	Disaster Management	3	1	0	4	Open Elective Course

4. Minor Specialization Courses Minor Specialization (Civil Engineering)

Course Code	Course Name	L	T	P	Credits	Course Type
CEM 1001	Structural Analysis and Design	3	1	0	4	Minor
CEM 1002	Intelligent Transportation System	3	1	0	4	Minor
CEM 1003	Water and Waste Water Engineering	3	1	0	4	Minor

Minor Specialization (Sustainable Infrastructure Engineering)

Course Code	Course Name	L	T	P	Credits	Course Type
CEM2001	Green Buildings	3	1	0	4	Minor
CEM2002	Environmental Impact Assessment	3	1	0	4	Minor
CEM2003	Road Safety	3	1	0	4	Minor

NAME OF DEPTT	: Department of Chemistry	
Course name	: Applied Chemistry I (CSE, ECE, EE, DS, VLSI, AI)	
Course Code	: CH2301	
Year	: 23-24 (2 nd semester)	
Credits	: 4	
L T P	: 3 0 2	Total No. of Lecture-42

Objective: To teach the fundamentals and applications of Chemical Sciences essential for the development of electrical and electronic materials and technologies. Students will be learning various analytical techniques for the characterizations of electronic organic/inorganic materials.

<i>Lecture wise breakup</i>		<i>No. of Lectures = 42</i>
1	Fundamentals for Applied Chemistry Molecular orbital theory, Jahn-Teller Effect in Crystal Field Theory, Solid state chemistry: Crystal defects and line imperfections, Reaction mechanism in organic chemistry: Principles and methods of determination, Chemical Kinetics: Langmuir–Hinselwood Mechanism, acid-base equilibria in non aqueous media, Introduction to Computational chemistry and open source softwares	(10)
2	Polymeric Materials Mechanism and methods of polymerization, structure-activity relationship, Conducting Polymers: types (n- or p- doping) and applications, Polymeric fibre materials	(6)
3	Spectroscopic Methods for structural Analysis: Principle and applications (UV, IR, NMR, SEM and TEM)	(9)
4	Energy Storage and Sensing Devices: Fundamentals of Electrochemistry, types of electrodes, Reference electrodes, Ion-selective electrodes, Fuel cells, Batteries (Lithium-ion Batteries and EV Batteries), Renewable energy (Artificial photosynthesis), Solar cells, Sensors for IoT	(8)
5	Chemistry of Electronic and Electrical Materials Semi-conductor and super conducting Materials, Carbon materials, Optical Materials (OLED), 2D Materials, Magnetic materials.	(9)

Outcomes: 1. To be able to apply the fundamentals of chemistry towards developing new Technologies based on new materials.

2. To attain the essential analytical skills and designing of materials for electrical and electronic applications.

3. Application of software as important tools in technological applications.

Books:

1. Concise Inorganic Chemistry, by J. D. Lee, 5th Edition, 2003 (Chapman & Hall).
2. Organic Chemistry by S. M. Mukherji, and S. P. Singh, 2017 (Newagepublishers).
3. Principles of Physical Chemistry by Puri, Sharma and Pathania, 2008 (W.H. Freeman& Co).
4. Atkin's Physical Chemistry by Peter Atkins, Julio de Paula, 7th Edition (Oxford University Press).
5. Principle of Polymerization by G. Odian, 4th Edition, (John Wiley & Sons, Inc.).
6. D. S. Pavia, G.M. Lasmpman and G.S. Kriz: Introduction to Spectroscopy, 4th Edition,(Thomson learning, Indian Edition).
7. Computational chemistry: Introduction to theory and applications of molecular and quantum mechanics: Lewars Errol G. (Springer)
8. NPTEL web lectures: Chemistry of Materials, Engineering Chemistry I & III.

Practicals	
Sr. No.	Name of Experiment
1.	To find the strength of the given sodium hydroxide solution by titrating it against standard solution of hydrochloric acid using pH meter.
2.	To determine the ferrous content in the supplied sample of iron ore by titrimetic analysis against standard $K_2Cr_2O_7$ solution using potassium ferricyanide $[K_3Fe(CN)_6]$ as external indicator.
3.	To find the strength of Sodium hydroxide (NaOH) solution by titrating it against 0.1 N hydrochloric acid conductometrically.
4.	Determination of reaction rate constant of acid catalyzed Hydrolysis of Ester.
5.	Verify Beer-Lambert's law for $KMnO_4$ colorimetrically.
6.	Synthesis of Polyaniline based conducting polymer.
7.	To determine the value of rate constant (k) for the inversion of sucrose by polarimeter.
8.	To prepare nickel dimethyl glyoxime complex, $[Ni(DMG)_2]$. Illustrate the structure of the complex using FTIR.
9.	Synthesis of iron oxide nanoparticles and characterization using FTIR.
10.	Predicting the best Molecular docking conformations of a protein with the help of Swiss Dock.

NAME OF DEPTT.	:	Department of Chemistry	
Course name	:	Applied Chemistry II (Mech.,Prod.,Metta,Aero.,Civil Engg.)	
Course Code	:	CH2302	
Year	:	23-24 (1 st semester)	
Credits	:	4	
L T P	:	3 0 2	Total No. of Lecture-42

Objective: To teach the fundamentals and applications of Chemical Sciences essential for the development of engineering materials and processing technologies. Students will be learning various analytical techniques for the characterizations of composites and hybrid materials.

Lecture wise breakup		No. of Lectures = 42
1	Fundamentals for Applied Chemistry Molecular orbital theory, Jahn-Teller Effect in Crystal Field Theory, Solid state chemistry: Crystal defects and line imperfections , Reaction mechanism in organic chemistry: Principles and methods of determination, Chemical Kinetics: Langmuir –Hinshelwood Mechanism, Acid -base equilibria in non aqueous media	(8)
2	Polymer Chemistry Mechanism and methods of polymerization, Structure-Activity relationship, High performance polymers and applications, Natural and synthetic fibers, biodegradable polymers	(5)
3	Fuels and Catalysis Petroleum processing, Solid and liquid Fuels for Propellants, Chemistry of combustion and equations, Catalytic convertors	(5)
4	Electrochemistry and Corrosion: Introduction to Electrochemistry, Fuel cells, Batteries (Lithium-ion Batteries and EV Batteries), Corrosion control and protective coatings	(7)
5	Spectroscopic Methods for structural analysis: Principle and Applications (UV, IR, NMR, AAS/ICP-AES, SEM, TEM, XRD).	(10)
6	Chemistry of Engineering Materials Ceramic and Cement Materials, Metals and Alloys ,Phase change materials, Bio-inspired materials, Composite materials, Smart materials: Chemical compositions and its applications	(7)

Outcomes:

1. To be able to apply the fundamentals of chemistry towards emerging materials to benefit the societal needs.
2. To attain the essential analytical skills and designing of materials for various applications.
3. To be able to identify the chemical compositions required for designing of high performance materials.

Books:

1. Concise Inorganic Chemistry, by J. D. Lee, 5th Edition, 2003 (Chapman & Hall).
2. Organic Chemistry by S. M. Mukherji, and S. P. Singh, 2017 (Newagepublishers).
3. Principles of Physical Chemistry by Puri, Sharma and Pathania, 2008 (W.H. Freeman & Co).
4. Atkin's Physical Chemistry by Peter Atkins, Julio de Paula, 7th Edition (Oxford University Press).

5. Principle of Polymerization by G. Odian, 4th Edition, (John Wiley & Sons, Inc.).
6. D. S. Pavia, G.M. Lasmpman and G.S. Kriz: Introduction to Spectroscopy, 4th Edition, (Thomson learning, Indian Edition).
7. NPTEL web lectures: Chemistry of Materials, Engineering Chemistry I & III.

Practicals	
Sr. No.	Name of Experiment
1.	Measurement of acid base equilibria by pH meter in water sample.
2.	To determine the ferrous content in the supplied sample of iron ore by titrimetric analysis against standard K ₂ Cr ₂ O ₇ solution using potassium ferricyanide [K ₃ Fe(CN) ₆] as external indicator.
3.	Synthesis of polyaniline based conducting polymer.
4.	To find the strength of given NaOH solution by titrating it against 0.05 N Hydrochloric acid solution using a conductivity meter.
5.	Synthesis of iron oxide nanoparticles and characterization using various techniques (UV/ FTIR/ SEM/XRD).
6.	To determine the value of rate constant (k) for the inversion of sucrose by polarimeter.
7.	To verify Lambert Beer's law for KMnO ₄ using UV-Visible Spectrophotometer.
8.	To determine the percentage of Cu in the copper alloy solution provided 0.1 N hypo solution.
9.	To prepare nickel dimethyl glyoxime complex, [Ni(DMG) ₂]. Illustrate the structure of compound using FTIR.
10.	Determination of reaction rate constant of acid catalyzed hydrolysis of ester.

Course Name	:	Introduction to Computer Programming
Course Code	:	ES 2301
Credits	:	4
L T P	:	3 0 2

Course Objectives:
- To develop logical skills so that students should be able to solve basic programming problems - To use programming knowledge to develop small projects including basic GUI design

Total No. of Lectures: 42

Lecture wise Breakup		No. of Lectures
Unit 1	INTRODUCTION TO PROGRAMMING Evolution of languages: Machine languages, Assembly languages, High-level languages. Software requirements for programming: System softwares like operating system, compiler, linker, loader; Application softwares like editor. Flowcharts. Algorithm, specification of algorithm. Industrial uses of programming in various domains	3
Unit 2	DATA TYPES and OPERATORS AND EXPRESSION Storing integers, numbers with decimals, characters and strings, typecasting. User input and output, use of command line arguments Operators: arithmetic operators, relational operators, logical operators, bitwise operators, miscellaneous operators. Expressions and their evaluation. Precedence and associativity rules.	7
Unit 3	ITERABLE CONTAINERS and STATEMENTS List, set, tuple and dictionaries; range function; difference between various iterable containers Decision making statements: if, if-else, nested if and if-else. Control statements: for & while loops, nested loops; Role of statements like break, continue	7
Unit 4	FUNCTIONS and CLASSES Advantage of modularizing program into functions, function definition and function invocation. Function arguments: default, keyword and positional arguments. Scope and lifetime of a variable. Recurrence relations and Recursion Advantage of using classes, defining class data members & functions and accessing using objects. Constructors and destructors in a class, parameterized constructors.	8

Unit 5	GUI design Introduction to tkinter library, use of TK & mainloop methods, use of widgets like Button, Canvas, Checkbutton, Entry, Frame, MenuButton, Listbox, Menu, Scrollbar, Text, Message, Pack, Grid, place etc. for GUI design.	5
Unit 6	SORTING AND SEARCHING Searching: Linear search, binary search and hash search. Sorting: Insertion sort, selection sort, bubble sort, quick sort, merge sort, heap sort, and Bucket sort. Time and space complexity of algorithms, comparing algorithms	9
Unit 7	Problem Solving Real-world programming problems	3

Total No. of Turns: 14

List of Experiments		Number of Turns
1	Implement programs to input/output various data types	2
2	Implement programs to use command line arguments and operators	2
3	Implement programs making use of conditional statements and loops	2
4	Implement programs making use of iterable containers	2
5	Implement programs making use of functions and recursion	2
6	Implement various searching and sorting algorithms	2
7	Project work including GUI design using tkinter	2

Course Outcomes: At the end of the course, students will be able to:	
1	Develop understanding of the fundamental concepts essential for programming.
2	Make efficient use of iterables, function and classes to programming problems
3	Develop simple GUI applications
4	Learn to compare algorithms and improve efficiency of algorithms

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1.	Python Programming: An Introduction to Computer Science by John M. Zelle, Franklin, Beedle& Associates Inc	Latest Edition
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1.	Think Python, How to Think Like a Computer Scientist, Version 2.0.17, Allen Downey Green Tea Press Needham, Massachusetts	Latest Edition
2.	Core python programming, Dr. R. Nageswara Rao, 2nd edition,	Latest

	Dreamtech press	Edition
--	-----------------	---------

Equivalent MOOCs courses:

Sr. No.	Course Links	Offered by
1.	Programming, Data Structures and Algorithms using Python, https://nptel.ac.in/courses/106106145	NPTEL
2.	Programming in Python, https://onlinecourses.swayam2.ac.in/cec22_cs20/preview	Swayam

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	3	1	2	1	3	1	-	1	-	1	-	1	2	-	-
CO2	3	2	2	2	3	1	-	1	-	1	-	1	2	-	-
CO3	3	2	3	1	3	1	-	1	1	1	1	-	1	-	-
CO4	3	2	2	3	2	1	-	1	-	-	-	1	1	-	-

1: Low, 2: Medium, 3: High

Course Name	Strength of Materials
Course Code	ES2306
Credits	4
L T P	3-0-2

Course Objectives:

At the end of this course, the student should be able to understand the basic concept regarding the strength of material. The course will prepare the students to apply these concepts to engineering and applied sciences problems.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
1	INTRODUCTION: Equations of static equilibrium, Concept of various forces/loads, stresses and strains developed due to these forces/loads, Uniaxial tensile test, Stress-strain diagrams for various types of ferrous and non-ferrous materials, isotropic and anisotropic materials, Compression test, impact test, fatigue test, hardness test, torsion and bending test as per ASTM standards	4
2	SIMPLE STRESS & STRAIN: Hooke's law, stress and elongation produced in various types of bars due to its own weight and applied axial force, Poisson's ratio, and relationship between elastic constants, stresses and elongation produced in simple & composite bars due to axial, thermal and combined loading.	6
3	2-D STATE OF STRESS ANALYSIS: Generalized 2-D state of stress accompanied by shear stress, stresses on an arbitrary plane under this state of stress, sign conventions for normal and shear stresses, complementary shear stress, principal stresses and principal planes, Different stresses determination through Mohr's stress circle approach in 2-D generalized state of stress.	6
4	SHEAR FORCE AND BENDING MOMENT IN BEAMS: Classification of beams, supports and loads, Shear force (SF) & Bending moment (BM) in beams and their sign conventions, Relation between rate of loading (w) with SF and BM. SF and BM diagrams of cantilevers, simply supported beams with or without overhang under different types of loading e.g. concentrated loads, uniformly distributed load, uniformly varying load, moment or its combinations, the point of contra-flexure	6
5	BENDING & SHEAR STRESSES IN BEAMS: Theory of pure bending, position of neutral axis, Bending equation, practical application of bending equation, review of moment of area concepts, variation of bending stress in various cross-sectional beams, shear stresses in beams, variation of shear stresses for different cross-sectional beams	6
6	TORSION OF CIRCULAR SHAFTS: Torsional equation of circular shafts, shear stress distribution, torsional rigidity, power transmitted by the shaft, comparisons of hollow & solid circular shafts, analysis of shafts in series and parallel mode, Equivalent bending moment and equivalent torque for a shaft subjected to bending moment and torque simultaneously	5
7	COLUMN AND STRUTS: Definitions, Euler's theory of columns buckling, Euler's equation for various end restraints, Rankine and other empirical formulae.	4
8	DEFLECTION OF BEAMS: Relationship between bending moment, slope and deflection, moment area method, method of integration, Macaulay's method, Use of these methods to determine slope and deflection for statically determinate and statically indeterminate beams under various loading conditions.	5

Sr. No.	List of Experiments	No. of Turns
1	To perform uni-axial Tensile Test on a given material and to determine its various mechanical properties under the uni-axial tensile loading.	2
2	To perform uni-axial compressive test on a given material and to determine its various mechanical properties under the uni-axial compression loading.	2
3	To perform shear test on a given material and to determine its shear stress of the material.	2
4	To perform torsion test on a given material and to determine its various mechanical properties under torsional load.	2
5	To perform Column test of a given material and to determine its Euler's buckling load and Young's modulus of elasticity of the material.	2
6	To perform Impact test on a given material and to determine its resilience.	2
7	To perform a test on close and open coil springs under axial loading on spring tester and determine its various mechanical properties.	2
8	To study and perform Fatigue test on a given material and to determine endurance strength and limit of the material.	2
9	To determine various Rockwell hardness of the given materials	2
10	To perform the Three point bending test on a given material and determine its Young's modulus of elasticity and bending strength.	2
11	To study the concepts of various strain gauges along with their areas of applications	2
12	To study the Creep test on the given material specimen and determine its creep strength.	2
13	To perform the various tests on the given wooden specimens by wood UTM and determine its various strengths.	2
14	To perform test on strain hardened specimen and to determine its effect on the Young's modulus of elasticity.	2

Course Outcomes: By the end of this course, the student will be able to:

1	Understand the concept of stresses & strains, various types of materials, its properties & testing processes as per ASTM standards.
2	Understand elastic constants and also be able to determine stresses & elongations in simple and composite bars under various types of loads.
3	Determine stresses on an arbitrary plane for a generalized 2-D state of stress accompanied by shear stress through analytical and graphical methods
4	Understand and be able to draw shear force and bending moment diagrams for different types of beams under various types of loading.
5	Understand bending and shear stress equations and its application in determination of bending & shear stresses in different cross-sectional beams at any point across its length.
6	Understand the torsional stresses for solid, hollow and composite circular shafts and its importance in power transmission.
7	Understand the Column and struts and determine the buckling load under various axial loadings.
8	Determine the deflections of various beams subjected to different loading by various methods.

Suggested Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	Strength of Material - G. H. Ryder (MacMillan)	1969
2	An Introduction to the Mechanics of Solids – Crandall & Dahl (Mc-Graw Hill)	2012
3	Engg. Mechanics of Solids - E. P. Popov (Pearson Education)	2003
4	Strength of Material - D S Bedi, Fifth Ed.	2010
5	Strength of Material - R K Rajput, Fifth Ed.	2012

Course Name		Engineering Drawing with CAD Software
Course Code		ES2302
Credits		3
L TP		2-0-2
Course Objectives:		
At the end of this course, the student should be able to understand the basic concepts of Engineering Drawing. The student should be able to visualize and draw the two- and three-dimensional objects. The student should also be able to understand the features associated with operations of the computer-aided design (CAD) software.		
Total No. of Lectures – 28		
Lecture wise breakup		Number of Lectures
1	Introduction to Engineering Graphics, Concept of points and lines, System of Projections, Orthographic projections, Dimensioning.	4
2	Introduction to different types of CAD Softwares e.g. SolidWorks/AutoCAD/CATIA etc., 2D-Sketching, Sketching Entities & Relation, 3D-Sketching, Editing and its Features, Dimensions, Sketch Tools, File handling.	7
3	Projections of planes / lamina on reference planes, classification of primary and secondary planes, use of auxiliary planes, Exercises using CAD software.	5
4	Classification of solids, Projections of solids on the basis of positions of the axis of various solids on reference planes and Sectioning of solids, Exercises using CAD software.	6
5	Introduction to Perspective projection, isometric views, Isometric lines & Axes, conversion of orthographic views to isometric views and vice-versa, Exercises using CAD software.	6
List of Experiments:		Number of Turns
Exercises to be done using CAD software		
1	2D & 3D Sketching using various sketching tools.	2
2	Projection of planes.	2
3	Developments of 3D-parts.	2
4	Projection of solids.	2
5	Projection of Sectioning of solids.	2
6	Isometric and orthographic views.	2
7	Generating drawings of 3D-parts.	2

Sr. No.	Course outcome By the end of this course, the students will be able to:	Knowledge Level (Blooms Level)
1	Understand the basic concepts of Engineering Graphics, drawing standards, conventions and symbols that are in common usage.	L2
2	Apply the concepts of engineering drawing to create orthographic projections of points, lines, planes, solids with conventional and CAD software.	L3 & L6
3	Visualize the actual objects and convert them in to readable drawings with conventional and CAD software.	L6
4	Create new designs/engineering models with conventional and CAD software.	L6

CO-PO & PSO Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	2	-	-	-	-	-	-	-	1	2	1
CO2	3	2	3	3	3	-	-	-	1	-	-	1	3	2
CO3	3	2	3	3	3	-	-	-	1	-	-	1	3	2
CO4	3	2	3	3	3	-	-	-	1	-	-	1	3	2

Suggested Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	Engineering Drawing, P. S. Gill, S.K. Kataria& Sons.	2012
2	Engineering Drawing, D.A. Jolhe, Tata McGraw Hill	2010
3	Engineering Graphics with SOLIDWORKS, David C. Plan chard, SDC Publications	2020

Course Name	:	Introduction to Mechatronics
Course Code	:	ES 2304
Credits	:	04
L T P	:	3-0-2
Course Objectives:		
The objective of the course content is to: 1.Impart knowledge and information about mechatronics system. 2. Understand the concepts of signal conditioning and data acquisition for intelligent systems. 3.Develop the basics for mechatronic product design		

Total No. of Lectures-42

Introduction to Mechatronics : Introduction, Elements of Mechatronics system, Classification of Mechatronics system, Mechatronic system intelligence, Components involved in intelligent system design and development, measurements and control system as a part of mechatronics system, Application of Mechatronic systems (6)

Sensors and transducers : Introduction, Performance characteristics of transducers, Transducer for displacement (Potentiometer, strain-gauge, Optical encoder, LVDT, Hall effect sensor); velocity (Tachogenerator), force (load cell), pressure (Piezoelectric sensors, Tactile sensor), liquid level (Floats, Differential pressure), Temperature (Bimetallic strips, RTDs, Thermistors, Thermocouples) and light sensor (Photovoltaic- transducer, LDR, Photodiode, Photo Transistor). (8)

Signal conditioning: Operational amplifier (Inverting, Non-inverting, Summing, Integrating, Differential amplifiers, comparator), protection, filtering, digital signals (R-2R ladder DAC and Successive Approximation ADC), Concepts of multiplexers (7)

Controllers: Basics of number system, binary, octal and hexadecimal systems with their conversion from one system to other. Boolean algebra, logic gates, ICs, flip-flops & counters. Microprocessor, Microcontroller, PLC & their Architectures, Working Principle, Software Programs (Assembly/High Level), Interfacing Aspects (7)

Actuators and mechanisms:

Pneumatic and hydraulic actuation systems: Directional control valves, Pressure control valves, cylinders
Mechanical actuation systems: Kinematic chain, cam, gear, clutches, ratchet and pawl, belt and chain drive, bearings

Electrical actuation systems: Relays, Solid-state Switches (Diode, Thyristor, Triac, BJT, FET), DC and AC motors, brushless dc motor, stepper motors, servomotors (8)

Robotics: Types of motions, Function, Governing Laws, Classification, Features and Components of Robots, System Automation (6)

Topics to be thought through Flipped Learning (NPTEL MOOCs) -

<https://nptel.ac.in/courses/112107298>

Mechanical actuation systems: Kinematic chain, cam, gear, ratchet and pawl, belt and chain drive, bearings (Unit 3 Lecture 11)

Pneumatic and hydraulic actuation systems: Directional control valves, Pressure control valves, cylinders (Unit 3 Lecture 12)

Basics of number, binary, octal and hexadecimal systems with their conversion from one system to other. Boolean algebra, logic gates, ICs, flip-flops. (Unit 5 Lecture 21 & 22)

PLC Controller (Unit 7 Lecture 33)

LIST OF EXPERIMENTS		
1	Experiment on Sensors & Transducers	Number of Turns
(i)	To study the characteristics of LVDT using linear displacement trainer kit & compare with ideal characteristics.	01
(ii)	To measure the strain of the metal strip using strain gauge trainer kit & compare with ideal characteristics.	01
(iii)	To measure the angular displacement of resistive & capacitive transducer using angular displacement trainer kit & compare with ideal characteristics.	01
(iv)	To obtain the characteristics of RTD, thermistor, thermocouple with hot and cold junction thermal trainer kit & compare with ideal characteristics.	01
2.	Experiments on Signal Conditioning.	
(a)	Experiments on Analog Devices	
(i)	PN Junction Diode	01
(ii)	Zener Diode	01
(iii)	Half wave rectifier	01
(iv)	Full wave rectifier	01
(b)	Experiments on Digital devices	
(i)	Logic Gates (AND, OR, NAND, NOR etc)	01
(ii)	Flip Flop - RS Flip Flop, JK Flip Flop, T Flip Flop and D Flip Flop.	01
3	Experiments on Controller	
(i)	To perform the basic sequence programming using PLC.	01
4.	Experiments on Actuators	
(i)	To perform AND and OR logic using pneumatic actuators.	01
5.	Project using Arduino	
(i)	To build a line follower robot using Arduino.	02

Course Outcomes:

By the end of this course, the student will be able:

CO1: To understand the basic concepts, applications and components of mechatronic system.

CO2: To analyze sensing, signal conditioning and data acquisition circuits.

CO3: To design product and systems theoretically as well as practically with Intelligence.

CO4: To apply the knowledge of mechatronic system for industrial applications.

Suggested Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	Mechatronics by W Bolton , 6 th edition, Pearson Education	2019
2	Mechatronics by Tilak Thakur 1 st edition, Oxford University Press	2016
3	Mechatronics by Dan Necsulescu, Pearson Education	2001
4	Mechatronics by H M T Limited, TMH	2017
5	Mechatronics Principles, Concepts & Applications by Nitaigour P Mahalik, TMH	2017

Course Name	Introduction To Electronics & Electrical Engineering
Course Code	ES 2305
Credits	4
L T P	3-0-2

Course Objectives:

To introduce to the students, the fundamental concepts of electronic devices, circuits and electrical systems for engineering applications.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
1	Semiconductor Devices and applications: Introduction to different semiconductor materials, familiarization with active and passive components, operation of p-n junction diode, applications of diodes, introduction to BJT and MOSFET, transistor as an amplifier, transistor as a switch, functional operation of OpAmp, concept of Oscillators, filters and their types	10
2	Digital Electronics: Number system, Binary arithmetic, Binary codes, introduction to logic gates, combinational circuits: adder, subtractor, multiplexer, demultiplexer, sequential circuit: flipflops.	9
3	Communication Systems: Basics of signals and systems, time and frequency domain analysis, various frequency bands used for communication, block diagram of Analog and Digital communication, need of modulation, Introduction to wired and wireless communication.	10
4	Fundamentals of Electrical Engineering: Introduction to circuit laws, amplitude, phase, phase difference, RMS value and average value of an AC signal, introduction and types of motors, Transformer: construction, working principle and applications	7
5	Microprocessor and Microcontroller: Introduction to microprocessor and microcontroller, architecture of 8086, functional diagram, register organization, memory segmentation	6

List of Experiments

S. No.	Name of the experiment	No. of turns
1	To familiarise with basic electronic components, oscilloscopes, power supply, multimeter etc.	2
2	To simulate and analyse the IV characteristics of PN junction diode	1
3	To simulate and analyse the functionality of opamp	2
4	To simulate and verify the truth tables of various logic gates	1
5	To simulate and analyse various combinational circuits	2
6	To simulate and verify the truth tables of various flip-flops	2
7	To familiarise with microprocessor and microcontroller kits	2
8	Write a MATLAB program for the generation of standard signals	1
9	To implement amplitude modulation and demodulation by using MATLAB	1

Course Outcomes: By the end of this course, the students will be able to

1	Express the understanding of semiconductor devices (p-n Diode, BJT, MOSFET etc), and their applications.
2	Demonstrate the functional operation of various analog and digital electronic circuits.
3	Solve basic electronic circuits using circuit laws.
4	Describe various signals, systems and fundamentals of communication systems.

5	Describe the basic principle and working of fundamental electrical systems, ac dc motors and transformer etc.
6	Explain the fundamentals of microprocessors and microcontrollers.

Suggested Books:

S. No.	Name of Book/ Authors/ Publishers	Year of Publication/ Reprint
1	Electronics Devices & Circuit Theory, RL Boylestead & L Nashelsky (PHI)	2009
2	Circuits and Networks: Analysis and Synthesis, Sudhakar and Shyam Mohan, TMH	2009
3	Electronic Communication Systems by G. Kennedy, Mc Graw Hill, 4 th Edition	2008
4	Digital Principles and Applications, 7 th Edition, A. Malvino and D. Leach	2011
5	Alexander, Charles K., and Sadiku, Matthew N. O., Fundamentals of Electric Circuits, 5 th Ed, McGraw Hill	2013
6	A course in Electrical and Electronic Measurements and Instrumentation, A K. Sawhney, 18 th Edition, Dhanpat Rai & Sons	2001
7	Microprocessors and Microcontrollers Architecture, Programming and Interfacing Using 8085, 8086 and 8051, Soumitra Mandal, Tata McGraw-Hill	2017

Course Name	:	Introduction to Product Design
Course Code	:	ES -2307
Credits	:	2
LTP	:	0-0-4

Course Objectives:

To demonstrate the concepts Product Design and Development skills through: Additive Manufacturing, Reverse Engineering, CNC machining, Laser engraving and Robotic

Course Outcomes:

1	Apply the fundamental concepts and principles of reverse engineering in product design and development.
2	Understand the process and applications of Additive Manufacturing for Product development
3	Understand the concepts and applications of CNC machines & Robot and its interface.

Practical's

Total: 56 hours

Sr. No.	Name	No of turns
1	Prepare a CAD model of given Product using NX Software	4X2
2	Reverse Engineering (3D Scanning, Repair and measurement) of given product for development of CAD model	4X2
3	Converting CT/MRI scan data using MIMICS Software to Develop the CAD model	4X1
4	Tessellation of various CAD models into STL file and Simulation of process parameters using software and its validation	4X1
5	Fabrication of CAD model using polymers based 3D Printer for Product development	4X1
6	To study various CNC machine codes and addresses and write a programme for a given profile	4X1
7	To execute the part program on CNC trainer kit followed by product fabrication on CNC machine tool	4X1
8	Demonstrate the various components of industrial robots and programming the Robot for pick and place application	4X1
9	To design a circuit using the PLC concept for automated devices	4X1
10	To sketching given design using CorelDRAW and fabricate the product using Laser engraving	4X1
11	To measure the surface integrity of given Product using advance measurement techniques	4X2

Course Name	:	Introduction to Environmental Sciences
Course Code	:	GS 2301 (Common to all branches)
Credits	:	1
L T P	:	1 0 0

Course Objectives:
This course aims to (i) acquaint the students with the basics of Environmental Science (ii) make them aware of the importance of Environmental Science

Total No. of Lectures – 14

Lecture-wise breakup		No. of Lectures
Unit 1	Multi-disciplinary Nature of Environmental Studies; Environmental Problems and their Causes, Concept of Sustainability; Sustainable Development, Sustainable Development Goals (SDG).	2
Unit 2	Types of Ecosystems - System Dynamics - Understanding Ecosystems, Ecosystem Degradation, Ecosystem Delivery, Habitat Classification.	2
Unit 3	Natural Resources and Associated Problems, Non Renewable Resources, Renewable Resources, Resource Utilization.	2
Unit 4	Energy and Environment – Fossil Fuel, Geothermal, Tidal, Nuclear, Solar, Wind, Hydropower and Biomass.	2
Unit 5	Environmental Pollution – Air Pollution, Water Pollution, Soil Pollution, Marine Pollution, Noise Pollution, Thermal Pollution, Nuclear Hazards ; Global Environmental Issues ; Solutions to Pollution Problems.	2
Unit 6	Cleaner Production and Life Cycle Analysis - LCA Methodology, Steps and Tools; EIA and Environment Audit.	2
Unit 7	Environment, Development and Society- Emerging Technologies for Sustainable Development and Environmental Management; Policies and Practices, Legislation; Disaster Management.	2

Course Outcomes (COs):	
At the end of the course, the students will be able to:	
1	Relate the importance of environmental science for sustainable development of society.
2	Apply the principles of environmental science and the concept of sustainable development in real life engineering problems.

Test Books		Year of Publication/ Reprint
S.No.	Name of Book	
1	“Environmental Science”, Miller G.T. and Spool, Ceonage Learning Publications.	2021
2	“Environmental Studies”, Banny Joseph, Tata Mcgraw Hill Publication.	2021
3	“Text book of Environmental Studies for U.G. Courses”, Erach Bharucna , University Press.	2021
4	“Environmental Studies – from criteria to cure”, R. Raogopalan, Oxford Univ. Press.	2022
5	“Principles of Environmental Science – Inquiry and applications”, Mary Ann Cunningham, William P. Cunningham, TMH Edition.	2020

Course Name	Universal Human Values
Couse Code	GS2302
Credits	1
L T P	1-0-0

Objectives:
The course has the following objectives- CO1-To equip students with foundation in ethical decision-making, teamwork, and social responsibility. CO2-To emphasize the integration of technical skills with ethical principles to prepare students for responsible and ethical professional practices. CO3-To shape students into well-rounded individuals with strong ethical foundation that guides their actions, decision and interactions with world around them.

Total Number of Lectures: 14

Lecture wise Breakup	Number of Lectures
Course Introduction: Need, Basic Guidelines, Content & Process for Value Education Understanding ethics, morals, and values Ethical theories and their application in engineering and technology	2
Ethical Decision-Making Steps in ethical decision-making Identifying and analysing ethical dilemmas in real-world scenarios	3
Teamwork and Collaboration Characteristics of effective teams Conflict resolution and managing differences Collaboration tools and techniques for virtual teams	3
Social Responsibility and Sustainability Role of engineers in promoting social good Environmental ethics and sustainable development Corporate social responsibility and ethical considerations in technology choices	3
Ethical analysis of Technological Innovations Engineering Ethics Balancing short-term gains with long-term consequences	3

Course Outcomes:
Some of the key outcomes that students can expect after completing this course: CO1-Increased awareness of ethical issues in both personal and professional contexts. CO2-Encourages critical thinking skills, allowing students to analyse complex situations from multiple angles and evaluate the ethical implications of different actions. CO3-Promotes an understanding of diverse cultures, beliefs, and backgrounds, fostering an inclusive mindset. CO4-Engage in self-reflection and introspection, leading to personal growth and a better understanding of their own values, strengths, and areas for improvement.

Suggested Books:		
S.No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Engineering Ethics" (Includes Human Values), Govindarajan M, Pearson Education Inc.	2017
2	"Professional Ethics and Human Values", Govindarajan M, Learning India Private Limited.	2020
3	"Professional Ethics and Human Values (JNTU-Kakinada)", B Raghavan, McGraw Hill.	2021
4	"The Moral Status of Technical Artefacts (Philosophy of Engineering and Technology)", Peter Kroes and Peter-Paul Verbeek	2020
5	"Ethics and Professionalism in Engineering (Broadview Guides to Business and Professional Ethics)" by Richard H McCuen and Kristin L Gilroy., Pubs: W.W. Norton & Company.	2020
6	"Engineering Ethics: Challenges and Opportunities" by W Richard Bowen, Cengage Learning	2019

Additional Reading-

S.No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	A Casebook in Interprofessional Ethics: A Succinct Introduction to Ethics for the Health Professions (SpringerBriefs in Ethics)", Jeffrey P Spike and Rebecca Lunstroth,2 Pearson Education Inc	2020
2	"Fundamentals of Ethics for Scientists and Engineers", Edmund G Seebauer and Robert L Barry,2020, Pearson Education Inc.	2020

SYLLABUS

Course Name	:	Communication Skills
Course Code	:	HS2351
Credits	:	3
L T P	:	2-0-2

Course Objectives
The objective of the course content is: CO1: To explain different aspects of communication process and enable the students to develop a strong theoretical base to handle various real-life communication tasks. CO2: To provide opportunities to the students to acquire and practice their LSRW skills in English for effective communication in professional life. CO3: To enable the students to articulate their ideas and perspectives using appropriate communication strategies and tools. CO4: To develop critical thinking and creative writing skills among the students and equip them with necessary analytical tools to achieve success in personal and professional domains.

Total No. of Lectures – 28

Lecture-wise Breakup		No. of Lectures
1	Introduction to Communication Concept, Process, Modes – verbal (oral and written) and non-verbal (kinesics, proxemics, chronemics), Types (formal, semi-formal, and informal), Channels (downward, upward, horizontal, diagonal), and Levels (extrapersonal, interpersonal, intrapersonal, organizational, and mass) of Communication, Barriers to Effective Communication with Solutions, Significance and Tools of Effective Communication.	(4)
2	Developing Effective Listening and Speaking Skills Hearing versus Listening, Listening Process (hearing, understanding, remembering, evaluating and responding), Note-taking, Barriers to Listening, and Strategies for Effective Listening. Tools and Techniques (linguistic and paralinguistic) of Effective Speaking at Various Levels (interpersonal, group, organization and society), Art of Conversation, Dialogue, Discussion, Public Speaking, Presentation, Negotiation, and Persuasion.	(7)
3	Developing Reading and Technical Writing Skills Concept, Strategies and Techniques (skimming, scanning, inferring, close reading) for Effective Reading and Comprehension, Understanding and Summarizing the Gist. Tips for Effective Technical Writing, Formal Letter Writing, Notice, E-mail Writing, Precis Writing, Statement of Purpose, and IMRD-Based Report.	(8)

4	Critical and Creative Thinking Skills Descriptive, Referential, Inferential, Discursive, Analytical, Evaluative, Creative and Lateral Using Texts and Various Media Forms like Books, Newspaper Articles, Films, and Social Visuals.	(4)
5	Job Preparation Sensitization to Building Portfolio, Job/Cover Letter and Resume, Interview Skills	(3)
6	Digital Media for Effective Communication Introduction, Current Trends in e-learning, Online Meetings, Digital Media Tools and Apps for Enhancing Communication Skills	(2)

Total No. of Practical Sessions: 14

Practical Session Wise Breakup		No. of Practical Sessions
1	Organizational Communication Verbal, Vocal and Non-Verbal Communication in Practice, Greetings and Self-Introduction, Speech, and JAM.	(2)
2	Speaking Techniques at Different Forums Group Discussion, Role-play, Making and Presenting Power Point Presentations.	(3)
3	Practice on Reading and Technical Writing Reading Comprehension, Formal Letter Writing, Précis Writing, Notice, Email Writing, Report Writing, and Statement of Purpose	(3)
4	Applying Critical Thinking Skills Book Review, Film Review, Social Visuals -Interpretation and Critical Analysis.	(3)
5	Towards Job Preparation Sensitization to Building Portfolio, Job/Cover Letter, Resume, Interview	(2)
6	Use of Digital Media for Effective Communication Training the students to use digital tools	(1)

OBJECTIVES OF THE COURSE

The objective of the course content is:

CO1: To explain different aspects of communication process and enable the students to develop a strong theoretical base to handle various real-life communication tasks.

CO2: To provide opportunities to the students to acquire and practice their LSRW skills in English for effective communication in professional life.

CO3: To enable the students to articulate their ideas and perspectives using appropriate communication strategies and tools.

CO4: To develop critical thinking and creative writing skills among the students and equip them with necessary analytical tools to achieve success in personal and professional domains.

OUTCOMES OF THE COURSE

By the end of this course, the students will be able to:

1. Understand the basic concepts of effective communication and learn the importance of communication skills for professional purposes.
2. Apply the four skills of language effectively by using latest learning strategies and digital tools to create and produce original pieces of technical writing and speech.
3. Analyse and evaluate critically what they listen, read and write to respond appropriately in different contexts of their personal and professional life.

Course Name	:	CALCULUS
Course Code	:	MA2301
Credits	:	4
L T P	:	3-0-2
Total No. of Lectures	:	42

Course Objectives:

At the end of the semester, the students should be able to

1	Understand the behavior of infinite series and their use.
2	Learn the concepts related to differential calculus of functions of several variables and their applications.
3	Learn the concept and methods of evaluating multiple integrals and their applications to various problems.

	Lecture wise breakup	No. of Lectures
1	INFINITE SERIES Limits of sequences of numbers, Theorems of calculating limits of sequences, Infinite series and convergence, alternating series, power series and convergence. Taylor's and Maclaurin's Series. (Scope as in Chapter 8, Sections 8.1 – 8.9 of Text Book 1).	12
2	DIFFERENTIAL CALCULUS Functions of several variables, Limits and continuity, Partial Derivatives, Euler's Theorem for Homogeneous functions; Differentiability, Linearization and Differentials; Chain rule; Extreme values and Saddle Points; Lagrange multipliers; Taylor's Formula. (Scope as in Chapter 12, Sections 12.1 – 12.6, 12.8 – 12.10 of Text Book 1).	14
3	INTEGRAL CALCULUS Parametrization of plane curves, Polar coordinates, Graphing in Polar coordinates, Cylinders and Quadric surfaces, Double integrals in Rectangular and Polar form, Triple integrals in Rectangular, Cylindrical and Spherical Coordinates, Substitutions in Multiple integrals. Applications to practical problems. (Scope as in Chapter 9, Sections 9.4, 9.6 and 9.7 ,Chapter10, Sections10.6 and 10.7 and Chapter 13, Sections 13.1, 13.3, 13.4,13.6 and 13.7 of Text Book 1).	16

Course Outcomes:

At the end of the semester, the students are able to

1	Test the behavior of infinite series.
2	Apply the concepts of differential calculus of functions of several variables.
3	Evaluate multiple integrals and apply them to practical problems.

Text Books:

1	Calculus and Analytic Geometry, Thomas and Finney, 9 th edition, Pearson Education Asia.	2006
---	---	------

Reference Books:

1	Advanced Engineering Mathematics, Wylie and Barrett, 6 th edition, Mc Graw Hill.	2003
---	---	------

Practical / Lab work to be performed using Mathematica/ Matlab

1. Study the convergence of sequences through plotting.
2. Analyze the convergence of infinite series by plotting their sequences of partial sums.
3. Study the convergence of infinite series using Cauchy's root test and Ratio test
4. Taylor and Maclaurin series of trigonometric, logarithmic, hyperbolic functions.
5. Plotting 2D curves in rectangular and polar coordinates.
6. Plotting 3D surfaces.
7. Find critical points and identify local maxima, local minima or saddle points
8. Draw the surfaces and analyze the existence of limits as they approach the specified points.
9. Check the continuity of functions
10. Draw the surfaces and find level curves at the given heights

Course Name	:	LINEAR ALGEBRA, DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS
Course Code	:	MA2302
Credits	:	4
L T P	:	3-0-2
Total No. of Lectures	:	42

Course Objectives:

At the end of the semester, the students should be able to

1	Learn the various concepts associated with real vector spaces and theory of matrices
2	Learn the methods to solve ordinary differential equations of various types.
3	Learn the various concepts of vector calculus and their applications to problems.

	Lecture wise breakup	No. of Lectures
1	ALGEBRA Vector spaces over reals, Linear dependence, Basis, Dimension, Co-ordinates with respect to a basis, Change of basis, Subspace, Linear transformation $R^n \rightarrow R^m$, Range space and Rank, Null space and Nullity, Rank and Nullity relation, Matrix representation of a linear transformation, Similar matrices, Invertible linear transformation, Eigenvalues and eigenvectors, Cayley Hamilton theorem, Diagonalization of a matrix.	16
2	ORDINARY DIFFERENTIAL EQUATIONS First order exact differential equations, Integrating factor, Orthogonal trajectories, Second and Higher order Linear Differential Equations with constant coefficients, Differential Operators, Methods of Variation of Parameters and Undetermined Coefficients, Euler Cauchy Equation, Wronskian.	12
3	VECTOR CALCULUS Gradient, Divergence and Curl – their physical interpretation, Line, Surface and Volume integrals, Green's theorem in the plane, Stoke's theorem, Divergence theorem, Applications to Science and Engineering.	14

Course Outcomes:

At the end of the semester, the students are able to

1	Solve the various problems related to real vector spaces and theory of matrices
2	Solve ordinary differential equations of various types
3	Apply various concepts of vector calculus to problems.

Text Books:

1	Introductory Linear Algebra with Applications, Kolman, B. and Hill, D.R., 7 th edition, Pearson Education	2001
2	Advanced Engineering Mathematics, Kreyszig, 8 th edition, John Wiley and Sons.	2005

Reference Books:

1	Differential Equations, S. L. Ross, John Wiley and Sons, India	2004
2	Advanced Engineering Mathematics, Wylie and Barrett, 6 th edition, Mc Graw Hill.	2003
3	Differential Equations, Frank Ayers, SI edition, Mc Graw Hill.	1972

Practical / Lab work to be performed using Mathematica/ Matlab

1. Perform basic Matrix operations.
2. Find rank, eigenvalues and eigenspace of matrices.
3. Check diagonalizability of matrices.
4. Solve ordinary differential equation.
5. Plotting of second order solution family of differential equation.
6. Plotting of third order solution family of differential equation.
7. Plotting of vector fields.
8. Find Gradient, Divergence and Curl.
9. Computation of line integrals and surface integrals.
10. Verify Green's theorem in the plane, Stoke's theorem, Divergence theorem.

Course Name	:	Electromagnetic Theory and Quantum Physics
Course Code	:	PY2301
Credits	:	4
L T P	:	3 0 2
Course Objectives:		
1. To familiarize the students with the concepts of electrostatics and boundary value problems. 2. To make the students able to understand the magnetostatics and their boundary value problems. 3. To make the students able to understand and apply the concepts of electromagnetic wave propagation. 4. To familiarize the students with the concepts and principles of Quantum Mechanics.		
Total No. of Lectures – 42		
Lecture wise breakup		Number of Lectures
1	VECTORS, FIELDS AND ELECTROSTATICS: Cartesian coordinate System, Cylindrical and Spherical coordinate Systems, Gradient, Divergence of a Vector and Divergence Theorem, Curl of a vector and Stoke's theorem, Gauss's law & its applications, Maxwell's 1st eqn. (Electrostatics), Electric Energy and potential, Potential gradient, the dipole fields, Energy density in an electrostatic field. Current and current density, Continuity of current, Metallic conductors, Dielectric materials, Electrostatic boundary-value problems, Introduction to Laplace's and Poisson's equations.	12
2	MAGNETOSTATICS: Biot-Savart's law, Ampere's circuital law, Magnetic flux and magnetic flux density, Scalar and vector magnetic potentials. Magnetic dipole, Force due to Magnetic field on a differential current element, force between two differential current elements, Force and torque on a closed circuit, magnetic materials, Magnetic boundary conditions, Inductors and inductances.	10
3	MAXWELL'S EQUATIONS AND ELECTROMAGNETIC WAVE PROPAGATION: Faraday's law, Displacement current, Maxwell's equations in point form, Maxwell's equations in integral form, EM waves in general, EM wave propagation in Lossy Dielectrics, Wave propagation in lossless dielectrics, Plane waves in free space, Plane waves in Good conductors, Power & Poynting Vector, Reflection of a plane wave at normal incidence.	10
4	QUANTUM PHYSICS: Need of Quantum theory, Photoelectric effect, The Compton effect; matter waves, group and phase velocities; Uncertainty principle and its application; time independent and time dependent Schrödinger wave equation; Eigen values and Eigen functions, Born's interpretation and normalization of wave function, applications of Schrödinger wave equation for particle in one dimensional infinite potential well. Introduction to nanoscience, Quantum materials, and Superconductivity.	10

List of Experiments:		Number of Turns
1	To study dielectric constant and Curie temperature of Ferroelectric ceramic BaTiO ₃ .	1
2	To determine coercivity of magnetic material using hysteresis loop tracer.	1
3	To study the Hall effect and to determine Hall Voltage (V_H) and Hall coefficient (R_H)	1
4	To determine the energy band gap of semiconductor (Ge) using Four Probe Method.	1
5	To design a method to draw equipotential lines with various geometries of electrodes kept at different potentials.	1
6	To determine the flashing and Quenching Potential of neon/Argon and also to find the capacitance of unknown capacitor.	1
7	To study the variation of magnetic field with distance along the axis of current carrying circular coil using Stewart and Gee's apparatus.	1

8	To plot I-V Characteristics of Solar cell.	1
---	--	---

Course Outcomes: By the end of the course		
1	Students will be equipped with the tools of electromagnetic theory.	
2	Students will be able to solve numerical problems based on electrostatics, magnetostatics, electromagnetic wave propagation.	
3	Students will be able to understand and apply the basic concepts of Quantum Mechanics.	

Suggested Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	Engineering Electromagnetics, William H Hyat, Jr., and John A. Buck, Tata McGraw Hill	2013 / 5 th edition
2	Elements of Engineering Electromagnetics, Matthew N.O. Sadiku, Oxford University Press	2012 / 4 th edition
3	Concepts of Modern Physics, Arthur Beiser, McGraw Hill Education (India) Pvt. Ltd., New Delhi.	2013
4	Modern Physics, J. Bernstein, P.M. Fishbane and S.G. Gasiorowicz, Pearson, Education India Pvt. Ltd., New Delhi	2009

Course Name	:	Mechanics and Optics
Course Code	:	PY2302
Credits	:	4
L T P	:	3-0-2

Course Objectives:

1. To inculcate the application of Mechanics concepts in engineering
2. To familiarize students with Statics, Kinematics, and Kinetics of rigid body.
3. To familiarize the students with Ultrasonics and their applications.
4. To familiarize students the basic concepts of LASER and Nanotechnology for possible industrial applications.

Total No. of Lectures – 42

Lecture wise breakup		Number of Lectures
1	STATICS AND DYNAMICS: Analysis of system of forces, Equation of equilibrium in space and its applications, Center of gravity, Centroid, mass, area and Polar moment of inertia of simple and compound bodies. Kinematics of a Particle: Introduction, Motion of a projectile, Kinetics of a particle: Force and acceleration, Work and energy, Impulse and momentum.	10
2	PLANAR KINEMATICS AND KINETICS OF A RIGID BODY: Rigid-body motion, Translation, Rotation about a fixed axis, Absolute general Plane Motion analysis. Relative-Motion Analysis: Velocity, Instantaneous center of zero velocity, Acceleration. Moment of Inertia, Planar Kinetic equations of motion, Equations of motion: Translation, Rotation about a fixed axis and General Plane motion, Kinetic Energy, Work of a Force, Work of a Couple, Principle of Work and Energy, Conservation of Energy, Linear and Angular Momentum, Principle of Impulse and Momentum, Conservation of Momentum, Eccentric Impact.	12
3	LASERS, OPTICAL FIBRES AND ULTRASONICS: Basics of Interference, Diffraction and Polarization, Michelson–Morley Interferometer, Raman effect, Laser and its characteristics, He-Ne laser, Ruby laser, Semiconductor lasers, Ultra-fast lasers, Applications of Lasers, Optical fibres; Numerical aperture, Classification of optical fibres, fibre Losses, fibre manufacturing, Applications of optical fibres. Production of ultrasonics, detection and uses of ultrasonics, reverberation.	12
4	NANOTECHNOLOGY: Introduction, Length Scale, Size Dependence, Synthesis of Nanoparticles: Mechanical Method, Sol-gel Technique, Physical Vapour Deposition, Chemical Vapour Deposition, Overview of Carbon-based nanostructures, X-ray Diffraction for nano-materials analysis, Applications of Nanotechnology, Introduction to Quantum Materials and Superconductors.	8

List of Experiments		No.of Turns
1.	Familiarization of students with basic instruments Vernier Calipers, Screw Gauge and Spectrometer.	1
2.	(i) To determine the wavelength of He-Ne laser using transmission grating. (ii) To determine the slit width using the diffraction pattern.	1
3.	To find the specific rotation of sugar solution using a Bi-quartz Polarimeter.	
4.	To determine the acceleration of gravity using Kater's pendulum	1
5.	To determine the Moment of Inertia of a Flywheel.	1
6.	To determine the range of the projectile as a function of angle of inclination and initial velocity.	1
7.	To determine the velocity of ultrasonic waves in a given liquid.	1
8.	To measure the centripetal force, F_c , and compare to $F_c = mv^2/r = m\omega^2 r$.	1

Course Outcomes:	
1	Students will be able to understand and implement the concepts of Mechanics, types of motions and characteristics of rigid body.
2	Students will learn about lasers and fibre optics which have important applications for societal needs.
3	Students are expected to develop capability to tackle problems in general and in the various areas covered in the course.

Suggested Books:	
Sr. No.	Name of Book/ Authors/ Publisher/Edition
1	Statics, R.C. Hibbeler, Pearson (11th Edition).
	Dynamics, R.C. Hibbeler, Pearson (11th Edition).
2	Dynamics, F.P. Beer et al., McGraw Hill (8th Edition).
3	Dynamics, Merriam and Kraige, Wiley and Sons (5th Edition).
4	Engineering Mechanics, G. Ramamurthy, R. S. Walia and Rajesh Kumar, I K International (3rd Edition).
5	Optics, Ajoy Ghatak, McGraw-Hill (3rd Edition).
6	Physics for Engineers, N.K. Verma, Prentice Hall India (3rd Edition).
7	Engineering Physics, Satya Prakash, Pragati Prakashan (9th Edition)

Course Name	:	CONDENSED MATTER PHYSICS
Course Code	:	PY2303
Credits	:	4
L T P	:	3 - 2/2

Total No. of Lectures – 42

Course Objectives:	
1.	To familiarize the students with basic concepts of the condensed phase of matter especially solids.
2.	To make the students able to understand the crystal structure, lattice vibrations, electronic properties, dielectric and the magnetic properties etc. in relation to engineering applications.
3.	To make the students able to understand the basics concepts of semiconductors, superconductivity, and their applications in various fields.
4.	To familiarize students the basic concepts of LASER and Nanotechnology for possible industrial applications.

Lecture wise breakup	No. of Lectures
-----------------------------	------------------------

1	Crystal structure and Band Theory of Solids: -Space lattices crystal structures (cubic and hexagonal cells), close packed morphology (Hexagonal and cubic close packing), single and polycrystalline structures, crystal Structure analysis, X-ray diffraction and Bragg's law, Classical and Quantum Theory of free electrons, Fermi-Dirac Distribution Function, Density of States, Motion of electrons in magnetic field (Hall effect), Energy bands in solids. Introduction of Phonons, Vibrations of one dimensional monoatomic and diatomic lattices, Momentum of Phonons. Electron in a Periodic Potential (Qualitative idea of Kronig Penney Model), Energy versus wave vector,	(14)
2	Dielectric and Magnetic Materials: Fundamental of dielectrics, active and passive dielectrics, various polarization mechanisms, Frequency and temperature dependence on polarization of dielectrics, Internal field, Dielectric Loss Tangent, Dielectric Breakdown. Review of basic formulas, magnetic susceptibility, classification of materials, anti-ferromagnetism and ferrimagnetism, ferromagnetism in metals, ferromagnetic domains, hysteresis, antiferromagnetism.	(7)
3	Semiconductors and Superconductivity: Introduction, Pure or Intrinsic Semiconductors, impurity or Extrinsic semiconductors, Drift velocity, mobility and Conductivity of Intrinsic Semiconductors, carrier concentration and Fermi level for Intrinsic and Extrinsic Semiconductors, Applications: P-N Junction diode, Zener diode and Tunnel diode. Introduction to superconductivity, Mechanism of superconductivity, Meissner Effect, Effect of current and magnetic field, Type I and Type II superconductors, Thermal properties, Isotope effect, London Equations, Qualitative idea of BCS theory, Applications of superconductivity.	(14)
4	Nanotechnology: Introduction, Length Scale, Size Dependence, Synthesis of Nanoparticles: Mechanical Method, Sol-gel Technique, Physical Vapour Deposition, Chemical Vapour Deposition, Overview of Carbon-based nanostructures, X-ray Diffraction for nano-materials analysis, Applications of Nanotechnology, Introduction to Quantum Materials.	(7)

S.No	List of Experiments	No. of
-------------	----------------------------	---------------

		turns
1.	To study dielectric constant and Curie temperature of Ferroelectric ceramic BaTiO ₃ .	1
2.	To determine coercivity of magnetic material using hysteresis loop tracer.	1
3.	To study the Hall effect and to determine Hall Voltage (V_H) and Hall coefficient (R_H)	1
4.	To determine the energy band gap of semiconductor (Ge) using Four Probe Method.	1
5.	To design a method to draw equipotential lines with various geometries of electrodes kept at different potentials.	1
6.	To determine the flashing and Quenching Potential of neon/Argon and also to find the capacitance of unknown capacitor.	1
7.	To study the variation of magnetic field with distance along the axis of current carrying circular coil using Stewart and Gee's apparatus.	1

Course Outcomes: By the end of the course	
1	Students will be able to understand the physics behind structural properties of the solids.
2	Students will be aware of latest developments in certain areas of condensed matter physics, which have important applications for societal needs.
3	Students are expected to develop capability to tackle problems in general and in various areas covered in the course.

Suggested Books:	
Sr. No	Name of Book/ Authors/ Publisher
1.	Introduction to Solid State Physics, Charles Kittel, Wiley India Pvt. Ltd., New Delhi (8 th Edition)
2.	Solid State Physics, S.O. Pillai, New Age International, Pvt. Ltd., New Delhi (5 th Edition)
3.	Solid State Physics, M. A. Wahab, Narosa Publishing House, Pvt. Ltd. New Delhi (3 rd Edition).
4.	Solid State Physics R.K. Puri, V.K.Babbar, S. Chand & Company, Pvt. Ltd. New Delhi (3 rd Edition).

Course Name	:	Geoinformatics	DCC
Course Code	:	CEN3001	
Credits	:	4	
L – T – P	:	3 – 0 – 2	

Course Objectives:

At the end of this course, the student should be able to:

- To make the students learn the basic concepts of surveying.
- To apply the knowledge in the preparation of maps
- To appreciate the use of remote sensing and GIS in Civil Engineering practice.

Total No. of Lectures – 42

Lecture-wise breakup		No. of Lectures
Unit 1	INTRODUCTION TO SURVEYING AND MAPPING Principles, generation of maps, Importance, application, conventional and advanced methods	4
Unit 2	METHODS OF MEASUREMENTS Compass, Theodolite, Contouring, Plane table	6
Unit 3	TRAVERSING, TRIANGULATION, CONTOURING, CURVES Selection of Traverse Station, GALE'S Traverse Table, Calculation of Independent Co-ordinates, Curves, Triangulation	6
Unit 4	LEVELLING Basic Definition, Methods of Levelling, Classification of Levelling, Level & Levelling staff, Basic Principles of Levelling	6
Unit 5	AERIAL PHOTOGRAMMETRY Basics of Photogrammetry, applications, types, and geometry of aerial photograph, flying height, and scale, Relief displacement.	5
Unit 6	TOTAL STATION, GPS SURVEYS, DRONE SURVEYS Introduction to the total station, Advantages of TS, Introduction to GPS surveys, Segments of GPS, Applications of GPS, DGPS, Drone	5
Unit 7	REMOTE SENSING (RS) Basic RS, Interaction with atmospheric and earth surface, RS Data Products, Resolution, Applications of Remote Sensing	5
Unit 8	GEOGRAPHICAL INFORMATION SYSTEM (GIS) Definition of GIS, Vector and raster data, database creation, Digital Elevation Model (DEM), Applications of GIS	5

List of Experiments		No. of Turns
1	Working of Chains, Compass	02
2	Estimation of angular measurements using theodolite	03
3	Estimation of RLs using Levels	03
4	Mapping using handheld GPS	02
5	Mapping using Total Station	04

Course Outcomes (COs):

At the end of the course, the students will be able to:

1	Apply the concepts and analytical methods related to Surveying
----------	--

2	Use the various surveying equipment.
3	Plan and execute surveying projects
4	Identify the potential use of Remote Sensing and GIS in Civil Engineering

Text Books		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	“Geomatics Engineering,” Arora & Badjatia, Nem Chand & Bros., Roorkee.	2011
2	“Surveying Vol. I & II”, B.C. Punmia, A.K. Jain & Jain. Luxmi Publications (P) Ltd., New Delhi.	2006
3	“Remote Sensing & Image Interpretation”, Wiley.	2011
4	“Application of Geomatics in Civil Engineers”, Ghosh and D.A. Silva, Springer.	2020
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	“Surveying Vol. II”, S.K.Duggal: Tata McGraw Hill, New Delhi.	2009
2	“Surveying and Levelling,” Subramanian, Oxford university press.	2012
3	“Surveying and Levelling,” N N Basak, McGraw Hill.	2014
4	“Surveying Vol II & III,” K.R. Arora, Standard Book House, New Delhi.	2010
5	“Theory and Principles of Geoinformatics”, PK Garg, Khanna Publishers	2019
6	“Introduction to UAV”, P K Garg, New Age International Publishers.	2021

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	https://nptel.ac.in/courses/105/104/105104101/ NPTEL video course on Surveying	NPTEL
2	https://nptel.ac.in/courses/105/107/105107122/ NPTEL web course on Surveying	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	1	3	2	1	1	1	1	2	2	2	1	2	3
CO2	1	1	1	1	2	1	1	1	2	2	3	1	2	3	1	2
CO3	3	2	3	1	2	1	1	1	2	3	3	2	1	3	2	3
CO4	2	2	3	2	2	1	1	1	1	2	1	2	3	2	1	2

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation

Course Name	:	Structural Analysis - I	DCC
Course Code	:	CEN3002	
Credits	:	4	
L T P	:	3 0 2	

Course Objectives :

At the end of this course, the student should be able to:

- Analyze various civil engineering determinate structures subjected to static loading.
- Analyze moving loads on determinate structures.
- Analyze the structures using latest software tools.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	TRAVELLING LOADS Maximum Shear Force and Bending Moment diagrams for simply supported beams carrying following moving loads: - 1. A single concentrated load 2. Uniformly distributed load 3. Two concentrated loads fixed distance apart 4. Series of concentrated loads Equivalent UDL for bending moment and shear force.	7
Unit 2	INFLUENCE LINES Influence lines for reactions, BM & SF for simply supported beam. Influence lines for forces in trusses, use of influence lines for calculating design forces due to dead load and moving live loads.	7
Unit 3	ARCHES Three hinged arches, BM, radial shear and normal thrust at any section of a parabolic and segmental arch due to different static loadings. Moving loads on three hinged arches. Effect of change in temperature.	6
Unit 4	SUSPENSION BRIDGES Un-stiffened suspension bridges, maximum tension in the suspension cable and backstays, pressure on towers.	4
Unit 5	DEFLECTION OF BEAMS Deflection of determinate beams by double integration method, Macaulay's method, moment area method, conjugate beam method.	6
Unit 6	DEFLECTION OF PERFECT TRUSSES Deflection of joints in trusses subjected to static loading by Castigliano's strain energy theorem method, Unit load method.	4
Unit 7	SPACE FRAMES Analysis of space frames by equilibrium method and method of tension coefficients.	5
Unit 8	INDETERMINATE STRUCTURES Introduction to Indeterminate Structures, Determination of kinematic and static indeterminacy of beams, frames and trusses.	3

Course Outcomes:

At the end of this course, the student will be able to:

- | | |
|----------|--|
| 1 | Analyze determinate structures for static loads. |
| 2 | Analyze determinate structures for moving loads. |

3	Determine deflections of various determinate structures
4	Determine indeterminacy of structures.

List of Experiments:		Number of Turns
1	To study the behavior of axially loaded column for various support conditions.	1
2	To determine the carryover factor for a beam.	1
3	Verification of Maxwell Reciprocal theorem.	1
4	To determine the moment required to produce a given rotation at one end of the beam for fixed and pinned ends.	1
5	To study the behavior of portal frame under different loading conditions.	1
6	To determine the flexural stiffness beams.	1
7	To study the behavior of 2 hinged and 3 hinged arches.	2
8	Introduction to software and its commands.	3
9	To analyze a building frame using software.	3

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Theory of Structures (Vol. II)”, B C Punmia, Laxmi Publications.	2017
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Structural Analysis”, R.C. Hibbeler, Prentice Hall of India Pvt. Ltd.	2021
2	"Structural Analysis", T.S. Thandavamoorthy, Oxford University Press.	2011
3	“Theory of Structures”, S. Ramamrutham & R. Narayan, Dhanpat Rai.	2014

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	(https://archive.nptel.ac.in/courses/105/105/105105166/)Structural Analysis I, IIT Kharagpur	NPTEL
2	(https://archive.nptel.ac.in/courses/105/101/105101086/)--- Structural analysis , IIT Bombay	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	2	3	2	2	2	2	2	3	3	3	2	3
CO2	3	3	3	3	2	3	2	2	2	2	2	3	3	3	2	3
CO3	3	3	3	2	3	2	2	2	2	2	2	3	3	3	2	3
CO4	3	3	3	3	2	3	2	2	2	2	2	3	3	3	2	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Fluid Mechanics	DCC
Course Code	:	CEN3003	
Credits	:	4	
L – T – P	:	3 – 0 – 2	

Course Objectives: The Students should be able to;

At the end of this course, the student should be able to:

- Understand the behavior characteristics of fluids.
- Apply conservation of mass and conservation of momentum expression for fluid in motion.
- Analyze the viscous and non viscous flow.
- Develop dimensional relationships between flow variables.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION Properties of fluids - Viscosity, Capillarity, Surface Tension, Compressibility, Types of fluids, Continuum principle.	2
Unit 2	FLUID STATICS Pascal's law of measurement of pressure, Manometers and gauges, Types of forces on a fluid system, Hydrostatic Forces on subsurfaces, Buoyancy, Stability of floating and submerged bodies.	6
Unit 3	FLUID KINEMATICS Lagrangian and Eulerian methods, Types of fluid flow, Types of flow lines, Description of properties in a moving fluid, Local and convective acceleration, Laplace equation, Stream function, Velocity potential, stream function and flow-nets, Circulation and Vorticity	6
Unit 4	FLUID DYNAMICS System and control volume approaches, Equation of conservation of mass, Differential form of continuity equation, Euler's equation of motion in Cartesian Coordinates, Bernoulli's equation and application, Linear momentum and angular momentum equations and their applications.	6
Unit 5	FLOW MEASUREMENT Venturimeter, Orifice meter, Pitot tube, Orifices, Mouth pieces, Notches, Weirs, Current meter.	4
Unit 6	VISCOUS FLOW Pressure gradient in steady uniform flow, Flow between parallel plates, Qualitative aspects of viscous flows, Hagen-Poiseuille's flow, Transition from laminar to turbulent flow, Turbulent flow in circular pipe,	5
Unit 7	DIMENSIONAL ANALYSIS AND SIMILITUDE Buckingham-pi Theorem, Non-dimensional groups, Dimensionless numbers, Geometric, kinematic, and dynamic similarity, Similitude.	4
Unit 8	FLOW THROUGH PIPES Laminar flow through pipes, Darcy-Weisbach equation, Moody's diagram, Energy and hydraulic grade line, Head losses in pipes and pipe fittings, Pipes in series and parallel, Hardy-Cross method.	5
Unit 9	DRAG AND LIFT Streamlined and bluff body, Types of drag, Drag on a sphere, flat plate, cylinder and airfoil, Development of lift on immersed bodies - circular cylinder and airfoil.	4

List of Experiments		No of turns
1	Basic Introduction to Experiments	4
2	Experimental analysis of stability of Floating Bodies.	2
3	Determination of Discharge coefficient of a Venturimeter	2
4	Determination of Discharge coefficient of a Orificemeter	2
5	Determination of C_v , C_c and C_d for an orifice	2
6	Determination of friction factor in pipe flow	2
7	Experimental determination of head loss in sudden expansion, contraction and due pipe bend	4
8	Drawing flow nets	2
9	Determination of Discharge coefficient in V Notch	2
10	Verification of Bernoulli's Theorem	4
11	Measurement of velocity head with a pitot tube	2

Course Outcomes (COs):

At the end of the course the students will be able to:

1	Understand the properties of fluids and their estimation
2	Apply kinematics and dynamics of flow for solving real life engineering problems.
3	Analyze the viscous flows and flow through pipe networks.
4	Formulate dimensional relationship between fluid and flow variables and familiarize with flow measurement techniques.

Text Book:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Introduction to Fluid Mechanics and Fluid Machines", (3 rd Edition), Som S.K., Biswas Gautam, McGraw-Hill.	2019
2	"Textbook Of Fluid Mechanics & Hydraulic Machines (6 th Edition)", Rajpoot R.K., S.Chand.	2016

Reference Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Introduction to Fluid Mechanics (8 th Edition)", Fox, R.W. and McDonald, A.T. John Wiley and Sons.	2011
2	"Engineering Fluid Mechanics", Garde, R.J., and Mirajgaoaker A.G., Scitech Publications.	2010
3	"Fluid Mechanics (4 th Edition)", Cengel, Y. A., and Cimbala, J. M., McGraw-Hill.	2019
4	"Fluid Mechanics (9 th Edition)", Streeter V.L., Benjamin, W.E., and Bedford, K. W., McGraw-Hill.	2010

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
---------	--------------	------------

1	https://nptel.ac.in/courses/105/103/105103192/-	NPTEL
2	https://nptel.ac.in/courses/105/101/105101082/ -	NPTEL
3	https://nptel.ac.in/courses/105/103/105103095/	NPTEL
4	https://nptel.ac.in/courses/105/105/105105203/	NPTEL
5	https://ocw.mit.edu/courses/civil-and-environmental-engineering/1-060-engineering-mechanics-ii-spring-2006/lecture-notes/	MIT Open Course Ware

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	3	1	2	3	1	2	3	1	2	3	3	1	3
CO2	3	3	3	3	3	1	2	1	3	3	1	3	3	3	1	3
CO3	3	3	3	3	3	3	3	1	3	3	3	3	3	3	1	3
CO4	3	3	3	3	3	1	3	1	2	3	3	3	3	3	1	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Building Materials and Building Construction	DCC
Course Code	:	CEN3004	
Credits	:	4	
L T P	:	3 0 2	

Course Objectives:

At the end of this course, the student should be able to:

- understand different materials used for building construction and their properties.
- Perform quality tests for different building materials.
- Identify various construction techniques for building components.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	STONE, BRICKS AND TIMBER Stones: Introduction, quarrying, classification, requirements of good building stone, testing and applications. Bricks: Introduction, manufacturing of bricks, classification, defects, testing and applications. Introduction to timber and seasoning of wood, defects in wood.	9
Unit 2	BRICK MASONRY Terminology, bonds in brick masonry. hollow block masonry, types of walls, viz., load bearing and non-load bearing walls.	4
Unit 3	LIME CEMENT AND CONCRETE Manufacturing of cement, hydration of cement, classification of cement, testing, application, introduction of lime and its applications. concrete: Introduction, characteristics and classification of aggregates, admixtures, classification of different grades of concrete, applications, properties of fresh and hardened concrete, testing of concrete.	10
Unit 4	DAMP PROOFING Causes of dampness in buildings, ill effects and remedial measures, methods of damp proofing.	3
Unit 5	DOORS, WINDOWS, LINTEL AND ARCHES Terminology, types of doors and windows, types of lintel and arches.	4
Unit 6	ROOFS, ROOF COVERINGS AND FLOORS Terms used, R.C.C. And wooden roofs, pitched and sloping roofs, trussed roofs, roof covering materials. Types of floorings and their construction details.	7
Unit 7	STAIRS AND STAIRCASES Terms used, different types of staircases, proportioning, doglegged, half-turn and quarter turn stairs.	3
Unit 8	FINISHING WORKS AND SCAFFOLDING Plastering and pointing, painting and distempering, introduction to scaffolding.	2

List of Experiments:

		Number of Turns
1	To determine fineness and specific gravity of cement.	01
2	To determine standard consistency of cement.	01

3	To determine initial and final setting time of cement.	02
4	To determine soundness of cement.	01
5	To examine water absorption and efflorescence in brick.	02
6	To determine the compressive strength of brick.	01
7	Determination of tensile and compressive strength of steel.	02
8	To carry out sieve analysis of coarse and fine aggregates and to find fineness modulus.	02
9	To determine specific gravity of coarse and fine aggregates.	02

Course Outcomes:

At the end of this course, the student will be able to:

1	Select the appropriate building materials for a particular civil engineering structure.
2	Perform quality tests on different building materials.
3	Understand the construction procedure of different building components.

Text Book:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Building Construction Punmia B.C, Arun Jain & Ashok Jain, "Building Construction" Eleventh Edition, Laxmi publication	2016

Reference Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Building Construction", Rangwala, Charotar Publishing House Pvt. Ltd	2016
2	"Building Materials," P.C. Varghese, PHI Learning Pvt Ltd.	2015
3	"Building Technology," S.S. Bhavikatti, Vikas Publishing House Pvt. Ltd.	2013
4	"Building Materials," S.K. Duggal, Fifth Edition, New Age International Publishers.	2019
	Relevant IS codes (Latest Revision)	

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	(https://nptel.ac.in/courses/105103093)---Construction Planning and Management, IIT Guwahati.	NPTEL
2	(https://archive.nptel.ac.in/courses/105/102/105102088)--- Building material & construction	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	2	3	2	3	2	3	2	3	3	3	1	3	3
CO2	3	2	2	2	3	2	3	2	3	2	3	3	3	1	3	3
CO3	3	3	3	2	3	2	3	2	3	2	3	3	3	1	3	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Transportation Engineering	DCC
Course Code	:	CEN3005	
Credits	:	4	
L-T-P	:	3 – 0 – 2	

Course Objectives:

To make the students learn the basic aspects of road infrastructure including planning, design and maintenance.

- To understanding of highway planning and development,
- To analyze the geometry design of highways
- To analyze the traffic studies and analysis
- To analyze the pavement design and analysis, highway material, drainage, bridges, and maintenance

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION Highway planning - basic principles, road development and planning in India; Highway alignment and Surveys	4
Unit 2	GEOMETRIC DESIGN OF HIGHWAYS Design of cross-section elements, sight distances, design of horizontal and vertical alignments	6
Unit 3	TRAFFIC STUDIES Traffic studies and their presentation, PCU and axle load survey, Capacity and Level of service, highway safety measures	8
Unit 4	HIGHWAY MATERIALS Aggregates and their characterization, Bituminous materials, tests on Aggregates and Bituminous materials	4
Unit 5	HIGHWAY CONSTRUCTION Construction Procedure for WBM, Bituminous Bound macadam and Cement Concrete roads	4
Unit 6	PAVEMENT DESIGN Classification of pavements and factors affecting design; Design methods of flexible pavements and of Rigid pavements	7
Unit 7	HIGHWAY DRAINAGE, HIGHWAY BRIDGE AND HIGHWAY MAINTENANCE Longitudinal and cross drainage, Surface and subsurface drainage, Types of bridges, components, selection criteria, General causes of pavement failure, failures in flexible and rigid pavements and their maintenance	9

List of Experiment		No. of turns
1	Impact Value test on Aggregates	2
2.	Specific Gravity of the Aggregates	2
3.	Crushing value test of the aggregates	2
4.	Flakiness and Elongation test of the Aggregates	2
6.	Specific Gravity of the Bitumen	2

7.	Softening Point of the Bitumen	2
8.	Ductility test of the bitumen	2
9.	Penetration test of the bitumen	2
10.	CBR value of the soil	4
11.	Traffic Volume Study	4
12.	Final Evaluation of students	2

Course Outcomes:

At the end of the course the students will be able to:

1	Identify the components of highways.
2	Plan, design and execute highway projects.
3	Choose the suitable material for highways, construction and maintenance
4	Realize the importance of traffic studies.

Textbooks:

Sr. No.	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Highway Materials and Pavement Testing” Khanna S.K., Justo C.E.G. and Veeraraghavan A., New Chand Publications, New Delhi.	2013
2.	“Principles of Transportation Engineering” Khitoliya R. K. and Gupta P. K, Dhanpat Rai Publishing, New Delhi.	2017

Reference Books:

Sr. No.	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Highway Engineering” Kadiyali L.R., Khanna Publishing, New Delhi.	2019
2.	“Principles & Practices of Highway Engineering” Kadiyali L.R, KhannaPublishers, New Delhi.	2004
3.	“Principles, Practice and Design of Highway Engineering” Sharma S. K. (2014), S.Chand Publication.	2014

Indian Standards:

Sr. No.	Code Name	Year
1	IRC:19-2005, Standard Specifications and Code of Practice for Water Bound Macadam https://law.resource.org/pub/in/bis/irc/irc.gov.in.019.2005.pdf	2005
2	IRC:27-2009, Specifications for Bituminous Macadam (First Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.027.2009.pdf	2009
3	IRC:37-2019, Guidelines for the Design of Flexible Pavements (Fourth Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.037.2019.pdf	2019

4	IRC:58-2015, Guidelines for the Design of Plain Jointed Rigid Pavements for Highways(Fourth Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.058.2015.pdf	2015
5	IRC:64-1990, Guidelines for Capacity of Roads in Rural Areas (First Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.064.1990.pdf	1990
6	IRC:66-1976, Recommended Practice for Sight Distance on Rural Highways https://law.resource.org/pub/in/bis/irc/irc.gov.in.066.1976.pdf	1976
7	IRC:73-1990, geometric design standards for rural (non-urban) highways https://law.resource.org/pub/in/bis/irc/irc.gov.in.073.1990.pdf	1990
8	IRC:86-2019, Geometric Design Standards for Urban Road and Streets https://law.resource.org/pub/in/bis/irc/irc.gov.in.086.2018.pdf	2019

Equivalent MOOC courses

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105105107 “Transportation Engineering”	NPTEL
2	https://nptel.ac.in/courses/105101087 “Transportation Engineering 1”	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	3	-	-	-	-	-	-	-	-	-	1	2	3
CO2	1	2	3	3	3	2	3	3	-	-	-	-	1	2	3
CO3	1	-	3	3	3	2	3	3	-	-	-	-	1	2	-
CO4	1	2	3	3	-	-	3	3	-	-	-	-	1	2	3

1 - Slight(low) Correlation

2 - Moderate(Medium) Correlation

3 - Substantial(High) Correlation

“-” indicates there is no Correlation

Course Name	:	Reinforced Concrete Design-I	DCC
Course Code	:	CEN4001	
Credits	:	4	
L T P	:	3 0 2	

Course Objectives:

At the end of this course, the student should be able to:

- Conduct quality tests on concrete and constituent materials of concrete.
- Design various RCC structural elements and assess their performance.
- Design various types of RC foundations.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	PROPERTIES OF CONCRETE Materials for concrete and their properties, concrete as composite material, properties of concrete in fresh and hardened state, workability, strength and durability, factors affecting workability, strength and durability, Indian Standard method of proportioning concrete mixes.	6
Unit 2	REINFORCED CONCRETE Design philosophies, Working Stress method, Ultimate load method and limit state method of design, characteristic strength, Design Loads, characteristic load, design values and partial safety factors, stress strain relationship for concrete and steel.	3
Unit 3	DESIGN AND DETAILING OF BEAMS Singly reinforced beams, modes of failure, moment of resistance and design of beams for flexure, analysis and design of doubly reinforced and flanged beams. Shear, bond, torsion. Limit state of serviceability, control of cracking, deflection and vibrations, design of continuous beams by co-efficient method as per IS code.	8
Unit 4	DESIGN AND DETAILING OF SLABS One-way slabs, two way slabs, continuous slabs.	5
Unit 5	DESIGN OF COLUMNS Types of columns, formulation of interaction charts, design of short columns, columns with helical reinforcement.	8
Unit 6	FOUNDATIONS Design of isolated and combined rectangular footings, design concepts of strip and raft foundation, design of masonry walls and their footings	7
Unit 7	STAIRCASES Design of different type of stairs.	5

List of Experiments:		No. of Turns
1	To determine compressive strength of cement.	2
2	To determine bulk density and voids of fine and coarse aggregates.	2
3	To determine workability of concrete.	1
4	To design concrete mixes.	3
5	To determine split tensile strength of concrete.	2
6	To determine flexural strength of concrete beam specimens.	2
7	To assess the quality of concrete by conducting Rebound hammer Test.	1
8	To assess the quality of concrete by conducting UPV Test.	1

Course Outcomes: At the end of the course the students will be able to:	
1	Assess the quality of concrete and concrete-making materials and design concrete mixes ensuring compliance with Indian standard codes and industry best practices.
2	Design Reinforced Concrete beams, slabs, columns and staircases that meet structural requirements and safety standards.
3	Design various types of RCC foundations.

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	"Reinforced Concrete Limit State Design" A.K.Jain, Nem Chand and Bros.	2012
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	"Reinforced Concrete Design", S.U. Pillai & Devdas Menon' Tata McGraw Hill Publications.	2005
2	"Plain & Reinforced Concrete (Vol. I & Vol-II)", Jain & Jai Krishan, Nem Chand and Bros.	2007
3	"Limit State Design of Reinforced Concrete", Punmia and Jain (Vol.II), Laxmi Publications.	2007
4	IS:456-2000 "Code of Practice for Plain and Reinforced Concrete Design", BIS Publication.	2000
5	"Concrete Technology", M. L. Gambhir, Tata McGraw Hill.	2017
6	Relevant IS codes (Latest Revision)	

Equivalent MOOCs courses

Sr. No	Course Links	Offered by
1	(https://archive.nptel.ac.in/courses/105/105/105105105/)---Design of Reinforced Concrete Structures, IIT Kharagpur	NPTEL
2	(https://nptel.ac.in/courses/105105104)--- Design Concrete Structures, IIT Kharagpur	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	1	3	3	1	3	2	1	3	3	2	3	3
CO2	3	3	3	2	3	3	3	1	2	2	1	3	3	3	3	3
CO3	3	3	3	2	3	3	3	1	2	2	1	3	3	3	3	3

- 1 – Slight (Low) Correlation
 2 – Moderate (Medium) Correlation
 3 – Substantial (High) Correlation
 “-” indicates there is no correlation.

Course Name	:	Structural Analysis - II	DCC
Course Code	:	CEN4002	
Credits	:	3	
L T P	:	3 0 0	

Course Objectives:

At the end of this course, the student should be able to:

- Analyze various civil engineering indeterminate framed structures.
- Analyze various civil engineering indeterminate trussed structures.
- Analyze the structures using matrix methods.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	SLOPE DEFLECTION METHOD Derivation of general slope deflection equation, its application to statically indeterminate beams, portal frames with and without sway, yielding of supports.	5
Unit 2	MOMENT DISTRIBUTION METHOD Introduction, stiffness factor, carry over factor, distribution factor, application to indeterminate beams, portal frames with and without sway, yielding of supports.	6
Unit 3	ANALYSIS OF FRAMES Approximate analysis of frames by Portal method, Cantilever method and Substitute frame method. Introduction to analysis of frames by using software.	6
Unit 4	INDETERMINATE TRUSSES Method of least work, Castigliano's 2nd theorem-its derivation and application, Analysis of plane indeterminate trusses, forces in various members of a plane redundant truss due to lack of fit of members.	7
Unit 5	TWO HINGED ARCHES Analysis of parabolic arch with secant variation of moment of inertia. Influence lines for different parameters, reaction locus and its application, effect of temperature, yielding of supports.	6
Unit 6	MATRIX METHODS Introduction to stiffness and flexibility method, application of flexibility method to beams, application of stiffness method to beams.	8
Unit 7	FINITE ELEMENT METHOD Introduction, Basic principle, discretization, Convergence criteria, Applications.	4

Course Outcomes:

At the end of the course the students will be able to:

1	Analyze various indeterminate structures using classical methods.
2	Analyze beams using matrix methods.
3	Analyze the behavior of structures experimentally.

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	"Theory of Structures", B C Punmia (Vol. II), Laxmi Publications.	2017
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	"Matrix Analysis of Framed Structures", Gere and Weaver, CBS Publishers and Distributors, Delhi-110032.	2004
2	"Advanced Structural Analysis", Ashok K. Jain, Nem Chand & Bros.	2015
3	"Structural Analysis", T.S. Thandavamoorthy, Oxford University Press.	2011
4	"Structure Analysis: A Matrix Approach", G. Pandit, S. Gupta Tata McGraw Hill Publications.	2008

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	(https://archive.nptel.ac.in/courses/105/101/105101086/) Structural Analysis II, IIT Bombay	NPTEL
2	(https://archive.nptel.ac.in/courses/105/106/105106050/) Structural Analysis, IIT Madras	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	2	3	2	2	2	2	2	3	3	3	2	3
CO2	3	3	3	3	2	3	2	2	2	2	2	3	3	3	2	3
CO3	3	3	3	2	3	2	2	2	2	2	2	3	3	3	2	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Soil Mechanics	DCC
Course Code	:	CEN4003	
Credits	:	4	
L - T - P	:	3 – 0 – 2	

Course Objectives:

The students should be able to:

- Understand the formation of soil, classification and its properties.
- Formulate and analyze engineering behaviour.
- Analyze development of stresses in the soil in the field situation.
- Explore the causes of failure in the soil.

Total No. of Lectures – 42

Lecture-wise break-up		No. of Lectures
Unit 1	INTRODUCTION TO SOIL Origin and formation of soils, types of soil, various soil deposits in India and their Characteristics. Identification and classification of soils; 3-phase system of soils particle size and plasticity of soils, limit of consistency and their determination, I.S. Soil classification system, weight, volume and unit weight relationships and their inter-relationships.	5
Unit 2	SOIL STRUCTURE AND CLAY MINERALS Introduction to Clay minerals, Basic concepts of soil structure and clay minerals, Clay Particle interaction, Soil Structure and Fabrics, Clay Water System, Clay Shapes and Surface area.	3
Unit 3	COMPACTION Compaction, Factors affecting Compaction, Standard and Modified Proctor's Test, Moisture-density relationship, Field Control of Compaction, field and laboratory compaction devices and their suitability of compaction process, effects of compaction on properties of soils.	5
Unit 4	STRESS DISTRIBUTION Concept of Total Stress, Effective Stress and Pore water Pressure, Stress below point load (Bousinessq's and Westergaard's equations), stresses below uniformly loaded circular and rectangular areas (exact method), approximate methods – point load method, Newmark's chart.	4
Unit 5	CONSOLIDATION Compressibility, Factors affecting compressibility, Consolidation, Terzaghi's 1-D consolidation theory, Time Factor, Degree of Consolidation, Predicting settlements and time rates, Consolidation tests, , Square root time fitting method and logarithmic time fitting methods, pre-consolidation pressure and over consolidation ratio, NC and OC Clays, Secondary consolidation.	7
Unit 6	PERMEABILITY Permeability; factors affecting permeability; Darcy's law, Laboratory tests and their suitability, Field method for determination of Permeability, flow nets for confined and unconfined flows, effective stress under hydrodynamic conditions, Seepage Pressure, quick sand condition, Filters, and Drains.	6

Unit 7	SHEAR STRENGTH Concept of shear strength in soils, Mohr's stress circle, Coulomb's failure envelope, stage of failure and plane of failure, cohesion intercept, angle of shearing resistance, shear strength equation in terms of principal stress, measurement of shear strength of soils by direct shear test, triaxial compression tests, unconfined compression test, vane shear test and presentation of their results. Performing shear strength tests under different drainage conditions (drained, undrained, and consolidated undrained), basic features of triaxial compression test apparatus, pore water pressure parameters. Sensitivity of clays, Thixotropy and critical void ratio.	6
Unit 8	EARTH PRESSURE Concept of Lateral Earth Pressure, Active and Passive Earth Pressure, Rankine's Theory and Coulomb's theories, effect of surcharge loading, water table fluctuations and soil stratification, computation of total lateral thrust and location of resultant earth pressure on earth retaining wall, estimation of depth of unsupported vertical cut in Cohesive backfills. Design consideration of earth retaining structures	6

List of Experiments:		Number of Turns
1	To determine grain size distribution by Sieve analysis.	1
2	To conduct Specific gravity test by Pycnometer.	1
3	To determine In-situ density using Sand Replacement method.	1
4	To determine the relative density of soil.	1
5	To determine the field Density test of soil by Core Cutter Method.	1
6	To Determine Liquid limit and Plastic limit.	1
7	To Determine shrinkage limit.	1
8	To determine coefficient of permeability by falling head method.	1
9	To determine coefficient of permeability by constant head method.	1
10	MDD and OMC by Standard Proctor compaction test.	1
11	To determine shear strength parameters by Direct Shear Test.	1
12	To perform Unconfined Compression Test.	1
13	Demonstration of Tri-axial Shear Test.	2

Course Outcomes:	
By the end of this course, the student will be able to:	
1	Characterize and classify soils.
2	Identify and analyze the properties of soil governing its behaviour.
3	Apply the principles of soil mechanics in the design of earth retaining structures.
4	Perform failure analysis of structures under shear failure and excessive settlement.

Text Book:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	"Basic and Applied Soil Mechanics," Gopal Ranjan & A.S.R. Rao. New Age International Publishers.	2006
2.	"Soil Mechanics and Foundations," Muni Budhu, John Wiley & Sons.	2010

Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Principles of Geotechnical engineering,” Braja M. Das, Cengage Learning, New Delhi.	2009
2.	“Soil Mechanics,” T. William Lambe & Robert V. Whitman, John Wiley & Sons, NY.	2008
3.	Soil Engineering,” Alam Singh, CBS Publishers, New Delhi.	2009
4.	“Soil Mechanics in Engineering Practice,” Terzaghi and Peck, John Wiley and Sons, New Jersey.	1948

Equivalent MOOCs courses

Sr. No	Course Links	Offered by
1.	https://nptel.ac.in/courses/105/105/105105168/ NPTEL video course on Geotechnical Engineering	NPTEL
2.	https://nptel.ac.in/courses/105/103/105103097/ NPTEL web course on Soil Mechanics	NPTEL
3.	https://nptel.ac.in/courses/105/106/105106142/ NPTEL web course on Geotechnical and Foundation Engineering	NPTEL
4.	https://nptel.ac.in/courses/105/101/105101201/ NPTEL video course on Geotechnical Engineering	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	3	1	1	-	-	-	1	1	-	3	3	-	3
CO2	3	3	3	3	2	2	1	-	-	1	1	-	3	3	-	3
CO3	3	3	3	3	2	3	2	-	-	1	1	-	3	2	-	3
CO4	3	3	3	3	3	2	2	1	-	1	1	-	3	3	-	2

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Irrigation Engineering and Open Channel Hydraulics	DCC
Course Code	:	CEN4004	
Credits	:	3	
L – T – P	:	3 0 0	

Course Objectives:

The student should be able to

- Understand concept of free surface flow.
- Apply principles of mass conservation and momentum conservation in open channel flow.
- Distinguish between rigid boundary and mobile boundary channel.
- Assess the crop water requirements and identify suitable irrigation system for command area.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION TO FREE SURFACE FLOWS Comparison between open channel flow and pipe flow, Types of Channels, Classification of flows, Velocity distribution, Pressure distribution, and Basic governing equations of open channel flows.	3
Unit 2	UNIFORM FLOW Establishment of uniform flow, Basic resistance equations in channel flow, Variation of roughness coefficient, Velocity and shear stress distribution, Conveyance of Channel, Section factor and normal depth computation, most efficient channel section, Maximum discharge and maximum velocity in channels.	4
Unit 3	CRITICAL FLOW Specific energy in an open channel, Properties of critical flow, Computation of critical depth, Hydraulic exponent for critical depth computation, Application of specific energy concepts.	3
Unit 4	GRADUALLY VARIED FLOW Governing equations, Classification and characteristics of water surface profiles and channel slopes, Computations of GVF profiles in prismatic channels.	3
Unit 5	RAPIDLY VARIED FLOW Momentum equation formulation for the jump, Specific force diagram, Hydraulic jump and its types, Characteristics of hydraulic jump in a horizontal rectangular channel, Loss of energy in hydraulic jump,	3
Unit 6	UNSTEADY FLOW Saint-Venant Equations, Gradually varied unsteady flows, Uniformly progressive wave, Rapidly varied unsteady flows – positive and negative surges.	3
Unit 7	CROP WATER REQUIREMENTS Classes and availability of soil water, Available moisture depth, Frequency of irrigation, Relationship between duty, delta and base period, Factors affecting duty, Methods of improving duty, Irrigation efficiencies, Command areas, Kharif, rabi and perennial crops, Crop rotation, Irrigation water requirement, Consumptive use.	3
Unit 8	IRRIGATION METHODS Types of irrigation schemes, Flooding methods, Border strip method, Check basin and furrow method, Assessment of irrigation water, Sprinkler	3

	irrigation and its design, Drip irrigation and its design.	
Unit 9	CANAL IRRIGATION Components of canal distribution system, Alignment and losses of channels, Kennedy's and Lacey's theories and design procedure, Garrets and Lacey's diagrams, Canal lining and design of a lined canal.	3

List of Experiments		No of Lectures
1	Basic Introduction to Experiments	4
2	Estimation of Manning's and Chezy's roughness coefficient in open channel flow	2
3	Analysis of Hydraulics of Broad Crested Weir	2
4	To Determine the Elements of a Hydraulic Jump	2
5	Analysis of Flow over a Hump in open channel flow	2
6	Verification of Stokes's Law	2
7	Seepage analysis	4
8	Estimation of Soil moisture using Thiessen polygon	4
9	Drawing the Details of Bhakra Nangal Dam in Drawing sheets	6

Course Outcomes (COs): At the end of the course the students will be able to:	
1	Formulate mathematical expression of open channel flow.
2	Analyze the flow patterns in open channels.
3	Design the rigid and mobile boundary channels.
4	Design the canal irrigation system for estimated crop water requirement of command area.

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Open-channel hydraulics", Chow, V. T., The Blackburn Press.	2009
2	"Flow in open channels (5 th Edition)", Subramanya, K., McGraw-Hill Education.	2019

Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Flow through open channel", Ranga Raju, K. G., McGraw-Hill Education.	2013
2	"Open Channel Flow (3 rd Edition)", Chaudhry M. Hanif, Springer.	2022
3	"Flow Through Open Channels", Srivastava Rajesh., Oxford University Press India.	2008
4	"Irrigation and Water Resources Engineering", Asawa, G. L., New Age International, New Delhi.	2014
5	"Irrigation Engineering and Hydraulic Structures" Garg, S. K., Khanna Publishers, New Delhi.	2011

Equivalent MOOCs courses

Sr. No .	Course Links	Offered by
1.	https://nptel.ac.in/courses/105/107/105107059/ - Web Course on ‘Advanced Hydraulics’	NPTEL
2.	http://www.nptelvideos.in/2012/11/hydraulics.html - Video Course on ‘Hydraulics’	NPTEL
3.	https://ocw.mit.edu/courses/civil-and-environmental-engineering/1-060-engineering-mechanics-ii-spring-2006/lecture-notes/ -	MIT Open Course Ware
4.	https://nptel.ac.in/courses/105/105/105105110/ - ‘Water Resources Engineering’	NPTEL
5.	http://www.nptelvideos.in/2012/11/water-resources-engineering.html	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	3	1	2	3	1	2	3	1	2	3	3	1	3
CO2	3	3	3	3	3	1	2	1	3	3	1	3	3	3	1	3
CO3	3	3	3	3	3	3	3	1	3	3	3	3	3	3	1	3
CO4	3	3	3	3	3	1	3	1	2	3	3	3	3	3	1	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Environmental Engineering I	DCC
Course Code	:	CEN4005	
Credits	:	3	
L - T - P	:	3-0-0	

Course Objectives:

The student should be able to

- compare raw water quality characteristics from different sources
- plan and design water treatment scheme
- imbibe the fundamentals of water distribution networks and their design
- identify causes of air pollution and strategies for mitigation

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION Components of water supply scheme, Water demand, Population forecasting methods	4
Unit 2	WATER QUALITY Drinking Water Quality Standards	6
Unit 3	SOURCES OF WATER Surface Water: Intakes for collecting surface water- types and location; Ground water: Aquifer, aquiclude, movement of ground water and its velocity, yield of wells; Infiltration wells & galleries. Quality comparison of water from different sources	8
Unit 4	TRANSPORTATION AND PUMPING OF WATER Various types of Pipes, Materials, Joints Various types of Pumps	4
Unit 5	WATER TREATMENT Aeration, Sedimentation – Type I & II, Coagulation Flocculation, Filtration, Disinfection, Specific treatment.	10
Unit 6	WATER DISTRIBUTION SYSTEM Components and Design, Various Layouts	5
Unit 7	AIR Composition and properties of air, Air pollution, Urban air pollution automobile pollution, Chemistry of combustion, Air quality standards, Control measures for Air pollution.	5

Course Outcomes:

By the end of this course, the student will be able to:

1	Analyze and interpret quality parameters of water
2	Identify the process design aspects of water supply
3	Design water distribution system
4	Describe the strategies to control air pollution

Text Book:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Environmental Engineering”, H.S. Peavy, D.R. Rowe, G. Tchobanoglous, McGraw Hill, NY.	1985
2.	“Manual on Water Supply and Treatment”, CPHEEO, MOUD.	1999
	“Environmental Engineering: Fundamentals and Application”, S.Verma, V. Kanwar, S. John, Routledge, Taylor and Francis group	2022
Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Water Works Engineering”, S.R. Qasim, E.R. Motley, G. Zhu , PHI (EEE Edition), New Delhi.	2010
2.	“Water and Wastewater Technology”, Hammer & Hammer, Prentice Hall of India.	2008
3.	“Introduction to Environmental Engineering”, P.A. Vesilind, S.M. Morgan, Thomson Books, NY.	2011
4.	IS: 10500. Drinking water specifications	2012

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	https://nptel.ac.in/courses/105/105/105105201/ - Water Supply Engineering	NPTEL
2.	https://nptel.ac.in/courses/105/107/105107207/ - Water and Wastewater Treatment	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	3	2	2	1	2	1	3	3	2	2	3
CO2	3	3	3	2	3	3	2	2	1	2	1	3	3	2	2	3
CO3	3	3	3	2	3	3	2	2	1	2	1	3	3	2	2	3
CO4	3	3	3	2	3	3	2	2	1	2	1	3	3	2	2	3

1 – Slight (Low) Correlation
 2 – Moderate (Medium) Correlation
 3 – Substantial (High) Correlation
 “-” indicates there is no correlation.

Course Name	:	Railway, Airport & Tunnel Engineering	DCC
Course Code	:	CEN4006	
Credits	:	3	
L-T-P	:	3 – 0 – 0	

Course Objectives :

At the end of the course the students will be able to:

- Students should be able to relate their understanding of the railroad industry, history, and principal components
- To analyze and design the components of railways.
- To know the basic components of airport, analyses and design them
- To know the tunneling systems, drainage and ventilation.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
A. Railways		
Unit 1	INTRODUCTION Indian Railways system, gauges, comparison of gauges, advantages of ungauged system, Ideal permanent way, Components of permanentway: Rails, Sleepers and Ballast.	3
Unit 2	GEOMETRIC DESIGN Geometric Design of the Railway Track, Curve, Cant, Cant deficiency	3
Unit 3	POINTS AND CROSSING Layout plans of simple crossovers, turnouts, diamond crossing, design aspects.	3
Unit 4	STATIONS AND YARDS Site selection, different types of stations and their layout, junctions and terminals, marshalling yards, level crossing	3
Unit 5	PERMANENT WAY CONSTRUCTION & MAINTENANCE Laying of tracks, Joints and fixtures, Creep, Maintenance	2
Unit 6	SIGNALLING & INTERLOCKING Objectives, different types of signals and their location in station yards, interlocking, requirements of interlocking system.	3
B. Airports		
Unit 7	INTRODUCTION Classification of airports, Aircraft characteristics, regulatory bodies, Terminology, Air travel demand forecasting, air traffic control systems	3
Unit 8	AIRPORT PLANNING Site Selection, Requirements, Zoning laws, Types of runway patterns, Runway layout conditions, length and width, Wind rose diagram, runway orientation, clearance, apron area, Taxiways, Airport capacity, Terminal area, Hangers, Airport markings, Lighting	3
Unit 9	AIRPORT PAVEMENT DESIGN Airport Pavement Design, Factors affecting pavement thickness, methods for flexible and rigid pavements	4
Unit 10	AIRPORT MAINTENANCE AND REHABILITATION Types of failures in air field pavements, pavement evaluation, repair techniques, design of overlays	3

Unit 11	AIRPORT DRAINAGE Surface and subsurface drainage, Runoff estimation, Structures, and materials for drainage	3
C. Tunnel Engineering		
Unit 12	Introduction, types of tunnels, Advantages, Disadvantages, Shapes of tunnel lining, Tunnel Lining materials	3
Unit 13	Soil classification and tunneling methods (Basic description only), drainage of tunnel, tunnel ventilations and dust.	6

Course Outcomes	
1	Know the history of development of railways.
2	Understand the working and design of different elements of railway track.
3	Understand the airport planning and design of various components for efficient development of airports
4	Get familiar with tunnelling systems.

Textbooks:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Airport Engineering: Planning & Design” Saxena S.C., CBS Publishers.	2019
2.	“A Textbook of Railway Engineering” Saxena S.C. and Arora S. P. , Dhanpat Rai and Sons, New Delhi.	2010
3.	“Principles of Railway Engineering” Rangwala S.C., Charotar Publishing House.	2015
Reference Book:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Airport Planning and Design” Jain S.S., Arora M.G. and Khanna S. K., NemChand and Bros.	1999
2.	“Railway Engineering” Chandra S. and Agarwal M.M., Oxford University Press, New Delhi.	2013
3	IRC:39-1986, Standards for Road-Rail Level Crossings (First Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.039.1986.pdf	1986
4	IRC:76-1979, Tentative Guidelines for Structural Strength Evaluation of Rigid Airfield Pavements https://law.resource.org/pub/in/bis/irc/irc.gov.in.076.1979.pdf	1979
5	IRC105:2019, Specifications for Dense Bituminous Macadam and Bituminous Concrete for Airfield Pavements (First Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.105.2019.pdf	2019

Equivalent MOOC courses

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105/107/105107123/ 'Introduction to airport engineering', 'Introduction to airport engineering'.	NPTEL
2	https://archive.nptel.ac.in/courses/105/105/105105212/ 'Rock Mechanics and tunneling'	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	-	2	-	-	-	-	-	-	3	2	1	1	-	1	3
CO2	3	2	3	1	-	1	2	1	3	2	1	3	3	3	1	3
CO3	3	2	3	1	-	3	2	1	2	2	1	3	3	3	2	3
CO4	3	1	3	-	1	-	-	-	-	3	2	1	1	-	1	3

1 - Slight(low) Correlation

2 - Moderate (Medium) Correlation

3 - Substantial (High) Correlation

“-” indicates there is no Correlation

Course Name	:	Design of Steel Structures- I	DCC
Course Code	:	CEN5001	
Credits	:	3	
L T P	:	3 0 0	

Course Objectives:

At the end of this course, the student should be able to:

- Know the basics of limit state design of steel structures, design bolted and welded connections.
- design compression and tension members.
- understand the basics of plastic design and able to design flexural members and laterally restrained beams.
- design steel roof truss.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION Introduction to steel and steel structures, Introduction to design, Design loads and load combinations, limit state design, allowable stresses in structural steel.	4
Unit 2	DESIGN OF BOLTED CONNECTIONS Terminology, Types of bolted connections, Modes of failure, Design of bolted connections for direct loads, for joints subjected to moment perpendicular to and in the plane of joint.	5
Unit 3	DESIGN OF WELDED CONNECTIONS Terminology, Types of welded connections, Modes of failure, Design of welded connections for direct loads, for joints subjected to moment perpendicular to and in the plane of joint	5
Unit 4	DESIGN OF TENSION MEMBERS Types of Tension members, strength of tension members, design of tension members, introduction to splicing and lug angles.	5
Unit 5	DESIGN OF COMPRESSION MEMBERS Types of compression members, Buckling classes of compression members, design of compression members, design of laced and battened columns.	6
Unit 6	BASICS OF PLASTIC DESIGN Introduction to plastic design, Moment curvature relationship, shape factor, plastic hinges, plastic moment of various sections.	4
Unit 7	DESIGN OF BEAMS Design of steel beams, web buckling, web crippling, design of built-up beams, laterally supported beams, introduction to torsional buckling.	7
Unit 8	ROOF TRUSS Introduction, Types of trusses, Roof coverings, estimation of wind load, design of steel roof truss subjected to dead, live and wind loads.	6

Course Outcomes:

At the end of this course, the student will be able to:

1	Understand the design philosophies of steel structures and design the bolted and welded connections of structural steel.
----------	--

2	Design compression and tension steel members.
3	Understand the basics of plastic design and design of flexural beams.
4	Design steel roof truss.

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	“Limit state design of Steel Structures”, S.K. Duggal, Tata McGraw Hill	2017
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	“Steel Structures: Design and practice”, N. Subramanian, Oxford Publishing Press.	2010
2	“Design of Steel Structures”, Arya & Awadhesh Kumar, Nem Chand and Bros., Roorkee.	2022
3	“Design of Steel Structures”, S.S. Bhavikatti, I.K. International Publishing House Pvt. Ltd.	2012
4	“Design of Steel of Structures”, Volume 1 Ram Chandra.S., Virendra Gehlot, , Scientific Publishers, New Delhi.	2016
5	Relevant IS Codes (Latest Revision)	

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	https://archive.nptel.ac.in/courses/105/106/105106112/ ---Design of Steel Structures I, IIT Madras	NPTEL
2	https://archive.nptel.ac.in/courses/105/105/105105162/ ---Design of steel structures, IIT Kharagpur	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	1	2	1	2	2	3	2	3	3	3	3	3
CO2	3	3	3	3	1	2	1	2	2	2	2	3	3	3	3	3
CO3	3	3	3	3	1	2	1	2	2	2	2	3	3	3	3	3
CO4	3	3	3	3	1	2	1	2	2	3	2	3	3	3	3	3

- 1 – Slight (Low) Correlation
 2 – Moderate (Medium) Correlation
 3 – Substantial (High) Correlation
 “-” indicates there is no correlation.

Course Name	:	Reinforced Concrete Design-II	DCC
Course Code	:	CEN5002	
Credits	:	3	
L T P	:	3 0 0	

Course Objectives :

At the end of this course, the student should be able to:

- Design specialized RCC structures.
- Design liquid retaining RC structures.
- Design basic PSC structures.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	RETAINING WALLS Types of retaining walls and their Components, stability criteria, Design and detailing of cantilever and counterfort retaining walls.	7
Unit 2	BEAMS CURVED IN PLAN Design of circular beams supported on symmetrically placed columns, semicircular beams supported on equally spaced columns.	4
Unit 3	DOMES Types of domes, analysis and design of RCC spherical and conical domes, openings in domes.	4
Unit 4	LIQUID RETAINING STRUCTURES Design criteria of liquid retaining structures, Design of tanks resting on ground, underground tanks and design concepts for overhead service reservoirs, its staging and foundation.	6
Unit 5	INTRODUCTION TO PRESTRESSED CONCRETE Introduction, materials for prestressed concrete, methods of prestressing.	3
Unit 6	LOSSES OF PRESTRESS Losses of Prestress, Loss due to elastic deformation, shrinkage and creep on concrete, relaxation of stress in steel, loss due to friction, anchorage slip, total losses.	4
Unit 7	ANALYSIS OF PRESTRESS MEMBERS Basics Assumptions, Analysis of prestress, Pressure Line, kern distance, Concept of Load Balancing, Stresses in Tendons.	6
Unit 8	LIMIT STATE DESIGN OF PRESTRESSED CONCRETE BEAMS Types of flexural failure, Design of PSC beams for flexure, shear, torsion and bond, deflection in PSC beams, IS codal specifications.	5
Unit 9	ANCHORAGE ZONE STRESSES End blocks, stress distribution in end blocks, transmission length, anchorage zone reinforcement.	3

Course Outcomes:

At the end of this course, the student will be able to:

1	Design specialized RCC structures as per IS codal specifications.
2	Design liquid retaining structures, complying to latest IS codes.
3	Analyze and Design basic Prestressed concrete structures as per IS codal specifications.

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Reinforced Concrete Limit State Design" A.K.Jain, Nem Chand and Bros.	2012
2.	"R.C.C. Designs (Reinforced Concrete Structures)" B.C. Punmia, Ashok Kumar Jain & Arun Kumar Jain, Laxmi Publications.	2015
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1.	"Plain & Reinforced Concrete", Jain & Jai Krishan (Vol. I & Vol-II), Nem Chand and Bros.	2007
2.	"Reinforced Concrete Structures", Syal and Goel, S. Chand & Company Pvt. Ltd.	2014
3.	"Reinforced Concrete Design", S.U. Pillai & Devdas Menon, Tata McGraw Hill.	2017
4.	"Plain & Reinforced Concrete (Vol. I & Vol-II)", Jain & Jai Krishan, Nem Chand and Bros.	2007
5.	"Design of Prestressed Concrete structures", Lin T.Y., Wiley Interscience, Publication, New Jersey	2010
6.	"Fundamentals of Prestressed Concrete", V Natrajan, B.I. Publications, New Delhi	1983
7.	"Prestressed Concrete", N. Raju Krishna, McGraw Hill Education	2019
8.	"Prestressed Concrete Structures 7Th Edition", Dayaratnam P., Medtech Publisher.	2017
9.	Relevant IS codes (Latest Revision)	

Equivalent MOOCs courses

Sr. No	Course Links	Offered by
1	(https://archive.nptel.ac.in/courses/105/105/105105105/)---Design of Reinforced Concrete Structures, IIT Kharagpur	NPTEL
2	(https://nptel.ac.in/courses/105105104/)--- Design Concrete Structures, IIT Kharagpur	NPTEL
3	(https://nptel.ac.in/courses/105106117/) --- Pre-stressed Concrete Structures, IIT Madras	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	3	3	1	2	2	1	3	3	2	2	3
CO2	3	3	3	2	3	3	3	1	2	2	1	3	3	2	2	3
CO3	3	3	3	2	3	3	3	1	2	2	1	3	3	2	2	3

- 1 – Slight (Low) Correlation
 2 – Moderate (Medium) Correlation
 3 – Substantial (High) Correlation
 “-” indicates there is no correlation

Course Name	:	Environmental Engineering II	DCC
Course Code	:	CEN5003	
Credits	:	4	
L - T - P	:	3-0-2	

Course Objectives:

The student should be able to

- identify various environmental issues related to wastewater
- plan and design wastewater treatment scheme
- analyze, design various components of wastewater collection, treatment and disposal
- identify various aspects of solid waste management

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION Wastewater quantity and quality characteristics	3
Unit 2	SEWERAGE SYSTEM components and design, Sewer appurtenances, Conveyance of sewage- Sewer design, Sewage pumping	10
Unit 3	EFFLUENT DISPOSAL Disposal on land, disposal in water, D.O. sag curve, self-purification of rivers	3
Unit 4	WASTEWATER TREATMENT Wastewater characteristics, preliminary, primary, secondary and tertiary treatment	12
Unit 5	SLUDGE Treatment and Disposal	3
Unit 6	DRAINAGE SYSTEMS Quantification and design of storm water drains	5
Unit 7	SOLID WASTE MANAGEMENT MSW: Components, collection, processing and material recovery, disposal	6

List of Experiments:		Number of Turns
1	Solids- total, suspended, dissolved, volatile, fixed	02
2	pH and turbidity	01
3	Acidity and alkalinity	01
4	Hardness	01
5	Jar Test	01
6	Chloride	01
7	Residual Chlorine concentration	01
8	Dissolved Oxygen	01
9	Biochemical Oxygen Demand	01
10	Chemical Oxygen Demand	01
11	Total Kjeldahl Nitrogen	01
12	Bacteriological count / Most Probable Number	01
13	Suspended Particulate Matter	01

Course Outcomes:

By the end of this course, the student will be able to:

1	Analyze and interpret water and wastewater characteristics
2	Interpret wastewater characteristics to plan and design for sewage and sludge treatment
3	Design sewerage system components
4	Design drainage system components
5	Describe the aspects of solid waste management

Text Book:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Environmental Engineering”, H.S. Peavy, D.R. Rowe, G. Tchobanoglous, McGraw Hill, NY.	1985
2.	“Manual on Water Supply and Treatment”, CPHEEO, MOUD.	1999
3.	“Environmental Engineering: Fundamentals and Application”, S.Verma, V. Kanwar, S. John, Routledge, Taylor and Francis group	2022
4.	“Sewage Disposal and Air Pollution Engineering”, S.K.Garg, Khanna Publishers.	2010

Reference Books:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Manual on sewerage and sewage treatment”, CPHEEO, MoUD.	2013
	“Manual on storm water drainage systems”, CPHEEO, MoUD.	2019
2.	“Water and Wastewater Technology”, Hammer & Hammer, Prentice Hall of India.	2008
3.	“Standard Methods for the Examination of Water and Wastewater”, AWWA.	2022
4.	“The STP guide : Design, operation & maintenance”, A.S. Kodavasal, Karnataka State Pollution Control Board.	2011

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	https://nptel.ac.in/courses/105/107/105107207/ - Water and Wastewater Treatment	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	2	3	3	2	2	2	1	3	3	1	2	3
CO2	3	3	3	2	2	3	3	2	2	2	1	3	3	1	2	3
CO3	3	3	3	2	2	3	3	2	2	2	1	3	3	1	2	3
CO4	3	3	3	2	2	3	3	2	2	2	1	3	3	1	2	3
CO5	3	3	3	2	2	3	3	3	3	3	3	3	3	3	3	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Hydrology and Hydraulic Structures	DCC
Course Code	:	CEN5004	
Credits	:	3	
L – T – P	:	3 – 0 – 0	

Course Objectives:

The students should be able-

- To understand the hydrological processes viz. precipitation and abstractions from precipitation like infiltration, evaporation, and evapotranspiration.
- To understand the concept of runoff, hydrographs, and flood routing
- To apply the basic concepts of hydrology in solving daily life problems related to water resources engineering.
- To understand and analyse the hydraulic structures like diversion and storage head work.

Total No. of Lectures-42

Lecture-wise breakup		No. of Lectures
Unit 1	INTRODUCTION TO WATER RESOURCES ENGINEERING Hydrological cycle, Types of hydrological models, Water-budget equation, Global freshwater resources, Importance and need of water resources projects, Sources of hydrologic data.	2
Unit 2	PRECIPITATION MEASUREMENT AND ANALYSIS Precipitation and its forms, Weather systems for precipitation, Measurement of precipitation, Rain gauge network, Hyetograph, and mass curve, Consistency of rainfall data, Infilling missing rainfall records, Areal precipitation, Depth-area-duration curves, Intensity-duration-frequency curves, Probable maximum precipitation.	5
Unit 3	HYDROLOGIC ABSTRACTIONS Infiltration, Factors affecting infiltration, Measurement of infiltration, Modelling infiltration capacity, Infiltration indices, Evaporation: its measurement and estimation, Evapotranspiration: its measurement and estimation.	5
Unit 4	RUNOFF AND HYDROGRAPH Measurement of stream flow, Area-velocity method, Stage-discharge (rating) curve, Hydrograph, Runoff characteristics of streams, SCS-CN method of estimating runoff volume, Flow duration curve, Factors affecting runoff hydrograph, Components of a hydrograph, Base flow separation, Unit hydrograph – its derivation, conversion, application and limitations, Synthetic unit hydrograph.	7
Unit 5	FLOOD ESTIMATION AND ROUTING Methods for peak discharge estimation, Recurrence interval and return period, Flood frequency analysis – Gumbel’s method, Flood routing – basic equations and data requirements, Hydrologic storage and channel routing.	5
Unit 6	DIVERSION HEADWORKS AND CROSS-DRAINAGE WORKS Components of diversion headwork and their functions, Layout plan, Failure of hydraulic structures on permeable foundations, Bligh’s and, Lane’s weighted creep theory, Khosla’s method of independent variables, Design of weir and barrages on permeable foundation. Types of cross-drainage structure – Aqueduct, Siphon aqueduct, Super passage, Canal	8

	siphon, Level crossing, Selection of suitable cross-drainage structure.	
Unit 7	STORAGE HEAD WORK Choice of type of dams, Site selection and investigation, Gravity dams – Forces acting on a gravity dam, Stability analysis of a gravity dam, Elementary and practical profile of a gravity dam, Earthen dams – Components of earthen dams and their functions, Estimation and control of seepage in earthen dam, Failure criteria of an earthen dam, Spillways - Type of spillways, Discharge characteristics of spillways.	10

Course Outcomes (COs):

At the end of the course, the students will be able to:

1	Understand the hydrological processes to estimate precipitation and abstractions.
2	Compute runoff, derive hydrographs, and carry out flood routing.
3	Understand the concepts of Diversion and storage Head Work.
4	Design important hydraulic structures and their components.

Text Book:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Applied Hydrology”, Chow, V. T., Maidment, D. W., and Mays, L. W., Tata Mc-Graw Hill Education.	2010

Reference Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Irrigation and Water Resources Engineering”, Asawa, G. L., New Age International, New Delhi.	2014
2	“Irrigation Engineering and Hydraulic Structures”, Garg, S. K., Khanna Publishers, New Delhi.	2011
3	“Irrigation Water Power and Water Resources Engineering”, Arora, K. R., Standard Publishers.	2011
4	“Engineering Hydrology”, Subramanya, K., Tata Mc-Graw-Hill, New Delhi.	2013

Indian Standards:

Sr. No.	Code Name	Year
1.	IS: 6512 – 1984 (reaffirmed 1998). Criteria for design of solid gravity dams. Bureau of Indian Standards, New Delhi	1998
2.	IS: 8826 – 1978 (reaffirmed 2002). Guidelines for design of large earth and rockfill dams. Bureau of Indian Standards, New Delhi.	2002
3.	IS: 10635 – 1993 (reaffirmed 1998). Freeboard requirements in embankment dams - Guidelines. Bureau of Indian Standards, New Delhi	1998

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	https://nptel.ac.in/courses/105/104/105104103/ - video course on ‘Water Resources Engineering’	NPTEL

2.	https://nptel.ac.in/courses/105/101/105101002/ - web course on 'Advanced Hydrology'.	NPTEL
3.	https://nptel.ac.in/courses/105/105/105105110/ - web course on 'Water Resources Engineering'.	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	3	1	2	3	1	2	3	1	2	3	3	1	3
CO2	3	3	3	3	3	1	2	1	3	3	1	3	3	3	1	3
CO3	3	3	3	3	3	3	3	1	3	3	3	3	3	3	1	3
CO4	3	3	3	3	3	1	3	1	2	3	3	3	3	3	1	3

1 – Slight (Low) Correlation
2 – Moderate (Medium) Correlation
3 – Substantial (High) Correlation
“-” indicates there is no correlation.

Course Name	:	Foundation Engineering
Course Code	:	CEN5005
Credits	:	3
L - T - P	:	3 – 0 – 0

Course Objectives:

The students should be able to:

- Understand different types of foundations, decide their suitability.
- Design the foundation under static and dynamic load.
- Analyze the stability of slopes.
- Apply the knowledge to improve the ground conditions.

Total No. of Lectures-42

Lecture-wise breakup		No. of Lectures
Unit 1	SITE EXPLORATION Objects of soil investigation for new and existing structures, soil exploration method with relative merits and demerits, depth and spacing of boreholes, soil sampling, SPT and plate load test, Borehole Logs.	4
Unit 2	BEARING CAPACITY Ultimate and allowable bearing pressure, general, local, and punching shear failures, bearing capacity theories, corrections for size, shape, depth, the eccentricity of loading, water table, etc., Presumptive bearing capacities, IS code Method, correlations with SPT.	6
Unit 3	SHALLOW FOUNDATIONS Design considerations-factors of safety, allowable settlements, location and depth of foundations, codal provisions, choice of shear strength parameters, total and differential settlement, stress distribution for settlement calculation, consolidation settlement in clays (with correction factors), settlement computation from N-values.	4
Unit 4	PILE FOUNDATIONS Types of piles, axial capacity of single piles – dynamic and Static formula, skin friction and end bearing in sands and clays, axial capacity of pile groups, group efficiency, settlement of single piles and groups, negative skin friction, uplift capacity of pile, pile load tests, codal provisions, laterally loaded piles, under reamed piles.	6
Unit 5	CASSION AND WELL FOUNDATIONS Types of Caissons, scour depth, parts of the well foundation, design criterion, estimation of allowable bearing pressure, lateral stability of well foundation, well sinking, tilt and shift	5
Unit 6	SLOPE STABILITY Finite and infinite slopes, critical failure surface, factor of safety, Swedish circle method, friction circle method, bishop's method, stability number, Taylor's stability chart, stability analysis of earthen dams.	5
Unit 7	SOIL DYNAMICS AND MACHINE FOUNDATIONS Engineering problems involving soil dynamics, dynamic loading, theory of vibrations, types of machine foundations, design criteria for machine foundations, codal provisions.	6

Unit 8	GROUND IMPROVEMENT Mechanical soil stabilization, mixing additives, grouting, dynamic compaction, Pre-loading with vertical drain, stone column, vibro-flotation, introduction to geosynthetics, reinforced soil.	6
---------------	---	---

Course Outcomes :

By the end of the course, the students will be able to:

1	Design suitable foundations for different structures under varied field conditions.
2	Analyze the stability of natural and manmade slopes
3	Analyze and design the machine foundations.
4	Improve the ground behaviour as per requirement.

Text Book:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Analysis and Design of Sub-Structure,” Swami Saran, CRC Press.	2006
2.	“Foundation Engineering,” P.C. Varghese, PHI Learning Private Limited.	2013
3.	“Principles of Foundation Engineering,” B. M. Das, Cengage Learning, New Delhi.	2013
4.	“Soil Mechanics and Foundations,” Muni Budhu, John Wiley & Sons.	2010
5.	“Soil Dynamics & Machine Foundation,” Saran, S., Galgotia Publication.	2006

Reference Books:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Foundation Analysis and Design,” J.E. Bowles, McGraw-Hill. (1996)	1996
2.	“Pile Foundation Analysis and Design” Poulos H.G. and Davis, E. H., John Wiley.	1980
3.	“Geotechnical-Earthquake Engineering,” Kramer, S.L., Pearson Education – Indian Low-Price Edition.	2004
4.	“Reinforced Soil and its Engineering Applications,” Swami Saran, I.K International Pvt Ltd.	2006
5.	“Soil Dynamics,” Prakash, S., McGraw Hill Book Company. (1981)	1981

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	https://archive.nptel.ac.in/courses/105/105/105105176/NPTEL video course on Foundation Engineering	NPTEL
2.	https://archive.nptel.ac.in/courses/105/105/105105207/NPTEL video course on Advanced Foundation Engineering	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	2	3	2	-	-	1	1	1	3	3	-	2
CO2	3	3	3	3	3	3	2	-	-	1	1	1	3	3	-	3

CO3	3	3	3	3	2	2	1	-	-	1	1	1	3	3	-	2
CO4	3	3	3	3	3	3	3	-	-	1	1	1	3	3	-	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

DEPARTMENTAL ELECTIVE COURSES-I

Course Name	:	Concrete Technology	DEC
Course Code	:	CEE1005	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives :

At the end of this course, the student should be able to:

- Understand the structure of concrete.
- Design concrete mix.
- Understand the use of special concretes.
- Evaluate distressed structures and apply retrofitting techniques.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	CONCRETE Introduction to concrete ingredients like Cement, Aggregates, Chemical Admixtures, Role of Supplementary Cementitious Materials, concrete production, RMC, compaction and curing.	4
Unit 2	CONCRETE MICROSTRUCTURE Cement hydration products, microstructure properties of concrete, Interfacial transition zone (ITZ), particle packing, Introduction of analytical techniques to examine microstructure of concrete.	5
Unit 3	RHEOLOGY OF CONCRETE Introduction to rheology of concrete, workability of concrete. Yield stress and viscosity of concrete, factors affecting rheology of concrete, testing methods.	5
Unit 4	CONCRETE MIX DESIGN General principles, IS codal guidelines, mix design procedure for medium and high strength concrete using various admixtures.	5
Unit 5	PROPERTIES OF HARDENED CONCRETE Mechanical and durability properties of concrete and test methods, Creep, Shrinkage.	6
Unit 6	SPECIAL CONCRETES: Self compacting concrete, fiber reinforced concrete, geo-polymer concrete, 3D-printed concrete. Introduction to pre-cast technologies.	5
Unit 7	DETERIORATION AND EVALUATION OF DISTRESSED STRUCTURES Deterioration: Embedded Metal Corrosion, Disintegration Mechanisms, Moisture Effects, Thermal Effects, Structural Effects, Faulty Construction. Evaluation: Visual Investigation, Destructive Testing Systems, Non-Destructive Testing Techniques, Semi-Destructive Testing Techniques.	6
Unit 8	REPAIR AND RETROFITTING TECHNIQUES Selection of Repair Materials, Surface Preparation, Bonding repair Materials to Existing concrete, Placement Methods, Introduction to structural members strengthening techniques.	6

Course Outcomes:

At the end of this course, the student will be able to:

1	Design of different types of concretes.
2	Understand the behavior of conventional and special concretes.
3	Evaluate distressed structures and apply retrofitting techniques.

Text Book:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Concrete-Microstructure, Properties and Materials" P.K. Mehta and P.J.M. Monteiro, McGraw Hill.	2014
2.	"Concrete Repair and maintenance illustrated" Peter H. Emmons, R.S means company INC.	2002

Reference Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1.	"Concrete Technology" by A.M. Neville and J.J. Brooks.	2019
2.	"Testing of Concrete in Structures," S. Bungey, G. Lillard, and M.G. Grantham, Taylor and Francis.	2006
3.	"Concrete Technology - Theory and Practice," MS Shetty, S. Chand.	2006
4.	"Concrete Repair and Maintenance," PH Emmons, Galgotia Publication.	2012
5.	"Handbook on Non-destructive Testing of Concrete," VM Malhotra and NJ Carino, CRC Press.	2004
6.	Relevant IS codes (Latest Revision)	

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	(https://nptel.ac.in/courses/105106202)--- Maintenance and Repair of Concrete Structures, IIT Madras	NPTEL
2	(https://archive.nptel.ac.in/courses/105/105/105105213/)--- Retrofitting and rehabilitation of civil infrastructure, IIT Kharagpur	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	3	3	2	3	2	2	3	3	2	2	3
CO2	3	2	3	2	3	3	3	2	2	2	2	3	3	2	2	3
CO3	3	3	3	2	3	3	3	2	3	2	2	3	3	2	2	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Multimodal Urban Transportation System	DEC
Course Code	:	CEE1006	
Credits	:	4	
L – T – P	:	3 – 1 – 0	

Course Objectives:	
To make the students learn about the recent trends in mass transportation system and the analytical techniques of multimodal transit.	
- Identify the sustainability principles in transportation - Introduce the concept of Travel Demand Management (TDM) - Disseminate the techniques of urban public transit planning, operations and management - Imbibe the concepts of non-motorized urban transport - Demonstrate the applications in intelligent transportation systems (ITS)	

Total No. of Lectures – 42

Lecture wise break up		No. of Lectures
Unit 1	OVERVIEW OF URBAN TRANSPORTATION Urbanization and Transport, Key issues in urban transportation, Challenges in urban transportation, Travel demand modelling overview, Vehicular Level of Service (LOS) overview.	6
Unit 2	PUBLIC TRANSPORTATION Introduction to public transportation, Basic operating elements of public transportation, Bus Transportation, Rail transportation, Intermediate Public Transportation, Measuring performance of transit systems.	10
Unit 3	NON-MOTORIZED TRANSPORTATION (NMT) PLANNING Introduction to NMT Systems, Non-Motorized Transportation (NMT) Planning, Basic NMT Characteristics, Non-Motorized Transportation (NMT) Planning, Bicycle Facilities and Level of Service (BLOS), Non-Motorized Transportation (NMT) Planning, NMT Design Principles, Design of Pedestrian Infrastructure, Design of Cycling Infrastructure	10
Unit 4	URBAN TRANSPORT AND SUSTAINABILITY Introduction to Intelligent Transportation Systems (ITS), ITS components, applications and communication, Public Bicycle Sharing (PBS) System with ITS, Multimodal transportation (MMT) environment, Design of multimodal transfer facilities, Introduction to Pedestrian Road Safety and associated Risk Factors, Sustainable strategies for Urban Transportation, Environmental Concerns of Urban Transport.	12
Unit 5	TRANSIT ORIENTED DEVELOPMENT (TOD) Introduction to TOD, Objectives of TOD, Design Principles of TOD, Benefits of TOD	4

Course Outcomes (COs):	
By the end of this course, the student will be able to:	
1	Apply the analytical techniques of multimodal transportation systems in the field.
2	To develop a basic understanding of public transportation system.
3	To understand the Management Issues in Multimodal Transport Planning
4	Understanding of Transportation Planning and Transit Oriented Development.

Textbooks:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Sustainable Mass Transit: Challenges and Opportunities in Urban Public Transportation” Thomas Abdallah, Elsevier.	2017
2.	“Transport Engineering and Planning” Papacostas C.S., Pearson Education India.	2015
Reference Book:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Urban Mass Transit – The Life Story of a Technolog”, Robert Post, Johns Hopkins University Press.	2010
2.	“Fundamentals of Intelligent Transportation Systems Planning”, Mashrur A. Chowdhury, Adel Wadid Sadek, Artech House, Inc. Boston.	2003
3.	“Sustainable Transport: Planning for Walking and Cycling in Urban Environments”, Rodney Tolley, Woodhead Publishing Ltd.	2003

Equivalent MOOC courses

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105/105/105105204/ “Introduction to Multimodal urban transportation systems”.	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	2	3	-	-	-	-	1	2	2	1	1	1	-	1	3
CO2	1	2	3	1	2	1	2	1	3	3	2	3	2	3	1	3
CO3	1	2	3	1	2	3	2	1	2	3	1	3	2	3	2	3
CO4	1	-	3	-	1	-	2	-	2	3	2	1	1	3	1	3

1 – Slight (low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” - indicates there is no Correlation

Course Name	:	Construction Project Management and Economics	DEC
Course Code	:	CEE1007	
Credits	:	4	
L – T – P	:	3 – 1 – 0	

Course Objectives:	
At the end of the course, the student should be able to learn the basic knowledge of Construction Project Management and Economics-	
• To understanding of Construction Management and Bar/ milestone charts • To analyze the PERT and CPM, CPM (Cost Model), • To explore construction Equipments and selection of construction equipment • To apply the concept of CPM updating and Resource allocation	

Total No. of Lectures – 42

Lecture wise breakup		No of Lectures
Unit 1	INTRODUCTION Project Management, Project Planning, Scheduling, Controlling, Method of planning and Programming, bar Charts and Milestone charts and Network Methods/techniques	4
Unit 2	PERT (PROGRAMME EVALUATION AND REVIEW TECHNIQUE) Definitions, Time estimates, frequency Distribution, mean ,variance and standard deviation, probability distribution, beta distribution, expected time, time computation, network analysis, critical path	6
Unit 3	CPM (CRITICAL PATH NETWORK) CPM Networks, Earliest event time, Latest allowable occurrence time. Earliest start time, earliest finish time, latest start time, latest finish time, float critical activity, critical path	6
Unit 4	CPM (COST MODEL) Project cost, direct project cost, indirect project cost, Total Project cost and optimum duration, slope of direct cost curve, steps in time cost optimization.	6
Unit 5	CONSTRUCTION EQUIPMENT: Classification of major equipment, Earth excavating equipment, earth cutting and hauling equipment, earth compacting and grading equipment, concreting plant and equipment.	6
Unit 6	SELECTION OF CONSTRUCTION EQUIPMENT: Task considerations, data required for updating, equipment engg. considerations, equipment acquisition options.	6
Unit 7	CPM (UPDATING) Updating process, data required for updating, steps in process of updating, when to update, PPT/BOT technique and its variance.	4
Unit 8	RESOURCE ALLOCATION Resource usage profiles: Histogram, Resources smoothing, Resource levelling, risk associated in construction project management.	4

Course Outcomes (COs):

By the end of this course, the student will be able to:

1	Proficient enough to apply the concepts of the construction project management.
2	Able to understand the concepts of PERT and CPM.
3	To obtain the different cost of the projects with total cost of the project
4	To understand the Selection of construction equipments, CPM updating and Resource

Textbooks:

Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Project planning and control with CPM/PERT”, Dr. B.C. Punmia, Laxmi Publication New Delhi.	2016

Reference Book:

Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Construction project management”, K.K Chitkara, Tata McGraw-Hills.	2015
2.	“Construction equipments”, by Mahesh Verma.	2016
3.	“Construction project management and an integrated approach”, Peter Femings, Yes Dec Publishing pvt. Ltd. Chennai.	2017

Equivalent MOOC courses

Sr. No	Course link	Offered By
1	Construction project management and Economics: https://archive.nptel.ac.in/courses/105/104/105104161/	NPTEL
2	Construction project management and Economics: https://ocw.mit.edu/courses/1-040-project-management-spring-2009/	MIT Open Course Ware http://ocw.mit.edu

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	1	1	3	3	2	3	3	3	3	2
CO2	3	3	3	3	3	1	3	3	3	2	3	3	3	3	3
CO3	3	2	1	2	3	3	2	1	3	3	3	3	3	2	3
CO4	2	3	2	3	1	2	3	1	3	3	3	3	3	3	2

1 - Slight(low) Correlation

2 - Moderate(Medium) Correlation

3 - Substantial(High) Correlation

“-” - indicates there is no Correlation

Course Name	:	Solid Waste Management	DEC
Course Code	:	CEE1008	
Credits	:	4	
L - T - P	:	3-1-0	

Course Objectives:	
The student should be able to	
• identify various aspects of solid waste management • characterize various types of wastes • plan and design processing/ treatment practices for different types of wastes • plan and design disposal of wastes	

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	SOLID WASTE Types of solid wastes, Quantity, characteristics, and composition of municipal solid wastes.	8
Unit 2	SOLID WASTE DISPOSAL Collection, processing, transport and disposal	8
Unit 3	ENGINEERED SYSTEMS FOR RESOURCES, ENERGY & MATERIAL RECOVERY Recovery from Biodegradable wastes, Recovery of Energy from Conversion Products, Material recovery, Recycling and upcycling	12
Unit 4	LANDFILLS Site consideration, operation and closure, Leachate: formation, composition, and control	6
Unit 5	LIFE CYCLE ASSESSMENT Concepts and Methodology, Sustainability Assessment of SWM- Circular Economy Analysis: concepts and design	4
Unit 6	LEGISLATION Relevant rules for the management of various types of wastes	4

Course Outcomes:	
By the end of this course, the student will be able to:	
1	Describe the functional elements of solid waste management
2	Identify methods for solid waste treatment/ processing
3	Plan and design disposal of wastes
4	Landfill design, closure and handling of leachate

Text Book:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Environmental Engineering”, H.S. Peavy, D.R. Rowe, G. Tchobanoglous, McGraw Hill, NY.	1985
2.	“Integrated Solid Waste Management”, G. Tchobanoglous, H. Theisen, S. A. Vigil, Mc Graw Hill International Ed.	1993

Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Manual on Solid Waste Management”, CPHEEO, MoUD & GIZ.	2016
2.	“Gazette notifications for various waste management rules”.	Latest
3.	“Textbook of Solid Wastes Management”, I.H. Khan & N. Ahsan, CBS Publishers.	2012

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	https://nptel.ac.in/courses/105/106/105106056/ - Solid and Hazardous Waste Management	NPTEL
2.	https://nptel.ac.in/courses/105/103/105103205/ - Municipal Solid Waste Management	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	1	3	3	3	2	2	2	1	3	3	1	2	3
CO2	3	3	3	1	3	3	3	2	2	2	1	3	3	1	2	3
CO3	3	3	3	1	3	3	3	2	2	2	1	3	3	1	2	3
CO4	3	3	3	1	2	3	3	2	2	2	1	3	3	1	2	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Engineering Geology	DEC
Course Code	:	CEE1009	
Credits	:	4	
L T P	:	3 0 2	

Course Objectives:			
- To analyze past and prevailing geological process operating on the Earth. - To explore processes of formation of minerals and rocks and their utility to Civil Engineers. - To develop knowledge as to how geological structures affect engineering projects. - To apply the importance of engineering geology in Civil Engineering projects.			

Total No. of Lectures- 42

Lecture wise breakup			No. of Lectures
Unit 1	GENERAL GEOLOGY Physiographic features of India, Branches of Geology and the importance of Engineering Geology; Weathering, erosion, transportation and deposition of Earth's material; Landforms associated with processes of erosion, transportation, and deposition in deserts, glaciated regions and river valleys.		10
Unit 2	STRATIGRAPHY Principals of Stratigraphy; Standard geological time scale and introduction to the concept of time in geological studies; Introduction to geochronological methods, and economic importance of Archean system, Cuddapah system & Vindhyan system.		6
Unit 3	STRUCTURAL GEOLOGY Stratification, dip, strike, and unconformities; Folds their classification, causes and Engineering consideration; Faults their classification, causes and Engineering consideration.		5
Unit 4	MINERALOGY AND PETROLOGY Crystals, crystal structures and Optical properties of Minerals. Physical and Chemical properties of minerals. Origin and Type of Rocks; Origin and Types of Soils. Stress- Strain behaviors of Soil and Rocks. Index testing of Soil and Rocks.		10
Unit 5	ENGINEERING GEOLOGY Characteristics of good building stones, their uses, and distribution in India. Geological Considerations in Tunnels, Highway, Foundations, Dams, and Reservoirs. Plate Tectonics and Earthquakes, Earthquake Hazard assessment, Geological Hazards: Seismicity and Volcanism, Seismic zones in India and Aseismic constructions, Landslides, Land Subsidence and Mass moments. Groundwater preliminaries and groundwater flow, Groundwater related engineering issues and Groundwater overutilization.		11

List of Experiments:			No. of Turns
1	To determine the physical properties of minerals		03
2	Examination of typical rock-forming and ore minerals		02
3	Examination of rocks used for engineering purposes		03
4	Study of structural models		02

5	Simple dip and strike problems	01
6	Map work: finding out out-crop, drawing cross-section and understand the geological history	02
7	Map dealing with the location of dam and tunnel sites	01

Course Outcomes:		
1	Analyze the basic concepts of geological processes and their importance in civil engineering.	
2	Develop skill of identification of rocks and minerals.	
3	Describes the importance of geological structures in civil engineering projects.	
4	To understand the geological risks connected to building different kinds of structures.	

Textbooks:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication/ Reprint
1.	“Engineering Geology,” Parbin Singh, N.D. Kataria.	1996
2.	“Geological Structures and Maps: a Practical Guide 2020 Edition,” Richard J. Lisle	2010
3.	“Structural Geology of Rocks and Region,” George H. Davis, Stephen J. Reynolds, Charles F. Kluth	2011
4.	“The manual of Mineral Science,” Cornelis Klein and Barbara Dutrow	2012
5.	“Geology of India and Burma,” 6 th edition, Krishnan, M.S.	2012
6.	“Holmes’ principles of physical geology,” 3 rd edition, Holmes, A.	2016
Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication/ Reprint
1	“Engineering Geology,” Hries and Watson, John Wiley.	1953
2.	“Geology of India” D.N.Wadia, Tata McGraw Hill, India.	1973
3.	“Geology for Engineers,” D. S. Arora, Mahindra, Chd., Sons, New York	1984
4.	“Petrology”, Tyrrell, B.L., N.D.	1994

Equivalent MOOC courses

Sr. No.	Course Links	Offered by
1.	Engineering geology https://nptel.ac.in/courses/105/105/105105106	NPTEL
2.	Engineering geology https://nptel.ac.in/courses/105/104/105104191	NPTEL
3.	Engineering geology https://nptel.ac.in/courses/105/104/105104147	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	1	2	2	1	1	1	1	1	2	1	3	1	1	3
CO2	1	2	1	2	3	1	1	1	1	1	1	1	3	1	1	3
CO3	3	3	2	3	3	1	1	1	2	1	1	3	3	3	1	3
CO4	3	3	3	3	1	1	2	1	3	1	1	3	3	3	2	3

- 1 – Slight (Low) Correlation
- 2 – Moderate (Medium) Correlation
- 3 – Substantial (High) Correlation
- “-” indicates there is no correlation.

Course Name	:	Ground Water Hydraulics	DEC
Course Code	:	CEE1010	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The students shall be able to

- Understand the basic concept of ground water hydraulics.
- Analyse ground water flow with governing ground water flow equations.
- Apply analytical methods to identify sites for artificial recharge.
- Develop methodologies to resolve issues related to ground water.

Total No. of Lectures- 42

Lecture-wise breakup		No. of Lectures
Unit 1	GROUND WATER: INTRODUCTION AND ASSESSMENT Groundwater vis-à-vis surface water, Role of groundwater in hydrological cycle, Hydrologic budget, Definition of groundwater Ground water resources status of India - Development and potential.	2
Unit 2	OCCURRENCE OF GROUNDWATER Vertical distribution of subsurface water, Saturated formation, Types of aquifers, Aquifer properties – porosity, void ratio, permeability, specific yield, specific retention, and storage coefficient, Measurement of water content and available water, Fence diagram.	3
Unit 3	GROUND WATER MOVEMENT Darcy's law and its validity, Hydraulic head and conductivity, Intrinsic permeability and transmissivity, Laboratory and field measurements of hydraulic conductivity, Homogeneous and isotropic systems, Groundwater flow rates, Groundwater flow directions - Flow nets, Governing equations of groundwater flow, Confined groundwater flow between two bodies, Steady unidirectional flow – confined and unconfined aquifers, Dupit's flow with and without recharge, Base flow to stream.	7
Unit 4	WELL HYDRAULICS Steady radial flow into a well – confined and unconfined aquifers, Unsteady radial flow into a well– confined and unconfined aquifers, Pumping and recuperation tests, Multiple well systems, Well losses, Specific capacity and safe yield, Hydraulics of open wells, Design of wells, Open wells versus borewells, Design problem of water well, Design of collector wells, Well construction.	12
Unit 5	ARTIFICIAL RECHARGE OF GROUND WATER Concept of artificial recharge, Recharge methods – merits, demerits and selection criteria, Artificial recharge strategy and identification of potential areas, Source water availability and assessment, Reuse and recycling wastewater, Rainwater harvesting, Innovative approaches for artificial recharge.	6
Unit 6	SALINE WATER INTRUSION IN AQUIFERS Occurrence of saline water intrusion, Ghyben–Herzberg relation between fresh and saline waters, Shape of the freshwater–saltwater interface, Effect of wells and tides on seawater intrusion, Upconing of saline water, Seawater intrusion in India, Control of saltwater intrusion.	6

7	GROUNDWATER MANAGEMENT Ground water basin management, Data collection for ground water management, Conjunctive use of surface and ground water, Management of coastal aquifers, Ground water management techniques.	6
---	---	---

Course Outcomes (COs):

At the end of the course the students will be able to:

1	Understand the important flow characteristics and behaviours of aquifer.
2	Analyze the ground water flow problems and design of water wells.
3	Identify sites for artificial recharge of groundwater and determine the consequences of artificial recharge.
4	Recognize the issues related to ground water and their remedial solutions.

Text Book:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Groundwater hydrology”, Todd, D. K., and Mays, L. W., John Wiley and Sons.	2004
2	“Hydraulics of groundwater”, Bear, J. Courier Corporation.	2012

Reference Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Groundwater hydrology”, Chahar, B. R., McGraw Hill.	2015
2	“Ground water”, Raghunath, H. M., New Age International Publishers.	2007
3	“Engineering hydrology”, Subramanya, K., McGraw Hill.	2014
4	“Groundwater hydrology: conceptual and computational models”, Rushton, K. R., John Wiley and Sons.	2004

Equivalent MOOC courses

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105/103/105103026/ - Web course on ‘Ground Water Hydrology’.	NPTEL
2	https://nptel.ac.in/courses/105/105/105105042/ -Video Course on ‘Ground Water Hydrology’.	NPTEL
3	https://ocw.mit.edu/courses/civil-and-environmental-engineering/1-72-groundwater-hydrology-fall-2005/ - Open Courseware Web Course on ‘Ground Water Hydrology’	MIT

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	3	1	2	3	1	2	3	1	2	3	3	1	3
CO2	3	3	3	3	3	1	2	1	3	3	1	3	3	3	1	3
CO3	3	3	3	3	3	3	3	1	3	3	3	3	3	3	1	3
CO4	3	3	3	3	3	1	3	1	2	3	3	3	3	3	1	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

DEPARTMENTAL ELECTIVE COURSES-II

Course Name	:	Design of Steel Structures- II	DEC
Course Code	:	CEE1011	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

At the end of this course, the student should be able to:

- Design Steel plate girder and gantry girder.
- Design trussed bridge.
- Design various elements of industrial buildings.
- Familiarize with plastic design and use of cold formed members.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	PLATE GIRDER Components of plate girder, Economical depth, post buckling behaviour of web, design of plate girder including design of web, flanges, stiffeners, web splices, flange splices and connections.	8
Unit 2	GANTRY GIRDER Introduction, Loads, position of moving load for maximum effects, Profile of gantry girder section, Limitation of vertical deflections, Design of gantry girder.	5
Unit 3	TRUSSED BRIDGE Introduction to trussed girder bridges, various loads on the bridges, assumptions for design of trussed bridge, design of compression and tension members, calculation of wind load on trussed bridge, top and bottom lateral bracings, portal and sway bracings, bearings.	15
Unit 4	INDUSTRIAL BUILDINGS Design of components of Industrial building, different types of bracings, stepped columns and column bases.	6
Unit 5	PLASTIC ANALYSIS AND DESIGN Design of continuous beams and design concept of portal frame using plastic design approach.	5
Unit 6	LIGHT GAUGE STEEL SECTIONS Introduction and brief description of various type of cold formed light gauge steel sections, local buckling	3

Course Outcomes:

At the end of this course, the student will be able to:

1	Design plate girder and gantry girder.
2	Design various components of the trussed bridge.
3	Design various elements in industrial buildings.
4	Carry out plastic analysis of beams and familiarize with the usage of cold formed sections

Text Book:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/
----------------	---	-----------------------------

		Reprint
1	“Design of Steel Structures”, Arya & Awadhesh Kumar, Nem Chand and Bros., Roorkee.	2022
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Limit state design of Steel Structures”, S.K. Duggal, Tata McGraw Hill.	2017
2	“Steel Structures: Design and practice”, N. Subramanian, Oxford Publishing Press.	2010
3	“Design of Steel of Structures”, S.S. Bhavikatti, I.K. International Publishing House Pvt. Ltd	2012
4	“Design of Steel Structures”, Volume 2 Ramchandra S., Virendra Gehlot, Scientific Publishers, New Delhi,	2018
5	“Comprehensive Design of Steel Structures”, Dr. B. C. Punmia, Ashok Kumar Jain. Laxmi Publications, New Delhi.	2015
6	Relevant IS codes (Latest Revision)	

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	(https://archive.nptel.ac.in/courses/105/106/105106113/)--- Design of Steel Structures II , IIT Madras	NPTEL
2	(https://archive.nptel.ac.in/courses/105/105/105105162/)--- Design of steel structures , IIT Kharagpur	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	1	2	1	2	2	3	2	3	3	3	3	3
CO2	3	3	3	3	1	2	1	2	2	2	2	3	3	3	3	3
CO3	3	3	3	3	1	2	1	2	2	2	2	3	3	3	3	3
CO4	3	3	3	3	1	2	1	2	2	3	2	3	3	3	3	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Road Safety	DEC/MSC
Course Code	:	CEE1012/CEM2003	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

To make the students identify and design the various components of road safety planning, design, maintenance and executing safety policies.

- To understanding of road safety in planning and geometric design,
- To analyze the role of urban infrastructure design in safety
- To analyze the fundamentals of traffic engineering
- To analyze the accident data and risk management data, traffic management systems for road safety

Total No. of Lectures – 42

Lecture-wise break-up		No. of Lectures
Unit 1	Unit I: Road Safety in Planning and Geometric Design: Vehicle and Human Characteristics, Road Design and Road Equipment's, Redesigning Junctions, Cross Section Improvements, Reconstruction and Rehabilitation of Roads, Road Maintenance, Traffic Control, Vehicle Design and Protective Devices, Post Accident Care	8
Unit 2	Unit II: Role of Urban infrastructure design in safety: Geometric Design of Roads; Design of Horizontal and Vertical Elements, Junctions, At Grade and Grade Separated Intersections, Road Safety in Urban Transport, Sustainable Modes and their Safety.	10
Unit 3	Unit III: Fundamentals of Traffic Engineering - Basic Characteristics of Motor-Vehicle Traffic, Highway Capacity, Applications of Traffic Control Devices, Traffic Design of Parking Facilities, Traffic Engineering Studies; Statistical Methods in Traffic Safety Analysis – Regression Methods, Poisson Distribution, Chi- Squared Distribution, Statistical Comparisons.	8
Unit 4	Unit IV: Accident Investigations and Risk Management, Collection and Analysis of Accident Data, Condition and Collision Diagram, Causes and Remedies, Traffic Management Measures and Their Influence on Accident Prevention, Assessment of Road Safety, Methods to Identify and Prioritize Hazardous Locations and Elements, Determine Possible Causes of Crashes, Crash Reduction Capabilities and Countermeasures, Effectiveness of Safety Design Features, Accident Reconstruction	8
Unit 5	Unit V: Traffic Management Systems for Safety, Road Safety Audits and Tools for Safety Management Systems, Road Safety Audit Process, Approach to Safety, Road Safety Improvement Strategies, ITS and Safety.	8

Course Outcomes:

At the end of the course the students will be able to:

1	Identify the components of road safety.
2	Plan, design and execute of road safety.
3	Choose the suitable statistical analysis for road safety

4	Realize the importance of road safety data studies and traffic management
---	---

Textbooks:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Traffic Engineering and Transportation Planning” L.R. Kadiyali, Khanna Publishers	2013
2.	Transportation Engineering – An Introduction, C.Jotin khisty, B. Kent Lall	2002
Reference Book:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Handbook of Road Safety measures”, Rune Elvik, Alena Hoyer, Truls Vaa, Michael Sorenson, second Edition, emerald	2004
2.	“Fundamentals of Traffic Engineering”, Richardo G Sigua	2008
3.	“Road Safety”, NCHRP	Latest

Equivalent MOOC courses

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105101008/582_Accident/point20/point.html Road Safety Audit	NPTEL
2	https://nptel.ac.in/courses/105105215 Traffic Engineering	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	-	-	-	-	-	3	-	-	3	-	1	2	3
CO2	1	2	3	3	3	2	3	3	-	-	3	-	1	2	3
CO3	1	2	3	3	3	2	3	3	-	-	3	-	1	2	3
CO4	1	2	3	3	3	2	3	3	-	-	3	-	1	-	3

- 1 - Slight(low) Correlation
2 – Moderate (Medium) Correlation
3 – Substantial (High) Correlation
“-” - indicates there is no Correlation

Course Name	:	Hydro Power Engineering	DEC
Course Code	:	CEE1013	
Credits	:	4	
L – T – P	:	3 – 1 – 0	

Course Objectives (COs):

The students should be able-

- To understand the different forms of energy, types of power plants, and hydropower development in India.
- To analyze the detailed knowledge of hydropower components like Intake structures, Trash racks, Penstocks, etc.
- To apply the knowledge of hydropower engineering in solving daily life problems.
- To understand and analyze the concepts of Water Hammer and surge in Hydropower development.

Total No. of Lectures – 42

Lecture-wise breakup		No. of Lectures
Unit 1	INTRODUCTION Sources of power, Role of hydropower in power system, Necessity and importance of harnessing small hydropower, Hydro power potential in India.	3
Unit 2	ESTIMATION OF WATERPOWER POTENTIAL Flow duration and power duration curves, Load curve, Load factor, Capacity factor, Utilization factor, Diversity factor, Firm and secondary power, and Prediction of load.	6
Unit 3	TYPES OF HYDROPOWER PLANTS Elements of Hydropower, Classification of hydropower plants, Run-of-river plants, General arrangement of run-of-river plants, Valley dam plants, Diversion canal plants, High head diversion plants, Storage and pond age, Pumped storage power plants, Base load and peak load plants in a power grid.	6
Unit 4	PENSTOCKS General classification, Design criteria, Economic diameter, Losses, Valves, Bends and Manifolds.	5
Unit 5	TRASH RACKS Types, Losses, Design, Stability.	4
Unit 6	INTAKES Types, Losses, Air entrainment, Anti-vortex device, Air vent, Power channels, Fore bay, Tunnel.	4
Unit 7	TURBINES Introduction, Types of turbines, Hydraulics of turbines, Velocity triangles, Draft tubes, Cavitation in turbines, Characteristics of turbines.	7
Unit 8	WATER HAMMER AND SURGES Introduction, Water hammer, Transients caused by turbine, Load acceptance, and rejection, Resonance in penstocks, Surge tanks, Channel surges.	7

Course Outcomes (COs):	
At the end of the course, the students will be able to:	
1	Solve the hydropower-related problems arising in Water Resources projects.
2	Estimate the waterpower potential in perennial rivers and should be able to design important hydraulic components of hydropower.
3	Estimate the hydropower potential of a catchment area
4	Analyze and solve the water hammer and surge phenomenon in hydropower generation.

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Water Power Engineering”, Dandekar, M. M., and Sharma, K. N., Vikas Publishing House.	2016
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“A Textbook of Water Power Engineering”, Sharma, R. K., and Sharma, T. K., S. Chand Publishing.	2003
2	“Irrigation and Water Power Engineering”, Punmia, B. C., Lal, P. B. B., Jain, A. K., and Jain, A. K., Laxmi Publications, Ltd.	2009
3	“Hydro Power Structures”, Varshney, R.S., Nem Chand and Bros.	2001
4	“Applied Hydraulic Transients”, Chaudhry, M. H., Van Nostrand Reinhold Company.	2022

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	1	1	3	1	3	3	3	2	3	3	1	3
CO2	3	3	3	3	3	1	3	1	3	3	3	3	3	3	1	3
CO3	3	3	3	3	3	1	3	1	3	3	3	3	3	3	1	3
CO4	3	3	3	3	3	1	3	1	3	3	3	3	3	3	1	3

- 1 – Slight (Low) Correlation
 2 – Moderate (Medium) Correlation
 3 – Substantial (High) Correlation
 “-” indicates there is no correlation.

Course Name	:	Unit Processes in Water and Wastewater Treatment	DEC
Course Code	:	CEE1014	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The student should be able to

- To understand the principles of various physio-chemical and biological processes
- Design the components for water supply
- Design the components wastewater treatment plants.
- Develop a knowledge of sludge handling

Total No. of Lectures – 42

Lecture-wise break-up		No. of Lectures
Unit 1	WATER TREATMENT Technologies for the removal of suspended, colloidal and dissolved solids and for disinfection, Design of aeration, Coagulation- flocculation- settling, Filtration, Membrane processes, Ion exchange, Adsorption, Chlorination, etc., Selection of suitable water treatment technology for given water.	14
Unit 2	WASTEWATER CHARACTERIZATION Quantification of sewage, Types of sewerage systems, Characterization of sewage, Treated sewage quality standards.	4
Unit 3	WASTEWATER TREATMENT Principles and Design of Preliminary treatment, Primary treatment, Secondary treatment technologies, Attached and suspended growth biological processes, Tertiary treatment, Decision making for suitable wastewater treatment technology, Concept of a common effluent treatment plant.	17
Unit 4	DISPOSAL OF TREATED EFFLUENT Disposal on land, Quality standards for irrigation water, Disposal in water bodies, Self-purification of river, Streeter-Phelps equation.	4
Unit 5	SLUDGE HANDLING Sludge Thickening, Sludge Stabilization (Aerobic and Anaerobic), Dewatering, Sludge Drying beds, Disposal of dried sludge.	3

Course Outcomes (COs):

At the end of the course the students will be able to:

1	Characterize ground and surface water quality.
2	Identify best practicable environmental option (BPEO) and design a water treatment plant.
3	Characterize wastewater and identify BPEO for wastewater treatment.
4	Identify BPEO for disposal of effluent and sludge.
5	Design a wastewater treatment plant.

Text Books:		
S.no.	Name of Books/Author/Publisher	Year of Publication s/Reprint
1	“Environmental Engineering”, Howard S. Peavy, D.R. Rowe, G. Tchobanglous, McGrawHill Book Co., New Delhi	2017
2	“Environmental Engineering (Volume 1 & 2) Water Supply Engineering”, Santosh Kumar Garg Khanna Publishers, New Delhi	2010
Reference Books :		
S.no.	Name of Books/Author/Publisher	Year of Publication s/Reprint
1	“Manual on sewerage and sewage treatment” Ministry of Urban Development, New Delhi.	2013
2	“Manual on Water Supply Engineering”, Expert Committee Ministry of urban Development, New Delhi	1999
3	“Environmental Engineering Volume 1 & 2” Standard Book House, New Delhi	2004
4	IS: 10500 – 2012. Drinking Water Specifications. Bureau of Indian Standards, New Delhi.	2012

Equivalent MOOCs Courses

Sr. No.		Offered by
1	https://nptel.ac.in/courses/105/107/105107207/ - NPTEL video course on ‘Water and Wastewater treatment’.	NPTEL
2	https://www.classcentral.com/course/swayam-wastewater-treatment-and-recycling-14353 or https://nptel.ac.in/noc/courses/noc19/SEM2/noc19-ce32/ Web course on ‘Wastewater treatment and recycling’.	NPTEL
3	https://sswm.info/sites/default/files/reference_attachments/UN%20HABITAT%202008%20Constructed%20Wetlands%20Manual.pdf Constructed Wetland Manual	NPTEL
4	https://nptel.ac.in/noc/courses/noc20/SEM1/noc20-ce23/ - NPTEL video course on ‘Water Supply Engineering’	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	-	3	2	-	2	1	1	-	2	3	2	2	1	2
CO2	3	3	2	1	2	-	1	2	1	1	-	3	3	1	1	2
CO3	3	2	3	2	2	1	1	1	-	-	2	2	3	1	2	1
CO4	3	3	2	2	3	2	3	1	2	1	2	1	3	2	1	1

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation

Course Name	:	Disaster Management	DEC
Course Code	:	CEE1015	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

- To analyze the causes and effects of various natural and man-made disasters.
- To explore hazard and risk assessment.
- To enable to plan various mitigation measures and capacity building.
- To develop infrastructure for disaster management.
- To apply applications of Remote Sensing in disaster mitigation.

Total No. of Lectures- 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION TO DISASTER MANAGEMENT Natural Disasters, Man-made hazards, causes, consequences, earthquakes, Floods, Drought, Coastal Hazards, Landslides, rockslides and Forest Fires, Tsunamis, Oil spillage, Gas leakage, etc.	5
Unit 2	DISASTER MITIGATION AND PREPAREDNESS Mitigation measures for natural disasters- Earthquakes, Tsunamis, Cyclones, Floods, Landslides, etc.; Mitigation measures for various industrial hazards/disasters; Preparedness for natural disasters.	7
Unit 3	HAZARD AND RISK ASSESSMENT Assessment of capacity, vulnerability and risk; vulnerability and risk mapping; stages in disaster recovery and associated problems.	5
Unit 4	EMERGENCY MANAGEMENT SYSTEMS Emergency medical and essential public health services, response and recovery operations; Reconstruction and rehabilitation.	5
Unit 5	CAPACITY BUILDING Disaster management approach to inculcate new skills and sharpen existing skills of government officials, voluntary activists, development of professional and elected representative for effective disaster management, an overview of disaster management in India.	8
Unit 6	INTEGRATION OF PUBLIC POLICY Planning and design of infrastructure for disaster management; Community based approach in disaster management; Methods for effective dissemination of information; Ecological and Sustainable development models for disaster management.	6
Unit 7	ROLE OF REMOTE SENSING IN DISASTER MITIGATION Importance of Remote Sensing in disaster mitigation, Case studies	6

Course Outcomes:

1	To annotate the causes and effects of various disasters.
2	To plan risk management/disaster mitigation techniques.
3	To evaluate the mitigation measures (Case study).
4	To demonstrate role of Remote Sensing in Disaster mitigation.

Textbooks:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication / Reprint
1.	“Elements of Earthquake Engineering,” Jai Krishna, Chandrasekharan and B. Chandra, South Asian Publishers, New Delhi.	2000
2.	“Disaster Management,” R.B. Singh (Ed), Rawat Publications.	2006
3.	“Disaster Management –Future Challenges & Opportunities,” Jagbir Singh, I.K. International Publishing House.	2007
Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication / Reprint
1.	“Natural Hazards in the Urban Habitat,” Iyengar, C.B.R.I., Tata McGraw Hill. Publications.	1997
2.	“Natural Disaster Management,” Jon Ingleton (Ed), Tudor Rose, Leicester.	1999
3.	“Earthquake Resistant Design of Structures,” Pankaj Agarwal and Manish Shrikhande, Prentice Hall of India Pvt. Ltd.	2006

Equivalent MOOCs courses:		
Sr. No.	Course Links	Offered by
1.	Natural Hazards https://onlinecourses.nptel.ac.in/noc20_ce07/preview	NPTEL
2.	Disaster Management https://archive.nptel.ac.in/courses/105/104/105104183/	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	1	2	3	2	1	1	1	1	1	1	1	1	1	3
CO2	2	2	1	2	2	2	3	1	2	2	2	3	3	2	2	2
CO3	1	2	3	1	2	1	3	1	2	3	3	2	1	1	2	1
CO4	3	3	3	3	1	1	2	1	3	1	1	3	3	3	3	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

DEPARTMENTAL ELECTIVE COURSES-III

Course Name	:	Seismic Design of Structures	DEC
Course Code	:	CEE1001	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

At the end of this course, the student should be able to:

- Understand the basic concepts of structural dynamics.
- To analysis and design RC structures for seismic forces as per IS Codal guidelines.
- Perform ductile detailing of RC and Masonry Buildings.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION Introduction to seismology. Basic terminology used.	4
Unit 2	THEORY OF VIBRATIONS Analysis of SDOF for undamped free vibration, damped free vibration, Forced vibration, transient vibration, Introduction to response spectrum.	10
Unit 3	DUCTILITY PROVISIONS IN EARTHQUAKE RESISTANT DESIGN OF RC BUILDINGS Introduction, Importance of Ductility, Requirements for Ductility, Factors affecting Ductility, Ductile detailing considerations as per IS code.	10
Unit 4	SEISMIC ANALYSIS OF RC BUILDINGS IS Codal requirements for seismic resistant design, calculation of seismic load, load combinations, permissible stresses as per IS codal provisions using equivalent static method and dynamic analysis method. Lateral distribution of base shear, design of flexural and compression member. Introduction to design of shear wall.	10
Unit 5	DUCTILITY PROVISIONS FOR MASONRY BUILDINGS Categories of masonry buildings, seismic behaviour of unreinforced and reinforced masonry walls, Box action and bands, behaviour of infill walls, improving seismic resistance of masonry buildings as per IS Codal Provisions.	8

Course Outcomes:

At the end of this course, the student will be able to:

1	Understand the basic concepts of dynamic analysis of structures.
2	Design and detail earthquake resistant RC structural elements.
3	Incorporate ductile detailing for masonry building.

Text Book:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
----------------	---	-------------------------------------

1	"Dynamics of Structures," A.K. Chopra, Pearson Education.	2012
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	"Structural Dynamics (Theory & Computation)," Mario Paz.	2014
2	"Masonry and Timber Structures including Earthquake Resistant Design," A.S. Arya, Nem Chand & Bros.	2009
3	"Earthquake Resistant Design of Masonry Buildings," Miha Tomazevic, Imperial College Press.	2011
4	"Earthquake Resistant Design of Structures," S.K. Duggal, Oxford.	2017
	Relevant IS codes (Latest Revision)	

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	(https://onlinecourses.nptel.ac.in)- NPTEL web course on ‘Seismic Analysis of structures’. •	NPTEL
2	(https://onlinecourses.nptel.ac.in)- NPTEL web course on ‘Structural Dynamics	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	3	2	2	2	3	2	2	3	3	3	2	3
CO2	3	3	3	3	3	2	2	2	2	1	2	3	3	3	2	3
CO3	3	3	3	3	3	2	2	2	3	2	1	3	3	3	2	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Traffic Engineering	DEC
Course Code	:	CEE1016	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

- To make the students learn human and vehicular characteristics affecting transportation system, , brief knowledge of road signs, road markings and traffic signalling system, and basic principles of traffic regulation and management.
- To carryout various types of traffic studies, collection of data, and report them in required formats.
- To understand the various statistical concepts and utilize them for application in traffic engineering.
- To introduce the brief knowledge of road signs, road markings and traffic signaling system, and basic principles of traffic regulation and management
- To know the different computer applications for traffic simulation

Total No. of Lectures – 42

Lecture-wise break-up		No. of Lectures
Unit 1	TRAFFIC ENGINEERING FUNCTIONS AND TRAFFIC CHARACTERISTICS Functions of traffic engineering, Organization of trafficEngineering Department, Importance of Traffic engineering under Indian conditions. Road user Characteristics, “PIEV” theory, Vehicular characteristics, IRC Specifications on vehicular dimensions and weights.	4
Unit 2	TRAFFIC STUDIES Urban Road Classification, Traffic Volume Studies, Speed Studies, Origin- Destination Studies, Traffic Capacity and Level of service, Parking Studies, Accident Studies	9
Unit 3	STATISTICAL METHODS FOR TRAFFIC ENGINEERING Collection and presentation of Statistical Traffic Data, Mean, Standard Deviation and Variance, Poisson and Binomial Distribution, Normal Distribution, Sampling Theory and significance testing, Linear Regression and Correlation.	7
Unit 4	TRAFFIC ENGINEERING DESIGN Principle of road junction design, Design of intersections, Rotary intersections, Design of parking facilities.	5
Unit 5	TRAFFIC CONTROL DEVICES Need for international standardization, Traffic signs, Road markings and design of Traffic signals.	7
Unit 6	TRAFFIC REGULATIONS AND MANAGEMENT Basic principles of Regulation, Regulation of speed, Regulation of vehicles, Regulation concerning driver, Parking Regulations and Enforcement of Regulations, Traffic management measures	3

Unit 7	FUNDAMENTAL PRINCIPLES OF TRAFFIC FLOW Traffic flow elements, Flow density relationships, Mathematical relationships describing Traffic flow (Macroscopic and Microscopic models)	5
Unit 8	INTRODUCTION TO TRAFFIC SIMULATION SOFTWARES	2

Course Outcomes (COs):

At the end of the course the students will be able to:

1	Carryout different traffic surveys to collect the data, analyze and report in prescribed format.
2	Plan an urban highway utilizing concepts of traffic studies.
3	Design various geometric elements of a rotary, bus terminal, parking, traffic signals etc.
4	Apply the learned concepts and statistical methods in real life traffic engineering problems.

Textbooks:

Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Principles of Transportation Engineering”, Khitoliya R.K. and Gupta P.K, Dhanpat Rai Publishing, New Delhi.	2017
2.	“Text Book of Highway & Traffic Engineering”, Saxena S.C., CBS Publisher.	2020

Reference Book:

1.	“Traffic Engineering”, Roger P. Roess, Elena S. Prassas, Pearson Education.	2019
2.	“Traffic & Highway Engineering with MindTap”, Nicholas J. Garber.	2015

Sr. No	Code Name	Year
1	IRC:3-1983 Dimensions & Weights of Road Design Vehicles (First Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.003.1983.pdf	1983
2	IRC:54-1974 Lateral and Vertical Clearances at Underpasses for Vehicular Traffic https://law.resource.org/pub/in/bis/irc/irc.gov.in.054.1974.pdf	1974
3	IRC:64-1990, Guidelines for Capacity of Roads in Rural Areas (First Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.064.1990.pdf	1990
4	IRC:65-2017 Recommended Practice for Traffic Rotaries https://law.resource.org/pub/in/bis/irc/irc.gov.in.065.2017.pdf	2017
5	IRC:70-2017, Guidelines on Regulation and Control of Mixed Traffic in Urban Areas https://law.resource.org/pub/in/bis/irc/irc.gov.in.070.2017.pdf	2017
6	IRC:93-1985 Guidelines on Design and Installation of Road Traffic	1985

	Signals https://law.resource.org/pub/in/bis/irc/irc.gov.in.093.1985.pdf	
7	IRC:106-1990 Guidelines for Capacity of Urban Roads in Plain Areas https://law.resource.org/pub/in/bis/irc/irc.gov.in.106.1990.pdf	1990
8	IRC:SP:12-2015 Guidelines for Parking Facilities in Urban Roads” (First Revision) https://law.resource.org/pub/in/bis/irc/irc.gov.in.sp.012.2015.pdf	2015

Equivalent MOOCs courses

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105105107 “Transportation Engineering”	NPTEL
2	https://nptel.ac.in/courses/105101087 “Transportation Engineering 1”	NPTEL
3	https://archive.nptel.ac.in/courses/105/105/105105215/ “Transportation Engineering”	

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	1	3	1	1	2	3	1	1	3	2	1	3
CO2	3	3	2	1	1	2	-	-	2	3	1	2	2	2	2	3
CO3	3	3	3	3	1	3	1	1	3	1	2	3	3	3	1	3
CO4	3	3	3	3	1	3	-	-	3	3	-	3	3	3	3	3

- 1 - Slight(low) Correlation
2 - Moderate(Medium) Correlation
3 - Substantial(High) Correlation
“-” - indicates there is no Correlation

Course Name	:	Stochastic Hydrology	DEC
Course Code	:	CEE1017	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The students shall be able to:

- Understand the concepts of probability theory and stochastic processes.
- Analyse the distributional characteristics of hydrologic data.
- Apply statistical methods for analyzing hydrologic data
- Apply stochastic methodologies in forecast of hydrologic event.

Total No. of Lectures – 42

Lecture-wise breakup		No. of Lectures
Unit 1	BASIC CONCEPTS OF PROBABILITY AND STATISTICS Random variables, random experiments, Axioms of probability, Conditional, marginal, and joint probability, Total probability theorem and Bayes' rule, Discrete and continuous random variables.	3
Unit 2	BASIC STATISTICAL PROPERTIES OF DATA Descriptive statistics – measures of central tendency, dispersion, symmetry and tailedness, Concept of moments and expectation, Covariance and correlation, Parameter estimation – method of moments and maximum likelihood, Chebyshev inequality, Graphical representation of hydrologic data.	5
Unit 3	PROBABILITY DISTRIBUTION OF HYDROLOGIC VARIABLES Discrete probability distributions – Binomial, Geometric, and Poisson distributions, Continuous probability distributions – Uniform, Exponential, Normal, Lognormal, Gamma, and Extreme Value distributions, Flood frequency analysis.	6
Unit 4	HYPOTHESIS TESTING Hypothesis testing, Parametric and non-parametric estimation, Goodness-of-fit tests – Anderson-Darling test, Kolmogorov-Smirnov test, Chi-square test, Parametric tests and Non-parametric tests.	7
Unit 5	REGRESSION ANALYSIS Simple linear regression, Multiple linear regressions, Evaluation of regression model, Correlation and regression, Correlation and causality, Confidence interval.	5
Unit 6	TIME SERIES ANALYSIS Components of a time series, stationary and non-stationary time series, Homogeneity in a time series and its assessment, Trend analysis, Data generation techniques.	6
Unit 7	TIME SERIES MODELLING Properties of time series models, Auto-regressive (AR) models, Moving average (MA) models, Auto-regressive moving average (ARMA) model, Auto-regressive integrated moving average (ARIMA) model, Parsimony of time series models, Diagnostic check of models.	8

Unit 8	CASE STUDIES Case studies related to time series analysis and modelling for hydrologic applications.	2
---------------	--	---

Course Outcomes (COs):

At the end of the course the students will be able to:

1	Familiarize with the concepts of probability and statistical properties of hydrologic data.
2	Identify the distributional characteristics of hydrologic data and their applications.
3	Perform statistical hypothesis test for time series analysis and develop relationship between hydrologic variables.
4	Study the time series characteristics and development of statistical models for forecasting hydrologic time series.

Text Book:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	“Statistical Methods in Hydrology”, Hann, C.T., First East-West Press Edition, New Delhi.	2005
2	“Applied Hydrology”, Chow, V. T., Maidment, D. W., and Mays, L. W., Tata Mc-Graw Hill Education.	2010

Reference Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	“Statistical Models in Hydrology”, Clarke, R.T., John Wiley, Chinchester.	2004
2	“Stochastic Hydrology”, Reddy, P. J. R., Laxmi Publications.	2007.
3	“Statistical Methods in Hydrology and Hydroclimatology (Vol. 585)”, Maity, R., Springer.	2018
4	“Stochastic Water Resources Technology, Kottegoda, N. T., The Macmillan Press, New York.	2002

Equivalent MOOCs courses

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105/108/105108079/ - ‘Stochastic Hydrology’.	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	3	1	2	3	1	2	2	1	2	3	3	1	3
CO2	3	3	3	3	2	1	2	1	3	1	1	3	2	3	1	3
CO3	3	3	3	3	3	3	3	1	3	1	3	3	3	3	1	3
CO4	3	3	3	3	3	1	3	1	2	2	3	3	2	3	1	3

1 - Slight(low) Correlation

2 - Moderate(Medium) Correlation

3 - Substantial(High) Correlation

“-” - indicates there is no Correlation

Course Name	:	Environmental Pollution Management	DEC
Course Code	:	CEE1018	
Credit	:	4	
L T P	:	3 1 0	

Course Objectives	
The student should be able to • Imbibe the basic knowledge of the subject • Impart up-to date know of air, water pollution • Understand the basic principles of environmental pollution • Design various pollution control strategies	

Total No. of Lectures – 42

Lecture-wise break-up		No. of Lectures
Unit 1	WATER POLLUTION Water pollution sources, Water quality parameters, Types of water pollutants, Water quality modelling design of water and treatment facilities and, Ground Water Pollution and Remediation.	10
Unit 2	AIR POLLUTION Air quality index (AQI), Air pollution standards Influence of meteorological phenomena on air quality, Modelling and design Engineered systems for air pollution control.	15
Unit 3	NOISE POLLUTION AND CONTROL Definition, Psychological & Pathological effects of noise, Physical nature of noise. Sound, amplitude frequency, intensity loudness, noise pollution standards noise pollution control, Noise barriers, models	8
Unit 4	SOLID WASTE MANAGEMENT Generation, Collection, Processing and Disposal, and landfill design.	9

Course Outcomes (COs):	
At the end of the course students will be able to:	
1	Identify and evaluate the environmental pollution problems.
2	Design and suggest strategies for pollution control and management
3	Decide air and noise pollution control measures.
4	Propose method for Solid waste management.

Text Books		
S.no.	Name of Books/Author/Publisher	Year of Publications/
1	“Introduction to Environmental Engineering and Science”, Masters, G.M. and Ela WP, Pearson Education, Singapore	2021
2	“Environmental Engineering”, Peavy. H.S., Rowe, D.R. Tchobanoglous G, McGraw Hill, NY	2021

Reference Books		
1	“Air Pollution”, Rao, M. N. & Rao, H. V. N., Mc Graw Hill Ed.	2020
2	“Environmental Pollution Control Engineering”, Rao, C.S., New Age International Publishers	2021
3	“Environmental Engineering”, Gerard Kiely, Tata Mcgraw Hill.	2007

Equivalent MOOCs Courses

Sr.No.	Course Links	Offered by
1	https://nptel.ac.in/courses/105/104/105104099/ -NPTEL video course on Environmental Air Pollution	- NPTEL
2	https://nptel.ac.in/courses/105/105/105105178/ Treatment and Recycling - NPTEL video course on Wastewater	- NPTEL
3	https://www.edx.org/learn/environmental-science/world-bank-group-solid-waste-management	- NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO2
CO1	3	3	-	2	2	-	1	1	1	-	1	3	3	1	2	2
CO2	3	3	3	1	1	-	1	1	1	1	-	3	3	1	2	1
CO3	3	2	2	2	1	1	2	1	-	-	1	2	3	2	1	1
CO4	3	3	3	2	2	2	2	1	1	1	1	1	3	2	1	1

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Remote Sensing and GIS	DEC
Course Code	:	CEE1002	
Credits	:	4	
L – T – P	:	3 – 0 – 2	

Course Objectives:

- To make the students understand the concept of digital mapping
- To appraise the students with the recent advancements/software in surveying like Remote sensing, GIS, etc.

Total No. of Lectures – 42

Lecture-wise break-up		No. of Lectures
Unit 1	CONVENTIONAL MAPPING Conventional Mapping, Overview of conventional data collection techniques, Limitations, Need of interventions of technology	2
Unit 2	EVALUATION OF DIGITAL MAPPING AND PROCURING OF DATA Introduction to digital mapping, advantages, applications, data collection techniques, Present Status in India and abroad, Methodology of procuring of GIS/Remote Sensing data	5
Unit 3	AERIAL PHOTOGRAMMETRY AND DRONE MAPPING Types, Stereoscopy, Scale of a photograph, flight planning, Introduction to Drones applications, working and limitations	5
Unit 4	GEOGRAPHICAL INFORMATION SYSTEM (GIS) Introduction, advantages, objectives of GIS, Definitions of GIS, Components of GIS, Overlay analysis, DTM, DEM, Spatial, Non-Spatial, Raster, Vector	6
Unit 5	INTRODUCTION TO REMOTE SENSING (RS) Introduction, EMS, Ideal & Real RS System, Visual Image interpretation, active and passive remote sensing, Reflectance; SRC, Energy interaction in the atmosphere; energy interactions with the Earth's surface, Resolution	7
Unit 6	THERMAL AND MICROWAVE REMOTE SENSING Basics, Introduction, importance, need of thermal and microwave remote sensing	5
Unit 7	DIGITAL IMAGE PROCESSING (DIP) Introduction, Histogram, Rectification, Registration, Orientation, Radiometric errors, and Geometric errors. Image Enhancement Techniques, Image classification – Supervised and Unsupervised classification, Digital Image data Formats: BSQ, BIL, BIP, Change Detection	8
Unit 8	APPLICATIONS OF REMOTE SENSING IN ENGINEERING Applications of Remote Sensing, Case studies	4

Break-up of the Practical:		No. of Turns
1	Introduction to the basics of GIS and RS software	02
2	Digitization of features (linear, point, polygon) and editing	04
3	Addition of attributes, query generation	04

4	Methods of procuring and working of Remote Sensing Images	04
---	---	----

Course Outcomes (COs):

At the end of the course, the students will be able to:

1	Understand the evaluation of advanced surveying techniques
2	Understand the working of Remote Sensing and ancillary products
3	Understand the workings of GIS
4	Prepare digital maps using software

Textbooks:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Geomatics Engineering,” Manoj Arora and R C Barjatiya, Nem Chand Brothers, Roorkee.	2003
2	“Principles of GIS,” Peter A. Burrough, Rachael A., Oxford University Press.	2001
3	“Application of Geomatics in Civil Engineers”, Ghosh and D.A. Silva, Springer.	2020

Reference Books:

Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1	“Remote Sensing and Image Interpretation,” Lillesand and Kiefer, Wiley Publishers.	1999

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1	https://nptel.ac.in/courses/105/108/105108077/ NPTEL video course on Remote Sensing	NPTEL
2	https://nptel.ac.in/courses/105/107/105107160/ NPTEL video course on Digital Image Processing of Remote Sensing data	NPTEL
3	https://nptel.ac.in/courses/105/103/105103193/ NPTEL video course on Remote Sensing and GIS	NPTEL
4	https://nptel.ac.in/courses/105/107/105107201/ NPTEL video course on Remote Sensing Essentials	NPTEL
5	https://nptel.ac.in/courses/105/101/105101206/ NPTEL video course on Remote Sensing Principles and Applications	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	2	1	3	1	1	1	2	2	3	2	2	1	1	2
CO2	2	1	1	1	3	1	3	1	1	2	2	1	2	1	1	2
CO3	2	1	1	1	3	1	2	1	3	2	3	3	2	1	1	2
CO4	3	2	2	2	3	1	2	1	2	1	1	1	3	2	1	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Green Buildings Materials & Techniques	DEC
Course Code	:	CEE1019	
Credits	:	4	
L – T – P	:	3-1-0	

Course Objectives:

The student should be able to

- understand basics of green design and sustainable development concept.
- understand guidelines for development and certification of green designs.
- identify different types of green building materials.
- understand the principles of effective energy and resource management in buildings.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION TO GREEN BUILDINGS Definition of green buildings, definition of sustainability, typical features of green buildings, benefits of green buildings towards sustainable development, Case Studies.	3
Unit 2	GREEN BUILDINGS RATING SYSTEM & CODES Energy codes (ECBC) requirement, Certification & Rating system for green buildings, green rating for Integrated Habitat Assessment (GRIHA) and Leadership in Energy and Environmental Design (LEED), Indian Green Building Council (IGBC).	5
Unit 3	GREEN BUILDINGS MATERIALS Green building materials, Natural and renewable materials, Recycled materials, Environmental impact of building material, Material Efficiency, Embodied Energy in Building Materials, Operational energy in Building and life cycle analysis, methods to reduce embodied energy in building materials.	8
Unit 4	SITE SELECTION & PLANNING Criteria for site selection, planning for thermal comfort, day lighting, ventilation.	4
Unit 5	WATER CONSERVATION AND EFFICIENCY Rainwater harvesting methods for roof & non-roof, reducing landscape water demand by proper irrigation systems, water efficient plumbing systems, water metering, waste water treatment, recycle and reuse systems.	5
Unit 6	ENERGY CONSERVATION & EFFICIENCY Heating & Cooling Systems, Passive solar heating & cooling, Techniques for Maximizing energy performance, Use of BIPV and other renewable energy sources in buildings, power generation, Energy efficient building envelopes, Solar Heat Gain Coefficient, U-Values for facade materials, zero ozone depleting potential (ODP) materials, energy metering and monitoring, efficient lighting technologies, net zero energy buildings.	7
Unit 7	WASTE MANAGEMENT Solid waste management and techniques, Handling of construction & demolition waste materials.	5

Unit 8	INDOOR ENVIRONMENT QUALITY VOC emission issues & indoor air quality for sustainability, health hazards, measuring indoor air quality, ventilation system for improved air quality, Methods of improving indoor air quality.	5
---------------	---	---

Course Outcomes (COs):

At the end of the course the students will be able to:

1	Apply principles and practices of green buildings and its standards.
2	Identify appropriate green building material and technique.
3	use natural resources and energy for sustainable development.
4	Design sustainable and energy efficient buildings.

Text Book:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Sustainable Construction: Green Building Design and Delivery,” Charles J. Kibert, John Wiley & Sons	2012
2.	“Green Building Fundamentals,” Mike Montoya, Pearson, USA	2010

Reference Books:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Hand book of Green building Design and construction,” Sam Kubba, Elsevier Architecture Press.	2010
2.	“Green building: principals and practice in residential construction,” Abe Kruger and Carl Seville, Cengage Learning.	2012
3.	“Alternative building materials and technologies”, K.S. Jagadish, B.V. Venkatarama Reddy and K.S. Nanjunda Rao, New age international publishers	2007
4.	“Sustainable Construction and Design”, Regina Leffers, Pearson/ Prentice Hall, USA	2009
5.	IGBC Green New building rating system (Version 3.0)	2015
6.	Energy Conservation Building Code. Bureau of Energy Efficiency.	2018
7.	Sustainable Building Design Manual, Vol.1 and 2, TERI, New Delhi	2004

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	(https://archive.nptel.ac.in/courses/105/102/105102195/)- Sustainable Materials & Green Buildings, IIT Delhi	NPTEL
2.	(https://archive.nptel.ac.in/noc/courses/noc21/SEM1/noc21-ar03/)- Sustainable Architecture, IIT Roorkee	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	3	3	2	2	2	2	3	3	2	2	3
CO2	3	3	3	2	3	3	3	2	2	2	2	3	3	2	2	3
CO3	3	3	3	2	3	3	3	2	2	2	2	3	3	2	2	3
CO4	3	3	3	2	3	3	3	2	2	2	2	3	3	2	2	3

- 1 – Slight (Low) Correlation
- 2 – Moderate (Medium) Correlation
- 3 – Substantial (High) Correlation
- “-” indicates there is no correlation.

DEPARTMENTAL ELECTIVE COURSES-IV

Course Name	:	Bridge Engineering	DEC
Course Code	:	CEE1020	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The student should be able to

- develop an understanding of terminologies related to bridge engineering.
- analyse and design superstructure of various types of bridges.
- analyse and design substructure of bridges.
- Understand different construction techniques for bridges.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION Types of bridges, Investigation for bridges, Selection of suitable type of bridge, Design loads for highway and railway bridges as per IRC and IRS standards respectively.	8
Unit 2	RCC CULVERTS Design of RCC slab culvert.	4
Unit 3	RCC T-BEAM and SLAB BRIDGE Design of RCC T-Beam and slab bridge.	6
Unit 4	PRE-STRESSED CONCRETE BRIDGES Types of pre-stressed concrete bridges, losses in prestressing, Design considerations for pre-stressed concrete bridges.	4
Unit 5	STEEL PLATE GIRDER BRIDGE Design of steel plate girder railway bridge.	8
Unit 6	BRIDGE SUBSTRUCTURE Design considerations for bridge piers, abutment, and bridge foundations. Various types of bearings, Design of elastomeric bearings.	8
Unit 7	CONSTRUCTION METHODS Introduction to construction/erection methods for bridges.	4

Course Outcomes: By the end of this course, the student will be able to:

1	Carry out preliminary investigation for bridges.
2	Design superstructure of bridges.
3	Design substructure of bridges.
4	Classify different construction techniques.

Text Book:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Essentials of Bridge Engineering”, D. Johnson Victor, Oxford – IBH Publ. New Delhi.	2019
2.	“Design of Bridges”, N. Krishna Raju, Oxford-IBN Publisher.	2019

Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Design of Concrete Bridges”, M. G Aswani, V. N Vazirani, M. M Ratwani, Khanna Publishers.	2004
2.	“Bridge Deck Analysis”, R.P.Pama & A.R.Cusens, John Wiley & Sons.	
3.	“Design of Steel Structures”, Arya & Awadhesh Kumar, Nem Chand and Bros., Roorkee.	2022
4.	“Design of Bridge Structures”, T.R.Jayaram & M.A.Jagadeesh, Prentice Hall of India, New.Delhi.	2009
5.	Relevant IS and IRC codes (Latest Revision)	

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	(https://nptel.ac.in/courses/105105177/)- Reinforced Concrete Road Bridges, IIT Kharagpur	NPTEL
2.	(https://archive.nptel.ac.in/courses/105/105/105105216/)- Bridge Engineering, IIT Dhanbad	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	3	2	1	2	2	2	3	3	2	2	3
CO2	3	3	3	2	3	3	2	1	2	2	2	3	3	2	2	3
CO3	3	3	3	2	3	3	2	1	2	2	2	3	3	2	2	3
CO4	3	3	3	2	3	3	2	1	2	2	2	3	3	2	2	3

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

Course Name	:	Transportation Planning and Management	DEC
Course Code	:	CEE1003	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

- To make the students learn about basics of transportation planning process.
- To determine the travel demand using different techniques of trip generation, distribution, mode choice and network assignment and various land-use patterns
- To make the student to learn about basic concepts of traffic management techniques and fundamentals of transport management system.
- To introduce the student to transport policy and transportation-environment.

Total No. of Lectures – 42

Lecture-wise break-up		No. of Lectures
Unit 1	GENERAL CONCEPT AND PROCESS General Concept and Process, Hierarchical Levels of Planning, Passengers and Goods Transportation.	4
Unit 2	URBAN TRAVEL CHARACTERISTICS Urban Travel Characteristics, Private & Public Travel Behaviour analysis, Introduction to urban freight transportation.	4
Unit 3	TRAVEL DEMAND Travel demand Estimation, Forecasting methods and Models, Trip Generation methods, Model Split analysis, Trip Distribution- Growth factor method, Gravity models, Intervening opportunity	14
Unit 4	NETWORK CAPACITY AND DISTRIBUTION Network assignments, Capacity Restrained and simultaneous distribution, Direct demand models	5
Unit 5	TRANSPORT PLANNING Land-use Transport Planning, State-wide and Regional Transportation Planning	2
Unit 6	TRANSPORT POLICY Transport and energy, transport and environment, Transport management (policy, organization, legal provisions), integration and coordination	3
Unit 7	TRANSPORT MANAGEMENT Fundamentals of Traffic Management, Principles and Methodology, Traffic Systems Management, Technique of management, Exclusive Bus Lanes Traffic Management Techniques, Speed control and Zoning, Parking Control, Traffic Segregation and Channelization, Principles and Design of Traffic Signs, Their Placement and Visibility. Transportation System Management, Route and Network Management, Area Traffic Management, City wide Traffic Control and Management, Centralized Data Processing and Monitoring.	10

Course Outcomes:	
By the end of this course, the student will be able to:	
1	Apply the learned concepts of traffic planning.
2	Estimating travel demand based on different forecasting methods, trip generation, distribution, modal split and network assignment.
3	Ability to apply traffic management techniques.
4	To understand the transport-environment and transport policy.

Suggested Textbooks:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Transport Engineering and Planning”, Papacostas C.S., Pearson Education India.	2015

Reference Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication/ Reprint
1.	“Transport Planning & Traffic Engineering”, O’Flaherty Coleman A., CRC Press.	2018
2.	“Transport Planning and Traffic Safety: Making Cities, Roads, and Vehicles Safer”, Tiwari G. and Mohan D., CRC Press.	2016

Equivalent MOOC courses

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105/107/105107067/ ‘Urban transportation planning’.	NPTEL
2	https://nptel.ac.in/courses/video/105106058/ ‘Introduction to transportation planning’	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	-	2	2	1	1	3	1	3	3	-	-	3
CO2	3	3	3	-	1	3	-	1	1	2	1	2	3	3	1	3
CO3	3	2	3	2	1	3	1	2	1	2	1	3	3	3	1	3
CO4	3	-	-	1	-	3	3	1	1	-	1	2	3	2	1	3

1 - Slight(low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” - indicates there is no Correlation

Course Name	:	Water Resources Planning and Management	DEC
Course Code	:	CEE1021	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The students shall be able to:

- Understand the aspects related to planning and management of water resources.
- Analyse the water resources project with system approach.
- Apply the applications of system engineering.
- Develop methodologies for Water Resources Planning and Management.

Total No. of Lectures – 42

Lecture-wise breakup		No. of Lectures
Unit 1	WATER RESOURCES PLANNING Role of water in national development, Assessment of water resources, Planning process, Long-term planning, Functional requirements in multipurpose planning.	4
Unit 2	ELEMENTS OF WATER RESOURCES DEVELOPMENT Reservoir planning - dependable yield, Sequent peak algorithm, Sedimentation in reservoir, Reservoir capacity, Empirical-area reduction method, Reservoir operation – standard operating policy.	8
Unit 3	WATER RESOURCES PROJECT ANALYSIS Planning process and data needed for planning, Project analysis, Project optimality conditions and profitability analysis.	6
Unit 4	WATER RESOURCES SYSTEM ENGINEERING Concept of system engineering, Optimal policy analysis, Linear and dynamic programming, Integrated river basin development.	8
Unit 5	APPLICATION OF SYSTEM APPROACH IN WATER RESOURCES Application of system engineering in practical problems like hydrology, irrigation and drainage engineering, Mathematical models for forecasting and other water resources related problems.	10
Unit 6	ECONOMIC AND FINANCIAL ANALYSIS OF WATER RESOURCES PROJECTS Meaning and nature of projects economics theory, Micro and macroeconomics, Benefit cost analysis – cost and benefit curves, cost and benefit estimation, Discounting factors and techniques, Cost allocation and comparison of alternatives.	6

Course Outcomes (COs):	
At the end of the course the students will be able to:	
1	Understand the planning and operation of reservoirs.
2	Apply the systems approach in analyzing the water resources systems.
3	Explore the applications of system engineering to practical problems in water resources engineering.
4	Assess the economic and financial aspects of a water resources project.

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	“Water Resources Systems Engineering”, Hall, W. A., and Dracup, J. A., McGraw-Hill Book Company, New York.	2007
2	“Water resources Systems Planning And Management”, Jain, S. K., and Singh, V. P., Elsevier.	2003

Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	“Water Resources Systems Planning and Management– An introduction to methods, models and applications, Studies and Reports in Hydrology”, Loucks, D.P. and Eelco van Beek, UNESCO Publishing.	2005
2	“Economics of Water Resources Planning”, James, L. D., and Lee, R. R, McGraw-Hill.	2007
3	“Water Resources Systems Planning and Management”, Chaturvedi, M. C., Tata McGraw-Hill Publishing Company.	2010
4	“Water Resources Systems”, Vedula S., and Mujumdar P. P., Tata McGraw Hill.	2005

Sr. No	Course link	Offered By
1	https://nptel.ac.in/courses/105/108/105108081/ - ‘Water Resources Planning and Management’	NPTEL
2	https://nptel.ac.in/courses/105/108/105108130/ - ‘Water Resources Systems: Modelling Techniques and Analysis’.	NPTEL

Equivalent MOOC courses

	PO1	PO2	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	3	1	1	2	1	2	2	3	2	3	3	1	3
CO2	3	3	1	2	1	2	1	3	2	2	3	2	3	1	3
CO3	3	3	1	3	1	2	1	3	2	2	3	3	3	1	3
CO4	3	3	1	3	1	2	1	2	2	2	3	2	3	1	3

1 - Slight(low) Correlation

2 - Moderate(Medium) Correlation

3 - Substantial(High) Correlation

“-” - indicates there is no Correlation

Course Name	:	Environmental Impact Assessment	DEC/MS
Course Code	:	CEE1004/CEM2002	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The student should be able to

- identify various aspects of EIA processes
- interpret environmental management plans and EIA documents
- plan in accordance with the environmental legislation and acts, role of MOEFCC, pollution control boards/committees
- compare and evaluate different alternatives in a project/ activity

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION TO EIA EIA origin, development, purpose and aims, core values and principles of EIA.	4
Unit 2	LEGAL ASPECTS OF EIA Environmental legislation in India, Environmental Appraisal procedures in India, EIA Gazette notifications of the year 1994, 2006 & thereafter, Environmental clearance process.	6
Unit 3	PROJECT AND THE ENVIRONMENT DESCRIPTION Environmental components of EIA, Identification of key issues, Terms of Reference, Baseline studies, Prediction and assessment of impacts on Air, Noise, Water, Soil & Geological, Biotic, Socio- economic, Cultural and Aesthetic environment.	8
Unit 4	METHODS OF IMPACT ANALYSIS AND EVALUATION Ad Hoc Methods, Checklists, Matrices, Networks, Environmental index, Cost Benefit analysis, Simulation methods, Overlays and GIS, and professional judgements etc. Aims and objectives of involvement of public and other stakeholders, Public involvement methods, approaches for EIA reviewing, Economic efficiency and valuation methods.	8
Unit 5	ENVIRONMENTAL MANAGEMENT PLAN AND EIA DOCUMENTATION Principles, Anticipated environmental impacts, Mitigation measures, Identification, comparison and evaluation of different alternatives, Mitigation measures, Preparation of EIA documents.	8
Unit 6	CASE STUDIES OF EIA	8

Course Outcomes: By the end of this course, the student will be able to:

1	Decide the environmental requirements applicable to the environmental impact assessment, and the environmental clearance process of developmental projects
2	Evaluate the environmental impacts of developmental projects
3	Apply suitable methods and tools of prediction of environmental impacts
4	Recommend mitigation measures in environment management plan
5	Compare and evaluate different alternatives in a project

Text Book:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Environmental impact assessment”, L.W. Canter, McGraw Hill Co.	1996
Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Environmental impact assessment”, R.R. Barthwal, New Age International (P) Ltd. Publishers, New Delhi.	2002
2.	“Environmental impact assessment: Available Techniques and Emerging Trends”, S.A. Abbasi & D.S. Arya, Discovery Publishing Pvt. Ltd.	Latest
3.	EIA Gazette Notifications such as SO 60 (E) dated 27-01-1994, MOEF, GOI, New Delhi. SO 1533 dated 14-09-2006, MOEF, GOI, New Delhi. SO 1199 (E) dated 23-03-2020, MOEFCC, GOI, New Delhi.	Latest

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	https://nptel.ac.in/courses/120/108/120108004/ - Environment Management.	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	2	3	3	2	2	2	1	3	3	1	2	3
CO2	3	3	3	2	2	3	3	2	2	2	1	3	3	1	2	3
CO3	3	3	3	2	2	3	3	2	2	2	1	3	3	1	2	3
CO4	3	3	3	2	2	3	3	2	2	2	1	3	3	1	2	3
CO5	2	2	1	2	2	1	2	2	2	2	1	2	2	1	2	2

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation

Course Name	:	Reinforced Soil Analysis and Design	DEC
Course Code	:	CEE1022	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The students should be able to:

- Understand the concepts of reinforced soil.
- Understand the different types and various applications of geosynthetic.
- Apply the knowledge of reinforced soil in the Civil Engineering field.
- Design the reinforced soil walls.

Total No. of Lectures: 42

Lecture-wise breakup		No. of Lectures
Unit 1	INTRODUCTION Basic concept of the Reinforced soil, Mechanism Design principles, Materials used for Construction, Advantages of reinforced soil.	04
Unit 2	PRACTICAL APPLICATIONS Reinforced soil in Civil Engineering structures, Basic components and strength characteristics of reinforced soil, Reinforced soil construction detailing.	06
Unit 3	GEOSYNTHETICS An overview of Geosynthetics, Description of Geotextiles, Geogrids, Geonets, Geomembranes, Geo-composites, Geocells-properties and test methods, Functions, Design methods for separation, stabilization, filtration, Drainage.	05
Unit 4	RETAINING WALLS Types of walls; Earth pressures for gravity/counterfort walls; structural design of wall and its foundation; stability of the wall-soil system; Slip circle analysis	04
Unit 5	REINFORCED SOIL WALLS Stability Analysis and construction aspects of Reinforced Soil Walls, Effect of reinforced sloped backfill on Soil wall design, Drainage design procedure.	06
Unit 6	WALL WITH REINFORCED BACKFILL Theoretical analysis, Pressure-Intensity on the wall, stability against sliding and overturning, Design procedure, Limitations of the analysis.	05
Unit 7	FOUNDATIONS ON REINFORCED SOIL Brief overview, Analysis of strip footing, isolated- square and rectangular footing on reinforced soil bed, Determination of Pressure Ratio.	06
Unit 8	SOIL NAILING AND ANCHORS Applications of Soil Nailing, Its components, Advantages and Limitations, Design aspects.	06

Course Outcomes (COs): By the end of the course, the students will be able to:	
1	Learn the characteristics of the different types of reinforcing material.
2	Choose appropriate reinforcement material.
3	Design the structures using reinforced soil.
4	Construct the various structures using the appropriate materials.

Text Book:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Designing with Geosynthetics,” Robert M. Koerner, Prentice Hall.	2012
2.	“Engineering with Geosynthetics,” G.V Rao & GVS Suryanarayana Raju, Tata Mc GrawHill Publishing Co. New Delhi.	1990
3.	“Reinforced Soil and its Engineering Application,” Swami Saran, New Age Publication.	2006
Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Fundamentals of Geosynthetic Engineering,” Shukla, S.K. and Yin, J.H., Taylor & Francis.	2006
2.	“Geotechnical Engineering,” Gulati, S.K. and Datta, M. Tata Mc Graw Hill Publishing Co. New Delhi.	2005

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	https://nptel.ac.in/courses/105/106/105106052 /NPTEL video course on Geosynthetics and Reinforced Soil Structures	NPTEL
2.	https://nptel.ac.in/courses/105/101/105101143 /NPTEL video course on Geosynthetics Engineering: Theory and Practices	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	1	2	2	3	3	-	-	1	1	-	3	1	-	3
CO2	2	2	2	1	3	2	3	-	-	1	1	-	3	2	-	3
CO3	3	3	3	3	2	3	3	-	-	1	1	-	3	3	-	3
CO4	3	3	2	2	2	3	3	-	-	1	1	-	3	3	-	3

1 – Slight (Low) Correlation
2 – Moderate (Medium) Correlation
3 – Substantial (High) Correlation
“-” indicates there is no correlation.

Course Name	:	Clean Technology	DEC
Course Code	:	CEE1023	
Credit	:	4	
L T P	:	3 1 0	

Course Objectives
The student should be able to • To analyse the prevailing clean technologies • To understand their design aspect • To develop knowledge of the concept • To apply principles of clean technology

Total No. of Lectures- 42

Lecture-wise break-up		No. of Lectures
Unit 1	MASS AND ENERGY TRANSFER FUNDAMENTALS Material balance, energy fundamentals, Environmental chemistry, Mathematics for resource consumption and population growth Introduction to energy science and energy technologies	10
Unit 2	ENERGY AND THERMODYNAMICS Earth and thermodynamics system, Global energy situation and system, fossil energy, Solar energy fundamentals and applications, Solar Photovoltaic system, Geothermal and Wind energy, Urban waste to energy conversion, Biomass energy, Energy storage system, Fuel Cells, Hydrogen and Methanol, Hydro energy, Energy and Environmental Audit.	10
Unit 3	GREEN MATERIALS, GREEN BUILDINGS Principles of Sustainability and Green Buildings, Design Features, Rating Systems Material, Energy Saving Features, Smart Cities.	7
Unit 4	RISK ASSESSMENT Perspective and Perception of risk, Risk assessment, Exposure assessment and Comparative risk analysis	7
Unit 5	ENVIRONMENTAL LEGISLATIONS AND POLICIES Environmental Laws, Air act, Water act, Environment act, Forest act, Environmental Impact assessment and management systems, International and National efforts for Environment Protection, Current Environmental Issues	8

Course Outcomes (COs):	
At the end of the course students will be able to:	
1	Choose technology options for sustainable development
2	Decide strategies for waste minimization and elimination.
3	Design clean technology systems.
4	Analyse green engineering practices in energy & manufacturing sector

Text Books:		
S. No.	Name of Books/Author/Publisher	Year of Publications / Reprint
1	“Clean Technology” Johansson Allan, Lewis Publisher	2022
2	“Introduction to Environmental Engineering and Science” Masters, G.M. and Ela WP, Pearson Education, Singapore	2001
Reference Books :		
S.no.	Name of Books/Author/Publisher	Year of Publications / Reprint
1	“Alternative building Materials and Technologies” Jagadish, K. S. Venkataramareddy B. U. New Age International	2007
2	“Renewable Energy for sustainable future”, Godfrey Boyle, Oxford University Press	2012
3	“Energy Technology”, Rao, S. & Parulekar, B.B, Khanna Publishers	2020

Equivalent MOOCs Courses

Sr.No.		Offered by
1	(https://www.edx.org/learn/environmental-science) world-bank-group-e-learning-course-on-greentech-mainstreaming-technologies-in-green-blue-and-clean-operations	EDX
2	(https://www.edx.org/course/enviromental-technologies-in-buildings)- Course on Sustainable Building Design.	EDX
3	(www.ciwmb.ca.gov/Greenbuilding/basics.htm#what)---Green Building Basics, California Integrated Waste Management Board	CIWMB

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO2
CO1	3	2	-	3	2	-	2	1	1	-	2	3	2	2	1	2
CO2	3	3	2	1	2	-	1	2	1	1	-	3	3	1	1	2
CO3	3	2	3	2	2	1	1	1	-	-	2	2	3	1	2	1
CO4	3	3	2	2	3	2	3	1	2	1	2	1	3	2	1	1

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

OPEN ELECTIVE COURSES (OEC)

Course Name	:	Disaster Management	OEC
Course Code	:	CEO1001	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

- To analyze the causes and effects of various natural and man-made disasters.
- To explore hazard and risk assessment.
- To enable to plan various mitigation measures and capacity building.
- To develop infrastructure for disaster management.
- To apply applications of Remote Sensing in disaster mitigation.

Total No. of Lectures- 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION TO DISASTER MANAGEMENT Natural Disasters, Man-made hazards, causes, consequences, earthquakes, Floods, Drought, Coastal Hazards, Landslides, rockslides and Forest Fires, Tsunamis, Oil spillage, Gas leakage, etc.	05
Unit 2	DISASTER MITIGATION AND PREPAREDNESS Mitigation measures for natural disasters- Earthquakes, Tsunamis, Cyclones, Floods, Landslides, etc.; Mitigation measures for various industrial hazards/disasters; Preparedness for natural disasters.	07
Unit 3	HAZARD AND RISK ASSESSMENT Assessment of capacity, vulnerability and risk; vulnerability and risk mapping; stages in disaster recovery and associated problems.	05
Unit 4	EMERGENCY MANAGEMENT SYSTEMS Emergency medical and essential public health services, response and recovery operations; Reconstruction and rehabilitation.	05
Unit 5	CAPACITY BUILDING Disaster management approach to inculcate new skills and sharpen existing skills of government officials, voluntary activists, development of professional and elected representative for effective disaster management, an overview of disaster management in India.	08
Unit 6	INTEGRATION OF PUBLIC POLICY Planning and design of infrastructure for disaster management; Community based approach in disaster management; Methods for effective dissemination of information; Ecological and Sustainable development models for disaster management.	06
Unit 7	ROLE OF REMOTE SENSING IN DISASTER MITIGATION Importance of Remote Sensing in disaster mitigation, Case studies	06

Course Outcomes:	
1	To annotate the causes and effects of various disasters.
2	To plan risk management/disaster mitigation techniques.
3	To evaluate the mitigation measures (Case study).
4	To demonstrate role of Remote Sensing in Disaster mitigation.

Textbooks:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication/ Reprint
1.	“Elements of Earthquake Engineering,” Jai Krishna, Chandrasekharan and B. Chandra, South Asian Publishers, New Delhi.	2000
2.	“Disaster Management,” R.B. Singh (Ed), Rawat Publications.	2006
3.	“Disaster Management –Future Challenges & Opportunities,” Jagbir Singh, I.K. International Publishing House.	2007

Reference Books:		
Sr. No.	Name of Book/Authors/Publisher	Year of Publication/ Reprint
1.	“Natural Hazards in the Urban Habitat,” Iyengar, C.B.R.I., Tata McGraw Hill. Publications.	1997
2.	“Natural Disaster Management,” Jon Ingleton (Ed), Tudor Rose, Leicester.	1999
3.	“Earthquake Resistant Design of Structures,” Pankaj Agarwal and Manish Shrikhande, Prentice Hall of India Pvt. Ltd.	2006

Equivalent MOOCs courses:		
Sr. No.	Course Links	Offered by
1.	Natural Hazards https://onlinecourses.nptel.ac.in/noc20_ce07/preview	NPTEL
2.	Disaster Management https://archive.nptel.ac.in/courses/105/104/105104183/	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	1	2	3	2	1	1	1	1	1	1	1	1	1	3
CO2	2	2	1	2	2	2	3	1	2	2	2	3	3	2	2	2
CO3	1	2	3	1	2	1	3	1	2	3	3	2	1	1	2	1
CO4	3	3	3	3	1	1	2	1	3	1	1	3	3	3	3	3

1 – Slight (Low) Correlation
 2 – Moderate (Medium) Correlation
 3 – Substantial (High) Correlation
 “-” indicates there is no correlation.

Course Name	:	Green Buildings	OEC/MSC
Course Code	:	CEO1002/CEM2001	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The student should be able to

- understand basics of green design and sustainable development concept.
- understand guidelines for development and certification of green designs.
- identify different types of green building materials.
- understand the principles of effective energy and resource management in buildings.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	INTRODUCTION TO GREEN BUILDINGS Sustainability, Principles of sustainable development, Definition of Green building, Building envelope, design features, Case Studies.	2
Unit 2	GREEN BUILDINGS RATING SYSTEM & CODES Energy codes (ECBC) requirement, Certification & Rating system for green buildings, green rating for Integrated Habitat Assessment (GRIHA) and Leadership in Energy and Environmental Design (LEED), Indian Green Building Council (IGBC)	4
Unit 3	ENVIRONMENTAL IMPACT OF BUILDING MATERIALS Building materials: sources, methods of production and their environmental impact, Material Efficiency, Embodied Energy in Building Materials, Operational energy in Building and Life cycle energy.	4
Unit 4	INTRODUCTION TO GREEN MATERIALS Local building materials, natural and renewable materials: bamboo, timber, rammed earth, stabilized mud blocks, materials with recycled content such as blended cements, pozzolana cements, fly ash bricks, vitrified tiles, materials from agro and industrial waste, reuse of waste and salvaged materials.	6
Unit 5	THERMAL COMFORTS IN BUILDING Heat Transfer Characteristic of Building Materials, Effect of Geographical Locations, Building Techniques related to thermal comfort.	5
Unit 6	UTILITY OF SOLAR ENERGY IN BUILDINGS Incidence of Solar Heat on Buildings, Concepts of Solar Cooling and Heating of Buildings, Low Energy Cooling, Solar PV system.	5
Unit 7	DAYLIGHTING Principles of light, transmission, reflection and absorption, illumination, day lighting concepts and its requirements.	5
Unit 8	PLANNING FOR VENTILATION Functions of ventilation, provision for air movement, air flow through buildings, orientation, external features, cross ventilation, position of openings, size of openings, humidity control.	5
Unit 9	MANAGEMENT TECHNIQUES FOR GREEN BUILDINGS	

	Energy conservation techniques, Water & waste-water management, solid management principles & techniques, indoor environmental quality measurements & improvement techniques.	6
--	---	---

Course Outcomes (COs):

At the end of the course the students will be able to:

1	Apply principles and practices of green buildings and its standards.
2	Understand the importance of sustainable use of natural resources and energy.
3	Distinguish appropriate green building material and technique
4	Design sustainable and energy efficient civil engineering project.

Text Book:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Sustainable Construction: Green Building Design and Delivery,” Charles J. Kibert, John Wiley & Sons	2012
2.	“Green Building Fundamentals,” Mike Montoya, Pearson, USA	2010

Reference Books:

Sr. No.	Name of Book/Authors/Publisher	Year of Publication /Reprints
1.	“Hand book of Green building Design and construction,” Sam Kubba, Elsevier Architecture Press.	2010
2.	“Green building: principals and practice in residential construction,” Abe Kruger and Carl Seville, Cengage Learning.	2012
3.	“Alternative building materials and technologies”, K.S. Jagadish, B.V. Venkatarama Reddy and K.S. Nanjunda Rao, New age international publishers	2007
4.	“Sustainable Construction and Design”, Regina Leffers, Pearson/ Prentice Hall, USA	2009
5.	IGBC Green New building rating system (Version 3.0)	2015
6.	Energy Conservation Building Code. Bureau of Energy Efficiency.	2018
7.	Sustainable Building Design Manual, Vol.1 and 2, TERI, New Delhi	2004

Equivalent MOOCs courses

Sr. No.	Course Links	Offered by
1.	(https://archive.nptel.ac.in/courses/105/102/105102195/)- Sustainable Materials & Green Buildings, IIT Delhi	NPTEL
2.	(https://archive.nptel.ac.in/noc/courses/noc21/SEM1/noc21-ar03/)- Sustainable Architecture, IIT Roorkee	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	3	3	2	2	2	2	3	3	2	2	3
CO2	3	3	3	2	3	3	3	2	2	2	2	3	3	2	2	3
CO3	3	3	3	2	3	3	3	2	2	2	2	3	3	2	2	3
CO4	3	3	3	2	3	3	3	2	2	2	2	3	3	2	2	3

- 1 – Slight (Low) Correlation
- 2 – Moderate (Medium) Correlation
- 3 – Substantial (High) Correlation
- “-” indicates there is no correlation.

Course Name	:	Clean Technology	OEC
Course Code	:	CEO1003	
Credit	:	4	
L T P	:	3 1 0	

Course Objectives	
The student should be able to • To analyse the prevailing clean technologies • To understand their design aspect • To develop knowledge of the concept • To apply principles of clean technology	

Total No. of Lectures – 42

Lecture-wise break-up:		No. of Lectures
Unit 1	MASS AND ENERGY TRANSFER FUNDAMENTALS Material balance, energy fundamentals, Environmental chemistry, Mathematics for resource consumption and population growth Introduction to energy science and energy technologies	10
Unit 2	ENERGY AND THERMODYNAMICS Earth and thermodynamics system, Global energy situation and system, fossil energy, Solar energy fundamentals and applications, Solar Photovoltaic system, Geothermal and Wind energy, Urban waste to energy conversion, Biomass energy, Energy storage system, Fuel Cells, Hydrogen and Methanol, Hydro energy, Energy and Environmental Audit.	10
Unit 3	GREEN MATERIALS, GREEN BUILDINGS Principles of Sustainability and Green Buildings, Design Features, Rating Systems Material, Energy Saving Features, Smart Cities.	7
Unit 4	RISK ASSESSMENT Perspective and Perception of risk, Risk assessment, Exposure assessment and Comparative risk analysis	7
Unit 5	ENVIRONMENTAL LEGISLATIONS AND POLICIES Environmental Laws, Air act, Water act, Environment act, Forest act, Environmental Impact assessment and management systems, International and National efforts for Environment Protection, Current Environmental Issues	8

Course Outcomes (COs):	
At the end of the course students will be able to:	
1	Choose technology options for sustainable development
2	Decide strategies for waste minimization and elimination.
3	Design clean technology systems.
4	Analyse green engineering practices in energy & manufacturing sector

Text Books:

S. No.	Name of Books/Author/Publisher	Year of Publications /Reprint
1	“Clean Technology” Johansson Allan, Lewis Publisher	2022
2	“Introduction to Environmental Engineering and Science” Masters, G.M. and Ela WP, Pearson Education, Singapore	2001
Reference Books :		
S.No	Name of Books/Author/Publisher	Year of Publications /Reprint
1	“Alternative building Materials and Technologies” Jagadish, K. S. Venkataramareddy B. U. New Age International	2007
2	“ Renewable Energy for sustainable future”, Godfrey Boyle, Oxford University Press	2012
3	“Energy Technology”, Rao, S. & Parulekar, B.B, Khanna Publishers	2020

Equivalent MOOCs Courses

Sr.No.		Offered by
1	(https://www.edx.org/learn/environmental-science) world-bank-group-e-learning-course-on-greentech-mainstreaming-technologies-in-green-blue-and-clean-operations	EDX
2	(https://www.edx.org/course/enviromental-technologies-in-buildings)-Course on Sustainable Building Design.	EDX
3	(www.ciwmb.ca.gov/Greenbuilding/basics.htm#what)---Green Building Basics, California Integrated Waste Management Board	CIWMB

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO2
CO1	3	2	-	3	2	-	2	1	1	-	2	3	2	2	1	2
CO2	3	3	2	1	2	-	1	2	1	1	-	3	3	1	1	2
CO3	3	2	3	2	2	1	1	1	-	-	2	2	3	1	2	1
CO4	3	3	2	2	3	2	3	1	2	1	2	1	3	2	1	1

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.

MINOR SPECIALIZATION COURSES

Course Name	:	Structural Analysis And Design	MSC
Course Code	:	CEM1001	
Credits	:	4	
L T P	:	3 0 2	

Course Objectives :

At the end of this course, the student should be able to:

- Understand the concepts of material behavior and their application in structural analysis.
- Analyze fundamental structural elements such as beams, columns, and trusses.
- Design simple reinforced concrete and steel structures.

Total No. of Lectures – 42

Lecture wise breakup		No. of Lectures
Unit 1	SIMPLE STRESSES & STRAINS Concept of stresses and strains, relationship between elastic constants, extension of uniform bar under its own weight and due to load applied, stresses produced in compound bars due to axial to loads, thermal stresses.	4
Unit 2	SHEAR FORCE, BENDING MOMENT AND DEFLECTION IN BEAMS Shear force, bending moment, Relation between load, SF and BM, SFD , BMD and axial force diagram for determinate beams under various types of loading, Slope and Deflection in beams by double integration method, Macaulay's method under the action of various loading conditions.	8
Unit 3	BENDING AND SHEAR STRESSES IN BEAMS Pure bending ,bending stresses, Variation of shear stresses for various cross-sections of a beam.	5
Unit 4	BUILDING MATERIALS Introduction to various type of commonly used material- stones, bricks, cement, concrete, steel, wood, their properties and applications.	7
Unit 5	INTRODUCTION TO DESIGN OF RCC STRUCTURES Properties of Concrete, Design Philosophies, Design of simple beams, slabs, columns and isolated footings.	9
Unit 6	INTRODUCTION TO DESIGN OF STEEL STRUCTURES Steel as construction Material, Design concepts of tension, compression and flexural steel members and connections. Introduction to design of roof trusses.	9

List of Experiments:		Number of Turns
1	To determine standard consistency of cement and its Setting Time.	2
2	To determine Fineness of Cement.	1
2	To determine Specific Gravity of Cement.	1
3	To determine Compressive Strength of Cement.	2

4	To determine Water absorption and Efflorescence of Bricks	2
5	To determine Compressive Strength and Hardness of Bricks.	2
7	To determine bulk density and voids of fine and coarse aggregates.	2
8	To determine fineness modulus and grain size distribution of fine and coarse aggregates.	2
Course Outcomes:		
At the end of this course, the student will be able to:		
1	Analysis simple structures under different loading conditions.	
2	Understand the use of various construction materials.	
3	Design simple reinforced concrete elements.	
4	Design simple steel structural elements.	

Text Book:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	"Mechanics of Materials", Punmia and Jain, Laxmi Publications (P) Ltd.	2017
Reference Books:		
Sr. No.	Name of Book/ Authors/ Publisher	Year of Publication / Reprint
1	"Mechanics of Materials", R.C.Hibbeler, Pearson Higher Education	2018
2	"Structural Analysis", R.C. Hibbeler, Prentice Hall of India Pvt. Ltd.	2021
3	"Limit state design of Reinforced Concrete", Punmia and Jain (Vol.II), Laxmi Publications.	2016
5	"Limit state design of Steel Structures", S.K. Duggal, Tata McGraw Hill. Publications.	2017
6	"Reinforced Concrete Limit State Design" A.K.Jain, Nem Chand and Bros.	2012

Equivalent MOOCs Courses

Sr. No.	Course Links	Offered by
1	(https://archive.nptel.ac.in/courses/105/105/105105166/)---Structural analysis I, IIT Kharagpur	NPTEL
2	(https://archive.nptel.ac.in/courses/105/101/105101086/)--- Structural analysis II, IIT Bombay	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1	1	2	1	1	2	1	1	3	3	3	1	2
CO2	3	2	2	1	1	2	2	1	3	2	2	3	3	3	2	2
CO3	3	2	2	2	2	2	2	2	2	2	2	3	3	3	1	2
CO4	3	2	2	2	2	2	2	2	2	2	2	3	3	3	1	2

- 1 – Slight (Low) Correlation
2 – Moderate (Medium) Correlation
3 – Substantial (High) Correlation
“-” indicates there is no correlation.

Course Name	:	Intelligent Transportation Systems	MSC
Course Code	:	CEM1002	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:	
- To make the students understand the transportation system engineering processes along with describing the concepts of transport system architecture and their evolution and to make students understand the capability of key technologies. - Develop a solid understanding of the key concepts, components, and technologies that form the foundation of Intelligent Transportation Systems. - Study vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication systems, and their role in improving traffic safety and efficiency. - Learn to collect, process, and analyze transportation data from various sources, including sensors, cameras, and GPS devices, to make informed decisions.	

Total No. of Lectures - 42

Lecture-wise break-up		No. of Lectures
Unit 1	INTRODUCTION TO INTELLIGENT TRANSPORTATION SYSTEMS Intelligent Transportation Systems (ITS), Advanced Traveller Information Systems (ATIS)	4
Unit 2	COMPONENTS OF ITS ITS Standards and Architecture; Organizational structure	4
Unit 3	TELECOMMUNICATIONS ITS Telecommunications	4
Unit 4	ITS TECHNOLOGY Interactive Voice Recognition (IVR), automated highway systems (AHS); sensors, electronic toll collection (ETC); dedicated short range communication and standards.	4
Unit 5	APPLICATION ON PUBLIC TRANSPORT Expected Time of Arrivals, Electronic Ticketing Machines, Automated Vehicle Location.	5
Unit 6	APPLICATIONS TO HIGHWAY SAFETY ITS Highway Safety Perspective	4
Unit 7	APPLICATIONS TO ENVIRONMENT ASPECTS Environmental Aspects of ITS; Connected Vehicle Technology and Applications	5
Unit 8	SECURITY ITS and Security; ITS Policy Issues	3
Unit 9	ECONOMICS Economics of ITS – Revenue Generation Models	4
Unit 10	CASE STUDIES Applications in bus transport, metro and highways	5

Course Outcomes (COs):	
At the end of the course the students will be able to:	
1	Understand the impact of technology on different modes and movement.
2	Select appropriate ITS technology depending upon site-specific conditions.
3	Differentiate different user services.
4	Understand the systems engineering application in ITS and ITS architecture

Suggested Textbooks:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Intelligent Transport Systems” Sarkar P.K. and Jain A.K. PHI Learning	2018

Reference Book:		
Sr. No	Name of Book / Authors/Publisher	Year of Publication/ Reprints
1.	“Perspectives on Intelligent Transportation Systems” Sussman, J.M.() Springer,Berlin.	2010
2.	“Intelligent Transportation Systems: New Principles and Architectures” Ghosh, S., Lee, T.S., CRC Press.	2000
3.	Fundamentals of IntelligentTransportation Systems Planning, Mashrur A. Chowdhury, and Adel Sadek,Artech House, Inc.	2003

Equivalent MOOC courses

Sr. No	Course link	Offered By
1	https://www.civil.iitb.ac.in/~vmtom/nptel/591_ITS_1/web/web.html ‘Introduction to Intelligent transportation systems - I’.	NPTEL
2	https://www.civil.iitb.ac.in/tvm/nptel/592_ITS_2/web/web.html - Introduction to Intelligent transportation systems -II’.	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	-	2	-	-	-	-	1	-	2	2	1	1	-	1	3
CO2	1	2	2	1	-	1	2	1	2	2	1	3	2	3	1	3
CO3	2	2	3	1	2	3	2	1	3	2	1	3	3	3	2	3
CO4	3	1	3	2	1	2	2	-	-	3	2	1	1	2	2	3

1 - Slight(low) Correlation

2 - Moderate(Medium) Correlation

3 - Substantial(High) Correlation

“-” - indicates there is no Correlation

Course Name	:	Water and Waste Water Engineering	MSC
Course Code	:	CEM1003	
Credits	:	4	
L T P	:	3 1 0	

Course Objectives:

The student should be able to

- Impart up-to date know of water pollution
- Understand the basic principles of water and waste-water pollution
- Design various pollution control strategies for water
- Design various pollution control waste-water

Total No. of Lectures - 42

Lecture-wise break-up		No. of Lectures
Unit 1	WATER SUPPLY Introduction; Need for and importance of public water supply, water quality standards for drinking water (BIS & WHO) standards, ill effect of contaminated waters on human health.	4
Unit 2	TREATMENT OF WATER - I Objectives, various treatments methods, sedimentation – theory and principle involved, various types of settlings, design of type-I (discrete settling) sedimentation tanks, sedimentation aided with coagulation and flocculation, chemistry of coagulation, different types of coagulants, design of mixing, flocculation tank and clarifiers, filtration theory and mechanism, comparison between slow and rapid sand filters, design criteria, design of rapid sand filters. Water disinfection, various methods, disinfection by chlorination and its various forms, breakpoint chlorination. Miscellaneous treatment Water softening objective, different methods of water softening, lime & soda process, zeolite process etc.	15
Unit 3	WASTE-WATER TREATMENT Introduction and need of waste-water treatment, implications of disposal of waste on rivers and land. Waste-water treatment methods – preliminary, primary, secondary, and tertiary treatment methods, hydraulic flow chart of a conventional sewage treatment plants, design of screening chambers, grit chambers, skimming tanks and primary settling tanks, biological treatment and various methods, tertiary treatment. detail and design criteria, oxidation ponds and ditches.	15
Unit 4	SEWAGE DISPOSAL Disposal methods for sewage, land, river and lake disposal, self purification of rivers and streams, zone of pollution in rivers, D.O. deficit and oxygen sag, kinetics of D.O. deficit (Streeter – Phelps equation) and its applications; Disposal of sewage on land – disposal and its implications, sewage sickness, disposal in lakes and sea.	4
Unit 5	SLUDGE TREATMENT AND DISPOSAL	4

Course Outcomes (COs):	
By the end of this course the student will be able to:	
1	Propose the water supply system and treatment.
2	Design the waste water facilities.
3	Explain the concept of wastewater disposal and management.
4	Explain the concept of sludge treatment and management

Text Books :		
Sr. No.	Name of Books/Author/Publisher	Year of Publications/ Reprint
1	“Environmental Engineering”, Howard S. Peavy, D.R. Rowe, G. Tchobanglous, McGrawHill Book Co., New Delhi	1985
2	“Environmental Engineering -1 Water Supply Engineering”, Santosh Kumar GargKhanna Publishers, New Delhi	2010
Reference Books :		
Sr. No.	Name of Books/Author/Publisher	Year of Publications/ Reprint
1	“Environmental Engineering–A Design Approach”, A P Sincero, G A Sincero, PHI, New Delhi,(EEE Edition)	1996
2	“Manual on Water Supply Engineering”, Expert Committee Ministry of urbanDevelopment, New Delhi	1993
3	“Environmental Engineering-1 (Water Supply Engineering)”, B.C. Punia, A.K. Jain & A.K. Jain Laxmi Publication, New Delhi	1995

Equivalent MOOCs Courses

Sr. No.		Offered by
1	https://nptel.ac.in/courses/105/104/105104102/ - video course on ‘Water andWaste Water Engineering’.	NPTEL
2	https://nptel.ac.in/courses/105/106/105106119/ - video course on ‘Water andWaste Water Engineering’	NPTEL
3	https://nptel.ac.in/courses/105/105/105105178/ - video course on ‘Waste waterTreatment and Recycling’	NPTEL
4	https://nptel.ac.in/noc/courses/noc20/SEM1/noc20-ce23/ - video course on ‘Water Supply Engineering’	NPTEL

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO2
CO1	3	3	-	2	2	-	1	1	1	-	1	3	3	1	2	2
CO2	3	3	3	1	1	-	1	1	1	1	-	3	3	1	2	1
CO3	3	2	2	2	1	1	2	1	-	-	1	2	3	2	1	1
CO4	3	3	3	2	2	2	2	1	1	1	1	1	3	2	1	1

1 – Slight (Low) Correlation

2 – Moderate (Medium) Correlation

3 – Substantial (High) Correlation

“-” indicates there is no correlation.